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Instituto de Ciências Matemáticas e de Computação

Asymptotics of Partition Functions of Coulomb Gases

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Concentration Area: Mathematics

Advisor: Prof. Dr. Guilherme Lima Ferreira Silva

USP – São Carlos
July de 2026

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*"En remontant chez moi pour y passer la soirée à travailler de mon mieux,
je me disais que le monde n'est pas construit pour l'équilibre.*

Le monde est désordre.

L'équilibre n'est pas la règle, c'est l'exception.

Et je faisais le serment de travailler pour l'ordre et l'équilibre.

Que de serments! Tu vas sourire."

G.Duhamel, Maitres, 1937

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For all of those who were before and are now:

Your courage, patience, daring, and caring has brought me here.

To hoping that I become deserving of such act, for I feel loved and dear; of immense luck.

For all of those who will be after:

To promise that I can for you as it was done for me.

*“João, é bom ver que você aprendeu e fez tantas coisas bonitas!
Continue assim, muito feliz! Brinque e aprenda muito.”
Te amo muito, Papai.*

RESUMO

JOÃO V. A. PIMENTA. **Assintóticas de Funções Partição em Gases de Coulomb**. 2026. 126 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

As matrizes aleatórias podem ser vistas de duas formas: como uma função mensurável de um espaço de probabilidade num conjunto de matrizes ou, mais concretamente, como uma matriz cujas entradas são variáveis aleatórias com alguma distribuição conjunta. Um conjunto de tais matrizes com uma dada distribuição arbitrária é designado por ensemble. Para ensembles de interesse, pode-se estudar estatísticas como autovetores e distribuições de autovalores. As simetrias do conjunto escolhido irão refletir sobre quais modelos tais estatísticas podem representar. Por exemplo, as distâncias dos valores característicos de uma matriz simétrica com coeficientes aleatórios modelam as distâncias entre níveis sucessivos de energias de núcleos atômicos pesados. Mais interessante ainda, existe uma certa universalidade nas estatísticas encontradas entre classes de ensembles, sugerindo a relevância da teoria no estudo da universalidade nos fenômenos modelados em geral. Particularmente importante é o estudo de tais matrizes quando estas crescem infinitamente, considerando assintoticamente o seu comportamento estatístico. O nosso foco é uma técnica para derivar assintóticas para a chamada função partição de alguns ensembles interessantes de matrizes aleatórias.

Palavras-chave: Teoria de Matrizes Aleatórias, Ensembles de Matrizes Planares, Assintótica.

ABSTRACT

JOÃO V. A. PIMENTA. **Asymptotics of Partition Functions of Coulomb Gases** . 2026. 126 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

Random Matrices can be thought of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. A set of such matrices with a given arbitrary distribution is called an ensemble. One can, for ensembles of interest, study statistic such as the eigenvectors and eigenvalue distributions. The symmetries of chosen set will reflect on what models such statistics can model. For example, distances of the characteristic values of a symmetric matrix with random coefficients model distances between successive levels of energies of heavy nuclei. More interestingly, there is some universality in the statistics found among classes of ensembles, suggesting the relevance of the theory in studying the universality in general modeled phenomena. Particularly important is the study of such matrices when they grow infinitely large, taking asymptotically its statistical behavior. We focus on a technique for deriving asymptotics for the so-called partition function of some interesting ensembles of random matrices.

Keywords: Random Matrix Theory, Planar Matrix Ensembles, Asymptotics.

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INTRODUCTION

We start our story with one of the primordial sciences: Astronomy. Two infamous personalities occupy the setting for the first concepts we discuss, namely the astronomers Tycho Brahe (1546 – 1601) from Denmark and Johannes Kepler (1571 – 1630) from Germany. Tycho was perhaps the best observational astronomer we ever knew of, his observations were precise and all encompassing. His obsession and careful noting with the motion of the sky objects is (still unmatched) unheard. Kepler has a more mathematical approach to astronomy, there is a reason (to think of Kepler when we talk about the laws of orbiting bodies) that to think of Kepler is usually of think of his mathematical laws for orbits. Tycho actually met Kepler and asked him to work as his assistant, unfortunately, not much time passed between this event and Tycho's death. However, the incredible amount of observations made during his lifetime would go on to empower Kepler to correctly deduct and improve his theories. Isaac Newton (1643 – 1727), The English polymath, who doesn't require introductions as one the most prolific and successful physicists, astronomers and mathematicians in history, came later to derive by first principles the laws of gravitation and explain the observational theory. At this point, the motion of planets was well understood and there was a good idea on how to work with other heavenly bodies.

There was some difficulty, however, when it came to the motion of comets and like bodies from outer space. The observations were scarce and imprecise, these bodies made appearance once or only a couple times during a lifetime if one were lucky enough find it. The theory of gravitational laws appointed ellipsis and parabolas as contenders for trajectories, but the observational data were still lacking. How to, from a small set of error-prone data, derive the original curves of motion? The problem is better stated by Gauss (1777 – 1855).

To determine the orbit of a heavenly body, without any hypothetical assumption, from observations not embracing a great period of time, and not allowing a selection with a view to the application of special methods. (GAUSS, 1809)

It was Gauss who solved the problem. From our current understanding, Gauss used a type of *Maximum Likelihood Estimation* method, that is, he tried to derive parameters of the underlying curve from a set of observational points without any previous knowledge or assumption. The idea is to try to minimize some error function that is correlated with the distance of the observational points to the predicted curve. One might remember for example of the *least squares* method. As mentioned, observational data carried errors, to perform such methods Gauss would need to understand how to weight the distance of the observational point to the theoretical curve. Surely, points very far from a predicted curve should be few, this is assumption one: The error function must be decaying. Another assumption was that this error function was symmetric, an even function. These assumptions led Gauss to the normal curve, or what it sometimes called the Gaussian distribution.

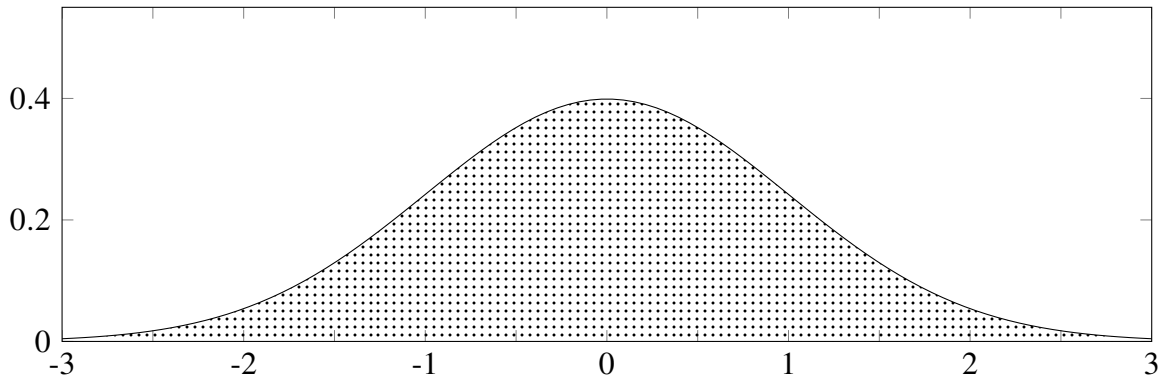


Figure 1 – Plot of $e^{(-x^2)/2}/\sqrt{2\pi}$, the Gaussian distribution.

Today, it is well-known among physicists and mathematicians alike that random independent phenomena share common characteristics when dealt in large quantities. The most important and known example is surely the convergence to the normal curve. Supposing you're working under some reasonable assumptions on the random variable at hand, if one looks at the distribution of the mean of such random variables when the number of samples increases one should see convergence to a known curve: The normal. The most important assumptions here is that you're dealing with a distribution with a somehow contained tail - finite moments, and independent samples. This may be the core result in much of applied statistical research. This is formally expressed in terms of the central limit theorem: Take $\{x_n\}_{n \geq 1}$ independent and identically distributed random variables, if we center and re-scale the variables $x_n \rightarrow y_n := (x_n - \mathbb{E}(x_n))/\sqrt{\text{Var}(x_n)}$, then we have the limit

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(\frac{\sum_{k=1}^n y_k}{\sqrt{n}} \leq t\right) = F^{(G)}(t; 0, 1)$$

where $F^{(G)}(t; 0, 1)$ is simply the cumulative distribution for the normal distribution. This is an amazing result as it appoints the Gaussian distribution as a universal distribution for a large number of uncorrelated random processes.

There is however, many phenomena where these assumptions don't hold. One can think for example of an interacting system of particles as an example of correlated phenomena. For

these cases another universality result must be considered, one where the Gaussian distribution is substituted with a more appropriate distribution. An important historical example appeared in physics in the last century. When working with heavy nuclei, physicist around the 1950's struggled a lot to properly understand what was going on with the system compared with the absolute success of quantum mechanics in simpler nuclei such as hydrogen. This type of work was condemned to be impossible: Theoretically, these atoms are huge and the interactions and intern structures were barely known. Dyson (1923 – 2020), a British-American mathematician and physicist, described the situation as follows.

We picture a complex nucleus as a 'black box' in which a large number of particles are interacting according to unknown laws. The problem then is to define in a mathematically precise way an ensemble of systems in which all possible laws of interaction are equally probable. (DYSON, 1962)

Physically, to understand a system one might want to write out its Hamiltonian - in a way, a expression of its energy. This takes the form of a matrix with size of the order of the agents you have in the system. For large nuclei, you would deal with huge matrices. Computationally, those were also useless, computers were not, and are not expected to be, at a point of evaluating meaningfully, even if one understood well the system, the evolution of such system. That was a typical experiment that was done to study properties of such nuclei and that physicists were very interested in understanding theoretically the results. The nuclei scatter is a technique of projecting neutrons in nuclei at multiple energies to detect resonance energies. These energies are related to the internal structure and of much use to researches. In a conference on nuclear physics in 1956, Eugene Wigner (1902 – 1995), a Hungarian-American theoretical physicist, presented his work regarding the theoretical statistics of the spacing distribution of adjacent neutron resonances in heavy nuclei. He said the following.

Perhaps I am now too courageous when I try to guess the distribution of the distances between successive levels (of energies of heavy nuclei). Theoretically, the situation is quite simple if one attacks the problem in a simpleminded fashion. The question is simply what are the distances of the characteristic values of a symmetric matrix with random coefficients. (WIGNER, 1956)

The fact is that Wigner had had a breakthrough that would change the approach in correlated systems and introduce Random Matrix Theory (RMT). He found a way to model scattering energies as eigenvalues of a specific model of random matrix - thought of simply as a matrix of random entries with some given distribution, correlated or not. To understand how and what, we must think of what it means to model a system by a random matrix. For that, Deift (1945), a South-African mathematician, has a very nice analogy.

This procedure can be viewed as follows: A scientist wishes to investigate some statistical phenomenon. What he has at hand is a microscope and a handbook of matrix ensembles. The data $\{a_k\}$ are embedded on a slide which can be inserted into the microscope. The only freedom that the scientist has is to center the slide, $a_k \rightarrow a_k - A$, and then adjust the focus $a_k - A \rightarrow \tilde{a}_k = \gamma_a(a_k - A)$ so that on average one data point $\tilde{a}_k$ appears per unit length on the slide. At that point the scientist takes out his handbook, and then tries to match the statistics of the $\tilde{a}_k$'s with those of the eigenvalues of some ensemble. If the fit is good, the scientist then says that the system is well-modeled by RMT. (DEIFT, 2006)

Wigner understood that the energies he was studying had the same statistic as the eigenvalues of random matrices from the so-called *Gaussian Ensemble* by the procedure described by Deift. This ensemble is nothing more than the set of hermitian matrices with independent random centralized and normalized Gaussian variables. The distribution of eigenvalues is given by the famous Wigner Semi-Circle.

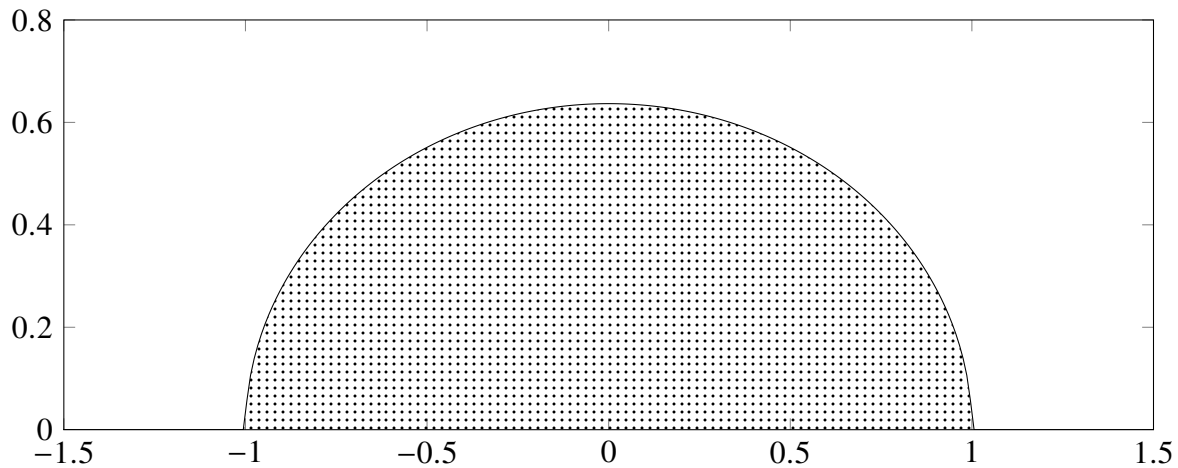


Figure 2 – Plot of the Semi-Circle.

In the bulk, the distribution can be described by the so-called Sine Kernel, the same statistic found in the atomic energies described by Wigner. This distribution also have other interesting kinks. For some point e in the edge of the distribution

$$p\left(e + \frac{x}{cN^{2/3}} \leq t\right) = F(x)$$

we have a special joint probability density function: the Tracy-Widom.

A very classical model for these distributions is the problem of the maximal increasing sub-sequence, know by the name of one of its researcher, the Polish mathematician Stanislaw Ulam (1909 – 1984), as *Ulam's problem* (ULAM, 1961). Suppose you take a sequence of integer number and shuffle them in what we call a permutation. There are many possible permutations, and we will consider for this application that they are all equally likely. Naturally one can get for

example

$$\pi = 54172386$$

as a permutation of the numbers $\{1, 2, 3, 4, 5, 6, 7, 8\}$. One can notice that the number are not necessarily increasing or decreasing globally, but that it may be the case locally and even more the case if one is willing to ignore the non-conforming numbers. That is, take for example the sub-sequence of π given by 1236. Note that these numbers are not sequential but nevertheless are an increasing sub-sequence of the original permutation. Suppose that we consider the distribution of sub-sequences lengths. For many reasons one might be interested in the maximal increasing sub-sequence l_N . Of course, calculating $l_N(\pi)$, for π , is a trivial exercise and for larger sequences, only requires patience and will power. However, in most applications one wants to understand the typical size of such maximal large sequences, i.e., what size one might expect to get from a random sampled permutation of the numbers. Some direct assumptions can be made: this size must depend on a parameter n , the size of the initial sequence. Of course, it cannot be bigger than n or smaller than 1; nevertheless it stands to reason that this size must have some scaling with n .

This problem is also highly linked with the problem of the *matching sequences* (MAJUMDAR, 2007), or more descriptively, the problem of the longest common sub-sequence. We deal again with a permutation of some alphabet of symbols, before we used the integers only allowing each number to appear once. Now, we could still use numbers but for a non-suspicious motive we will use the four letter alphabet *ATCG* and allow repetition of symbols in a sequence. Consider for example the sequences α, β given by

$$\begin{array}{lcl} \alpha & : & A \ C \ G \ C \ T \ A \ C \\ \beta & : & C \ T \ G \ A \ C \end{array}$$

Again, we want to look for, non necessarily adjacent, sub-sequences of symbols. For example, a sub-sequence of α could be *CGAC*, corresponding to the second, third, sixth and seventh letters. As it so happens, we chose a sub-sequence that is common to both α and β . Moreover, as β is of size 5, this might just be the longest (one of) matching sequences as long as β is not contained as a sub-sequence in α . For the biology interested reader, an application of this type of metric may have become clear. The fact is that DNA are sequences of a four lettered alphabet such as α and β . It is our understanding that genes responsible for specific proteins evolve with time and that, finding this longest common sequences between genes of different species may indicate what is conserved through evolution. One may also be interested in checking that two genes are the same, or if they are considerably different. Just by chance, of course, one must expect the longest matching sub-sequence to be relatively large for a few-lettered alphabet. However, if one understands whats the expectation of such length and how lengths deviate from this mean, it gets considerably easier to make statements that two sequences are evolutionary close or far apart.

Note that in both cases these variables can not be understood as independent random variables and that the classical central limit theorem is useless for its study in the enunciated form.

For the longest increasing subsequence problem, historically, [Ulam \(1961\)](#), discussed the idea of using Monte Carlo approaches to study the asymptotic behavior of $\mathbb{E}[l_N]$, the expectation of the longest increasing sequence length when $N \rightarrow \infty$. Based on his simulations, Ulam conjectured that it exists a constant c such that $\mathbb{E}[l_N] = \sqrt{N}c$ when N went to infinity. Later on, [Hammersley \(1972\)](#), introduced a Poissonized version of this problem and formally proved that the constant c exists. Five years afterwards, [Logan and Shepp \(1977\)](#) and, independently, [Vershik \(1977\)](#) showed that $c = 2$. Much later, [Odlyzko and Rains \(2000\)](#), performed analytical simulations indicating that

$$\lim_{N \rightarrow \infty} \frac{\text{Var}[l_N]}{N^{\frac{1}{3}}} = c_0 \equiv 0,819 \dots$$

and also that

$$\lim_{N \rightarrow \infty} \frac{\mathbb{E}[l_N] - 2\sqrt{N}}{N^{\frac{1}{6}}} = c_1 \equiv -1.748 \dots$$

At this point, the expectation of l_N was well known. However, the proper variable had not been taken face on. Finally, Ulam's problem was solved by [Baik, Deift and Johansson \(1999\)](#), who studied the convergence in distribution of the re-scaled variable $\chi_N := (l_N - 2\sqrt{N})/N^{\frac{1}{6}}$ and discovered that it converges in distribution to

$$\lim_{N \rightarrow \infty} \mathbb{P}\left(\frac{l_N - 2\sqrt{N}}{N^{\frac{1}{6}}} \leq x\right) = F_{(2)}(x).$$

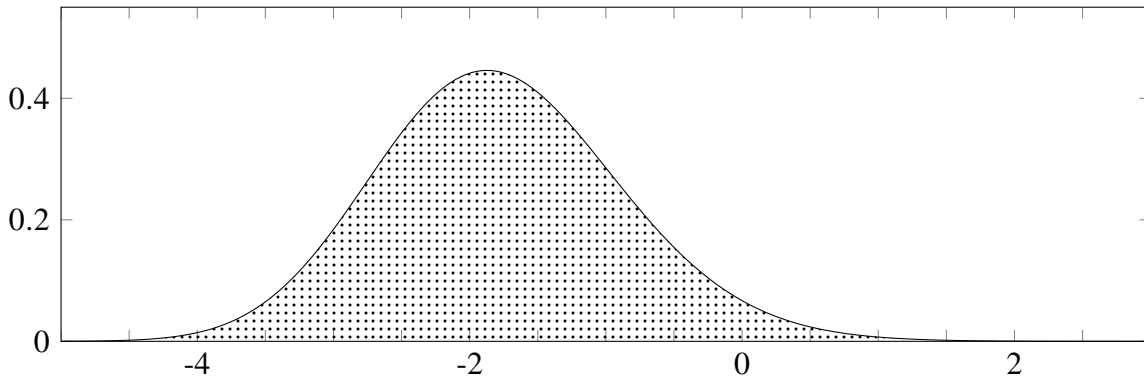


Figure 3 – Plot of $F_{(2)}(x)$.

The daunting resemblance to the central limit theorem should be noted. This cumulative distribution function $F_{(2)}(x)$ ended up being known now as the *Tracy-Widom*. By good measure we must also state a result from the matching sequence example. The result is only valid for a simplified model: The *Bernoulli matching model*; got if one ignores some correlation that appears in the modeling. This is not completely bad as these correlations tend to zero when the number of letters in the alphabet increases. Name the size of the longest matching sequence $L_{i,j}^{(BM)}$ where i, j are the size of the first and second sequences used. If $i = j = n$, for large n , one gets that

$$L_{n,n}^{(BM)} \rightarrow \gamma_c^{(BM)} n + f(c) n^{1/3} \chi$$

where $\gamma_c^{(BM)} = 2/(1 + \sqrt{c})$, χ is a random variable with a n -independent distribution and $p(\chi \leq x) = F_2(x)$ is precisely the Tracy-Widom distribution, again.

This special function appears many other times in many other different models and phenomena. It appears to be related to the universality of phase transitions - such as the physical transition of water or superconductivity. One of the indications of this development was the work of Majumdar and Schehr that shown that for large N , the Tracy-Widom to describe the crossover function between stable and unstable regimes of a random dynamical system - a phase transition, such as the ones observed in two-dimensional Quantum Chromodynamics ([KIEBURG; VERBAARSCHOT; ZAFEIROPOULOS, 2014](#)). Eventually physicists realized that the energy curves of certain strongly correlated systems have a kink at $\sqrt{2N}$. The associated peak for these systems is the Tracy-Widom distribution, which appears in the third derivative of the energy curve - that is, the rate of change of the rate of change of the energy's rate of change.

More than that, for a large class of random matrices called Wigner Matrices, independent of individual probability density of matrices, the eigenvalue distribution tends to the Wigner's Semi-Circle. It represents a class of universality in matrices. Interestingly enough, the eigenvalue dynamics can be often interpret as a physical gas of charged interacting particles submitted to some confining external potential know as a Coulomb gas. Note that such particles would repel each other, however unnatural this might sound for eigenvalues of random matrices, this is what is observed: Eigenvalues configurations with two extremely close eigenvalues are much less likely. This creates a bridge to a theory of dynamical systems and point processes that lead us to consider what happens when such gases are taken to the thermodynamic limit. That is, what happens when we increase the number of charged particles, the mass number of an atomic nuclei or even the length of the sequences considered before. This give us under some appropriate considerations, to asymptotic equilibrium configurations of the points and is a major field of study in RMT. It also motivates ways of simulating this equilibrium measures, namely, using usual particle dynamics methods such as in [Chafaï and Ferré \(2018\)](#).

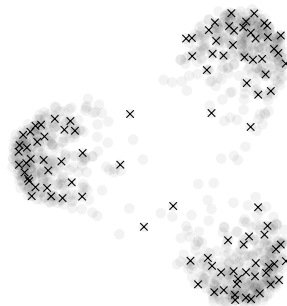


Figure 4 – Plot of a simulation of a Coulomb gas. In black crosses we plotted one sample from the simulation. In shadowy dots, any “averaged” samples indicating the equilibrium region.

A coulomb gas can be thought of as a measure μ - a mathematical probability distribution, especially it can be thought of as the the Gibbs Measure. If you take a random array of points $\xi = \{x_1, x_2, \dots, x_n\}$, then this measure gives you a probability that your system is in such configuration. This distribution takes the well known form

$$\mu(\xi) = \frac{e^{-\beta \mathcal{H}(\xi)}}{\mathcal{Z}_n^\beta}.$$

Here, β is just as usual in physics, a parameter that stands for the inverse temperature of the system. Higher β 's means the system is very cold, with less energy - the particles would be moving less. Alternatively, a lower β would give you an energetic system - the particles would likely spread more as they would be with higher energy. The function $\mathcal{H}$ is related to the system static energy, it is called the Hamiltonian in the physics literature. It is a composition of the internal energy of the system, given by the potential energy of the coulomb interactions between particles, and the potential energy of the particles by the application of an external field Q . in two dimensions,

$$\mathcal{H}(\xi) = n \sum_{i=1}^n Q(x_i) - \frac{1}{2} \sum_{i \neq j}^n \log(x_i - x_j).$$

This gives a mathematical interpretation for the idea that two particles oppose to being very close. Starting with a configuration with two particles at a point x_c or very close points x_c^+ , x_c^- , the log would evaluate the distance between these particles as a infinitely negative number. In this way, $\mathcal{H}$ would be very positively large and therefore μ would be extremely low, indicating low, or zero, probability of such configuration. The same reasoning applies to understanding why particles don't diffuse very far for a sufficiently fast growing external potential Q , if a particle is very far from its minimum, the configuration would be rendered non-likely.

There is a omission so far, we have not discussed the fact that a probability measure should add up, summing all micro-states, to 1. This is the role of the function $\mathcal{Z}_n^\beta$, it is a normalizing function called the *partition function*. The letter $\mathcal{Z}$ stands for the German word *Zustandssumme*, meaning the "sum over states". It works by basically any other normalization, summing over all states such that now every state probability is normalized to sum to 1. However this is a very physically interesting function; micro-state's energy is a function of many thermodynamic variables, with a model of the system one can calculate the energy for each state and therefore calculate the partition function. As such, the partition function holds in some way the macro expression of the micro-states energy distribution. This is more direct if one is familiar with the usual way of thermodynamics and know that many macro-state such as internal energy, entropy and the alike can be expressed as functions of the partition function.

Explicitly knowing the expression for a partition function is no easy task, the summation over *all* micro-states can be a quite elusive idea. A typical method in thermodynamics is to consider the thermodynamic limit, that is, the state of the system when the number of particles grows very large (the volume grows accordingly). The idea is that fluctuations of any quantity

will become negligible when considering a very large amount of particles, in this way, one can better say a systems has some mean quantity without much deviation. This is the same technique used in mathematics, we will evaluate such function when the number of particles is very large. Even then, the exact expression is a very hard task for many systems. We will perform an asymptotic analysis to get some approximate expression like

$$\ln(\mathcal{Z}_{N,\beta=2}^Q) = -I_Q N^2 + \frac{1}{2} N \ln N + E_Q N + G \ln N + F^Q + o(1).$$

Interestingly, these terms have very nice physical interpretations: The term I_Q is given by the energy of the associated equilibrium distribution and E_Q is an entropy term! Other coefficients can also be useful although with less straightforward interpretations.

Recapitulating, Wigner noted that modulo some symmetry considerations, all nuclei in the same symmetry class exhibited universal behavior; and that in all cases, the energies exhibited repulsion, or, more precisely, the probability that two energy levels would be close together was small. If one equips the space of symmetric matrices with a probability measure of smooth density, the probability of a matrix $\hat{M}$ having equal eigenvalues would be zero and the eigenvalues $\lambda_1, \dots, \lambda_n$ of $\hat{M}$ would be a random set with repulsion already built in. With this, Wigner had a model with repulsion. But why choose the Gaussian Orthogonal Ensemble, composed of real Gaussian random matrices? This is where the universality constraint comes into play. We quote from Wigner, where he mentioned the previously used matrices defined by [Wishart \(1928\)](#) to study covariance matrices in real world data.

“Let me say only one more word. It is very likely that the curve in [Figure 1] (referring to the Semi-Circle) is a universal function. In other words, it doesn't depend on the details of the model with which you are working. There is one particular model in which the probability of the energy levels can be written down exactly. I mentioned this distribution already in Gatlinburg. It is called the Wishart distribution. ...”

So in this way Wigner introduced a model with built-in repulsion and in which the energy level distribution could be calculated explicitly.

This type of approach became better know and phenomena in the same universality class started to pup-up many times. A well know story regards H. Montgomery (1944), an American mathematician, and Dyson. Montgomery works with analytical number theory and is better know for his *pair correlation conjecture* on the zeros of the Riemann zeta function $\zeta(s)$ - perhaps one of the best known analysis open problem today. The idea is that Montgomery found a way to re-scale the non-trivial zeros of the Riemann zeta function and find a pair correlation between these objects. That is, he found some statistical information on the distribution of such zeros, a very hard problem. During a visit in the 1970's to the Institute for Advanced Study in Princeton he was suggested to present his result to Dyson. Before Montgomery could state his result however, Dyson took out a pen and predicted some function that looked like

$$R(a, b) = \int_a^b 1 - \left(\frac{\sin(\pi r)}{\pi r} \right)^2 dr.$$

Of course, Montgomery should have felt nothing short of deep confusion as how his result should be that obvious for Dyson. The explanation for such a guess is that Dyson believed that the zeros of the zeta function might behave like the eigenvalues of a random Gaussian Unitary Ensemble (GUE) matrix (composed of unitary matrix with Gaussian entries), and, were that the case, this would be exactly the formula for the two-point correlation function. This was derived from the well known by the time, Sine Correlation Kernel.

This of course, was incredibly interesting and was looked for in all sorts of phenomena. Another interesting result is the work of [Abul-Magd \(2006\)](#), that investigated the gap distances between parked cars to find that these distances behaved as the distances of eigenvalues for the GUE. The explanation given was that drivers tended to park near each other but still tried to keep a reasonable distance to maneuver safely, in other words, they behaved just like a charged particle gas with a confining potential! Another instance of this type of result was in the 1990's when two Czech Physicists took interest in the transportation system of the city of Cuernavaca, in Mexico. In this work, [Krbálek and Seba \(2000\)](#) suspected that the peculiarities of the system public buses could result in a result expressed in terms of known objects in RMT. There were no public buses in the city, buses were private and did their own timetables only following the approved routes designed by the city. As there were no coordination for the buses times, sometimes buses were right before another line and could not take anyone. To avoid that, the city installed sensors and started to sell the information about buses leaving the stations. Now, with this information, buses started to avoid each other but of course, still wanted to do the route as soon as possible to avoid losing clients. This resulted in the same behavior noted before: buses were behaving as the aforementioned gas!

This modeling was repeated for many other highly correlated processes and was very successful as a general theory of random correlated systems. This can now be observed in the plenitude of posterior systems modeled with such statistics, for example: the previously cited Ulam's Problem ([ULAM, 1961](#)) and Matching Random Sub-sequences ([MAJUMDAR, 2007](#)) but also in the Zeros of the Riemann Zeta Function ([GOLDSTON; MONTGOMERY, 1987](#)), Tiling problems (Aztec Diamond for instance) and Interacting Random Walks, to name a few.

1.1 Overview of the dissertation

Our main goal with this work is to obtain, following the recent results of [Byun, Kang and Seo \(2023\)](#), asymptotic expressions, on the number of particles, for the normalizing function - called partition function - of the distribution of eigenvalues for the ensemble of matrices we denominate planar ensembles. There is much to unpack in this sentence. Namely, one can mention that such distributions of eigenvalues behaves as a fluid gas of interacting particles under some external potential; we call such system a Coulomb gas. Such gases carry some physical implications that we make good use in the mathematical theory. One can imagine that if the

potential is confining, with the increase of particles we would get the thermodynamic limit of the system, i.e., particles should stabilize on some equilibrium distribution that minimizes energy, or equivalently maximizes entropy. Those will be our equilibrium measures for the eigenvalues. Furthermore, these systems behave as a point dynamic called determinantal point process (DDP), i.e., the joint distribution of the points can be expressed as a determinant of a function called correlation kernel. These kernels are expressed as a function of special monic polynomials that are orthogonal in respect to a so-called weight function that depends on the confining potential on the gas system. Now, the partition function is simply the summation over all possible micro-states of the system. This can be expressed as an integral of the eigenvalues density over the possible configurations and therefore as a function of the same correlation kernel as before. To perform the asymptotic for the partition function is then to perform the asymptotic for the orthogonal polynomials or its norming constants in a sense. Luckily the integrals we need to perform asymptotic on are of the physics favorite form, the integrand is the exponential of a function with one or a few isolated maxima points multiplied by some increasing parameter. This is good because these integrals can be, given some conditions, approximated asymptotically by a Gaussian integral and some error bound. This asymptotic results are extremely interesting as the coefficients on each order of n , the size of the system, have physical interpretations such as an internal energy, entropy and others less straightforward quantities.

On [Chapter 2](#) we will introduce our main culprits: The Random Matrices in Random Matrix Theory (RMT). Our goal with this chapter is to formalize notation and language for the rest of the work while introducing the main ideas one need to understand when dealing with RMT. Random matrices can be of many types, we will divide the types of ensembles and introduce some that are closer to our work or that are central to the theory. Dealing with random matrices one must also decide on with quantities are of interest in respect to this objects, it may be the expected characteristic polynomial, the trace, the distribution of eigenvector or eigenvalues, among others. The ideal of orthogonal polynomials and how they fit into the story of random matrices are also briefly touched as needed for the development. This chapter is essential in understanding what goes on in [Chapter 5](#) and [Chapter 6](#) and important to be connect with the ideas developed in [Chapter 3](#).

On [Chapter 3](#) we explored the mathematical structure of the point dynamics we gain when dealing with eigenvalue statistics: Point Process (PP) and more specifically for us, Determinantal Point Process (DDP). On this chapter we reinterpret some quantities taken from the random matrices as function of the dynamic itself. This allow us to better understand what will be called the k -moments and correlation functions. This also lead us to the main structure in the distribution of eigenvalues of invariant ensembles; the determinant of the correlation kernel. We also show some properties for such kernels and explore a very important ideal of transformations and limit of these object. This connects nicely with the ideal of universality in RMT and the different local regime found in eigenvalue distributions, here represented by different scaling and transformations. This chapter is a good complement of [Chapter 2](#) and provides some theoretical

motivation for what comes in [Chapter 5](#) and [Chapter 6](#).

On [Chapter 4](#) we construct the necessary and appropriate background in asymptotic methods for integral functions, to be used when performing asymptotics of orthogonal polynomials and related objects. There are two separate ideas in this chapter. The first, about quadrature methods, is an excursion and done as technical requirement for [Chapter 6](#). This works on a way to approximate, asymptotically, integrals as summations using the Euler-Mclaurin formula. The second part refers to the main ideas of asymptotics we use: the Laplace technique. This is a way to approximate integrals in which a major part of the contribution comes from a few isolated points in the domain. This chapter is a bit out of place but a strict requirement for [Chapter 6](#) and a soft one for a few ideas in [Chapter 5](#).

On [Chapter 5](#) we more directly discuss the idea of Coulomb gases and how this idea lead us to equilibrium distributions for the eigenvalues. We begin by specifying the measure of Coulomb gases and noting that it is the same of invariant ensembles of matrices. Then we discuss quantities such as the energy of such system with and without an external confining field and how each case lead us to a minimization problem for the energy. The main results here refers to the Variational Equations that determine some techniques for identifying the equilibrium measure. This is also where we get to an generic expression for the partition function and where we motivate the asymptotic we will get on [Chapter 6](#). This chapter is better read after [Chapter 2](#). It server as motivation and justification for equilibrium measures and the search for the expression on [Chapter 6](#).

On [Chapter 6](#) we finally get to our main result: We prove the asymptotic expression of the partition function for some random normal ensembles. This is done following the work of [Byun, Kang and Seo \(2023\)](#). Firstly we state our model and discuss the characterization of the results we want to get. The following pages are dedicated to a dense computation of the results for the two relevant cases of equilibrium measures: with disk and wing shaped supports. This is the culmination of the theoretical work done on the previously chapters and the main point of the work.

I hope you have a good journey into this work and enjoy the details and cool examples as much as I did. Good read!

RANDOM MATRIX THEORY

Random Matrices can be thought of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. In some cases there is an explicit density formula for the eigenvalues of such a random matrix. We will be interested in these cases, which we call the *Invariant Ensembles*. Moreover, we will be particularly interest in a subset on which this density can be expressed as a *Kernel* - an matricial object that will hold the statistical characterization of our system. For us, upon a restriction, this will indicate a determinantal structure and will lead us to the concepts introduced in [Chapter 3](#), of Determinantal Point Processes (DDP).

The definitions and some structure in the chapter that follows is heavily based (to be very generous with my writing) on the incredibly organized and written Thesis by [Rahman \(2023\)](#). What follows is my best attempt in introducing the concepts in Random Matrix Theory (RMT) that are pertinent to the scope of this work and of general interest, albeit inside the limitations of time, effort and experience.

2.1 Random Matrix

Borrowing a concept from physics, an *ensemble* can be thought of as a virtual collection of all possible states that a system can be in, with more probable states appearing more often in the collection. There is an analogy to be made with ensembles in thermodynamics: When considering sets of matrices with equal probability - in some sense to be more clear soon - we consider the *Gibbs weight* for the state's distribution as we would in the physical *Canonical Ensemble*, of constant temperature. More concretely, in the context of Random Matrix Theory we have the following definition.

Definition 1. A *random matrix ensemble* $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is a set of matrices $\mathcal{S}$ of fixed size equipped with probability density function (p.d.f.) $\mathcal{P} : \mathcal{S} \rightarrow [0, \infty)$. If we say that a matrix $\hat{X}$ is drawn from the random ensemble $\mathcal{E}$, we are saying that it is a random variable assuming values in $\mathcal{S}$ with p.d.f. $\mathcal{P}$.

Example 1. The $m \times n$ real, complex, and quaternionic Ginibre Ensembles are respectively $\mathbb{M}_{m,n}(\mathbb{R}), \mathbb{M}_{m,n}(\mathbb{C}), \mathbb{M}_{m,n}(\mathbb{H})$ with p.d.f.

$$\mathcal{P}^{(Gin)}(\hat{G}) = \pi^{-mn\frac{\beta}{2}} \exp\left(-\text{Tr}(\hat{G}^* \hat{G})\right) = \prod_{i=1}^m \prod_{j=1}^n \pi^{\frac{\beta}{2}} \exp\left(-|G_{i,j}|^2\right),$$

defined in respect to the Lebesgue measure

$$d\hat{G} = \prod_{i=1}^m \prod_{j=1}^n \prod_{s=1}^{\beta} dG_{i,j}^{(s)}.$$

The coefficient β is called the Dyson index and counts the generic number of real components of each matrix entry $G_{i,j}$, so that $\beta = 1, 2, 4$ refers to real, complex and quaternionic entries. These are the Ginibre orthogonal, unitary and symplectic ensembles.

For most applications, there are three ideologies that stand out for defining ensembles: universality, uniformity and composition. We will mainly consider in this work the second one, although the first holds some illustrative power due to its simplicity. The third approach will not be expanded but consists of ensembles of matrices defined by the product of random matrices (POZNIAK; ZYCKOWSKI; KUS, 1998). We briefly describe the universality approach so that we can illustrate certain important results in the theory. This ideology consists of assuming very little of the system and focusing on the universal results, as the name suggests. More precisely, take a matrix $\hat{Y} = \{Y_N\}_{i,j}$ to be such that $Y_{i,j} = Y_{j,i}^*$ and satisfying the following properties: $\{Y_{i,j}\}_{1 \leq i \leq j \leq n}$ are independent; $\{Y_{i,j}\}_{i \neq j}$ are identically distributed and also $\{Y_{i,i}\}$ are identically distributed; $\mathbb{E}(Y_{i,j}) = 0$ and $\mathbb{E}(Y_{i,j}^k) < \infty$, i.e., $r_k = \max(\mathbb{E}(Y_{i,j}^k), \mathbb{E}(Y_{i,j}^k)) < \infty$. Then we can define the *Wigner ensemble*, central to this approach.

Definition 2. Let every $\hat{Y}$ be as before and set $r_2 < \infty$ with $\mathbb{E}(Y_{1,2}^2) > 0$. Then, we say $\hat{Y}$ is a **Wigner matrix**. Furthermore, if $\hat{X}_n = n^{-\frac{1}{2}} \hat{Y}_n$, $\hat{X}$ is what we call a **normalized Wigner matrix**.

Example 2. Let $\hat{G}$ be drawn from the $n \times n$ real, complex or quaternionic Ginibre Ensemble. Then, the random matrix $\hat{H} = \frac{1}{2}(\hat{G}^* + \hat{G})$ represents the corresponding Gaussian ensemble (also known as Hermite). Note that such a matrix is a Wigner matrix as it is the symmetrization of a matrix from the Ginibre Ensemble - given by centered independent

and identically distributed Gaussian random variables.

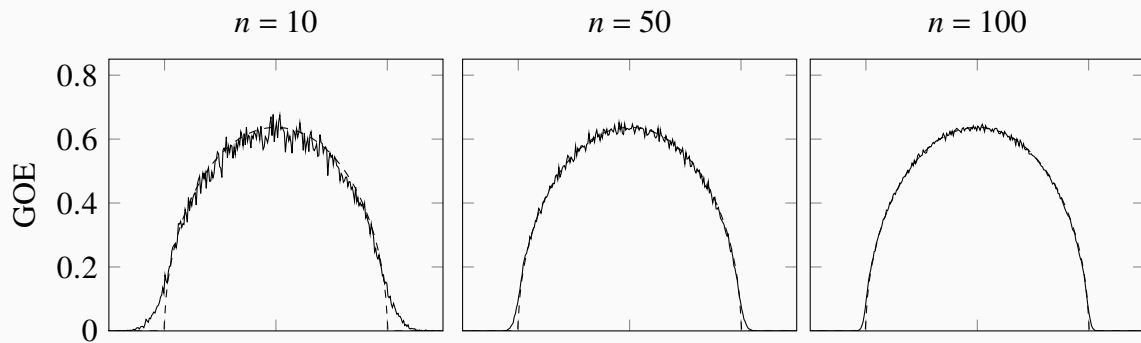
Many results hold for independently and identically distributed centered random variables - and those are the advantages of such approach. The universality name comes from a very important theorem from Wigner. Give the name of *semicircle distribution* to the measure $\mathbf{p}^{(SC)}$ defined by $\mathbf{p}^{(SC)}(dx) := 1/(R^2\pi)\sqrt{(R^2 - x^2)_+}dx$, where $\text{supp } \mathbf{p}^{(SC)} = [-2, 2]$ is the support of $\mathbf{p}^{(SC)}$.

Is is not hard to notice why the distribution takes such name; it is shaped such as the upper half of a circle. More interestingly, this distribution has a very important role when discussing Wigner matrices: It is an universal distribution. This means that, for Wigner matrices, for whichever random variable you choose for the entries given the restrictions, when taken to the limit all of them converge to this limiar shape. This is the *Wigner's Semicircle Law*.

Theorem 1. (*Wigner's Semicircle Law*) *The averaged eigenvalue distribution $\mathbf{p}_N$ for Wigner matrices converges weakly to the semicircle distribution $\mathbf{p}^{(SC)}$.*

This type of result is named *universality* in the RMT and can be better explained when we discuss the theory of *point processes* and an object called *Kernel*. However, in general terms, universality refers to the phenomena in RMT where different ensembles have the same limit statistic, generally in terms of the so-called Kernel, that is, the Kernel is some shared limit object. Wigner Matrices have the same limit distribution for eigenvalues, for example. Other classes of universality may share the same statistics at various scales and regimes: Eigenvalue spacing and maximum eigenvalue distribution are one of the statistics one might be interested in.

Example 3. (*Numerical analysis of Semicircle Law*) *We can numerically see the convergence theorem in action. If we take the Gaussian Orthogonal, Unitary and Symplectic Ensemble - correspondent in Example 2 to starting respectively with the real, complex or quaternionic Ginibre matrix, then we expect to see, with $N \rightarrow \infty$, the eigenvalue measure $\mathbf{p}_N$ approximating the semicircle distribution $\mathbf{p}^{SC}$. The numerics for this situation are exhibit below.*



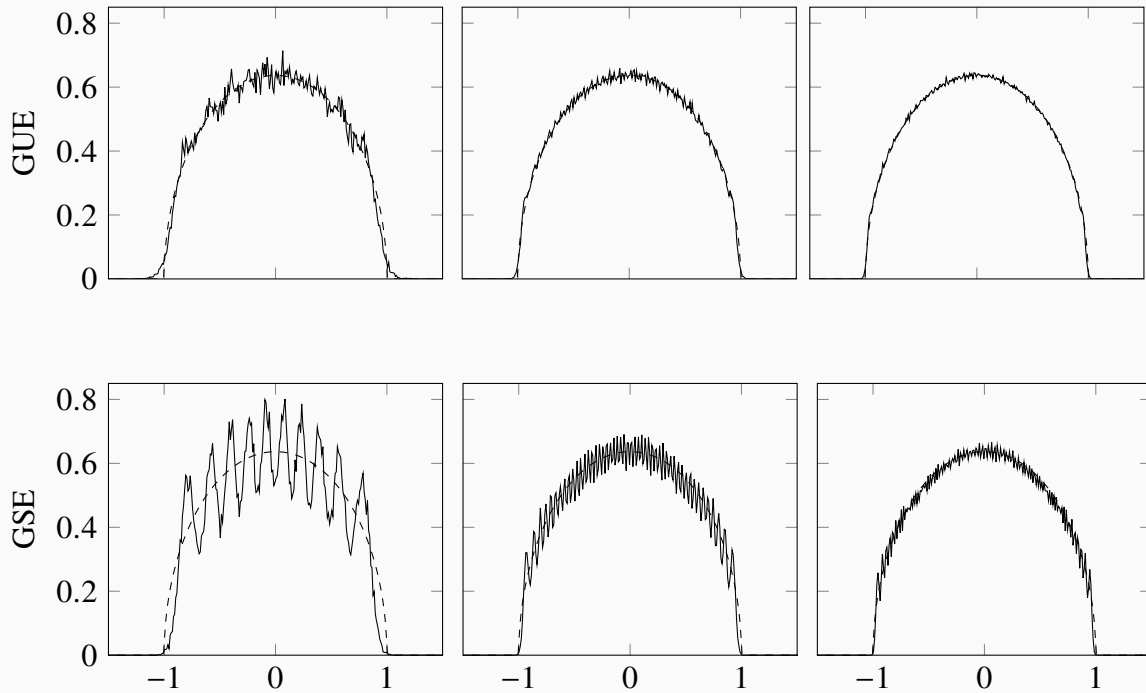


Figure 7 – For all graphs, we have in dashed line the Semicircle distribution. In black the average distribution of eigenvalues is displayed. The columns correspond to $n = 10, 50, 100$ from left to right. The lines correspond to the GOE, GUE and GSE from top to bottom. The matrices, and corresponding eigenvalues, were sampled with the ensemble distribution 10000 for each graph.

We now move on to the main approach we will deal with in this work. The second ideology to construct ensembles is equivalent to assume equal probability on each state of a system; a very common assumption in thermodynamics. This would be analogous to the Canonical Ensemble in physics. We consider each matrix to be equally likely, and more common states - similar matrices - appear more often in the set. Similar to the choice of uniform measure on the sphere being defined by rotational invariance, to construct such ensembles we will equip them with the *Haar Measure*.

Theorem 2. (*Haar, compact case*). *Given a compact group $\mathcal{S}$ (a topological group with compact topology), let $U \in \mathcal{S}$ be a generic variable. There exists a unique (up to normalization measure) $\mathrm{dp}(U)$ on $\mathcal{S}$, called the Haar measure such that*

$$\int_{\mathcal{S}} \mathrm{dp}(U) < \infty$$

and for any fixed $U_0 \in \mathcal{S}$,

$$\mathrm{dp}(U_0 U) = \mathrm{dp}(U) = \mathrm{dp}(U U_0),$$

that is, finite for compact sets and invariant by left and right actions. If $\mathrm{dp}(U)$ integrates to unity in $\mathcal{S}$ it is referred as the Haar probability measure.

This measure will allow us to define the ensembles we are interested in; nominally, it will lead us to [section 2.2](#) to discuss the *Invariant Ensembles*. Interestingly, there is only one ensemble in the intersection of both of these approaches: The Gaussian Ensembles. No other ensemble of Wigner matrices is also invariant.

Example 4. We define the following classical ensembles, all examples of invariant ensembles. We know by the spectral theorem that these self-adjoint matrices can be decomposed using a orthogonal, unitary or symplectic matrix which we consider to be Haar distributed.

- Let $\hat{G}$ be drawn from the $n \times n$ real, complex or quaternionic Ginibre Ensemble. Then, the random matrix $\hat{H} = \frac{1}{2}(\hat{G}^\dagger + \hat{G})$ represents the corresponding Gaussian ensemble (also known as Hermite). That is, $\mathcal{S}^{(G)} = \{\hat{H} \in M_{n,n}(\mathbb{F}) | \hat{H} = \hat{H}^*\}$ and

$$\mathcal{P}^{(G)}(\hat{H}) = \frac{1}{\mathcal{Z}_{n,\beta}^{(G)}} \exp(-\text{Tr}(\hat{H}^2)).$$

- Let $\hat{G}$ be drawn from the $n \times m$ real, complex or quaternionic Ginibre Ensemble. Then, the random matrix $\hat{W} = (\hat{G}^\dagger \hat{G})$ represents the corresponding (m, n) Laguerre ensemble (also known as Wishart). That is, $\mathcal{S}^{(L)} = \{\hat{W} \in \mathcal{S}^{(G)} | \hat{W} \text{ positive definite}\}$ and

$$\mathcal{P}^{(L)}(\hat{W}) = \frac{1}{\mathcal{Z}_{n,a}^{(L)}} (\det(\hat{W}))^a \exp(-\text{Tr}(\hat{W})).$$

- We say $\hat{Y}$ an element of the (m_1, m_2, n) real, complex and quaternionic Jacobi Ensemble if it is of the form $\hat{Y} = \hat{W}_1(\hat{W}_1 + \hat{W}_2)^{-1}$ where each $\hat{W}_i$ belong to the corresponding (m_i, n) Laguerre ensembles with $m_i \geq n$. That is, $\mathcal{S}^{(J)} = \{\hat{Y} \in \mathcal{S}^{(L)} | \hat{I}_n - \hat{Y} \text{ positive definite}\}$ and

$$\mathcal{P}^{(J)}(\hat{Y}) = \frac{1}{\mathcal{Z}_{n,a,b}^{(J)}} (\det(\hat{Y}))^a (\det(\hat{I}_n - \hat{Y})).$$

To study random matrices, we need to determine which quantities related to the matrix we want to measure. This may be the expectation of some function on the matrices or even on the eigenvalues and eigenvectors, if they exist. Assume a matrix is such that it is determined by a set of eigenvalues and eigenvectors. While eigenvectors may be of interest in some cases, eigenvalue statistics have been noticeably more explored. For some matrix $\hat{M}$, one of the quantities one might want to take on are the *average* or *expected characteristic polynomial*

$$P_n(x) = \mathbb{E}[\det(x\hat{I}_n - \hat{M})], \quad n \in \mathbb{N}.$$

The characteristic polynomial have roots that coincides with the eigenvalues. Therefore, morally, roots of an averaged characteristic polynomial could be thought of as the typical set of eigenvalues for the random matrix model chosen for $\hat{M}$. For random Hermitian matrices associated with

some probability distribution $\det(w(\hat{M}))$, then the average characteristic polynomial is simply the monic orthogonal polynomial $P_n^w(x)$ of degree n in respect to the weight function $w(X)$. (EDELMAN, 1989)

Example 5. (*Expected characteristic polynomial for Gaussian Unitary Matrices*) Hermite Polynomials are the average characteristic polynomials of the Gaussian Unitary Ensemble. That is, if $\hat{M}$ is an order n matrix which is hermitian and with joint density given by

$$\frac{1}{Z_n} e^{-\text{Tr} \hat{M}^2} dM$$

then the average characteristic polynomial satisfies

$$P_n(x) = xP_{n-1}(x) - \frac{n-1}{2}P_{n-2}(x).$$

If one notices that $P_0(x) = 1$ and $P_1(x) = x$, then one can also see that the recurrence equation give us the Hermite Polynomials. One can show numerically that the concentration of roots of the appropriately normalized Hermite polynomial and the eigenvalues coincide. The roots are the typical set of eigenvalues.

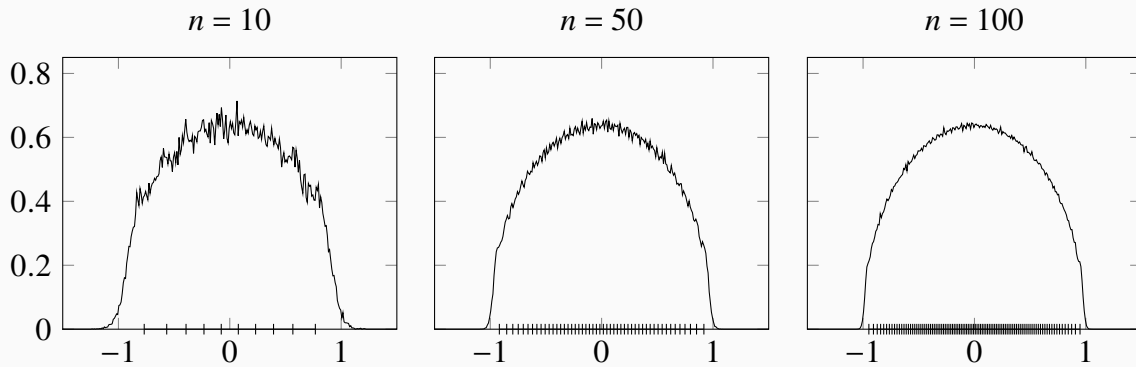


Figure 8 – For all graphs, the line is the average distribution of eigenvalues for the GUE. The columns correspond to the polynomial order (and matrix size) $n = 10, 50, 100$ from left to right. The dashes in the axis correspond to the roots of the n -th Hermite polynomial.

This results holds true only for the case presented. For real and symplectic symmetric matrices, for example, $P_n^w(x)$ can be calculated only for a few functions - an example would be the so-called *classical weight functions* (Example 7). Taking the limit for n - consequently the order of the expected characteristic polynomial - one might expect the distribution of roots to coincide with the distribution of eigenvalues. This can be seen in Example 5. Taking the asymptotic on the orthogonal polynomials reveal something about the nature of the distribution of eigenvalues! This is not necessarily true however in general and especially not true for ensembles with non-real roots. In these cases, the orthogonal polynomial associated with the weight function has no clear reason to be the expected characteristic polynomial. However, this a powerful idea that motivates future results in this work.

By this point is clear that we are naturally interested in the eigenvalue joint probability density function (j.p.d.f.) $p_n(\lambda_1, \lambda_2, \dots, \lambda_n)$ of the eigenvalues $\{\lambda_i\}_{i=1}^n$ of $\hat{M}$. This is especially true because we will be dealing with ensembles where the distribution is uniform in the eigenvectors, that is, all that matters is the distribution of eigenvalues. The eigenvalue j.p.d.f. p_n is defined to be such that

$$\int_{[a_1, b_1] \times \dots \times [a_n, b_n]} p_n(\lambda_1, \lambda_2, \dots, \lambda_n) d\lambda_1 \dots d\lambda_n$$

is the probability that $a_i < \lambda_i < b_i$ for every i . Sometimes we may also want to talk about the expected number of eigenvalues in a single interval of the real line - as in concentration or measure for the eigenvalues. Therefore we define, related to the eigenvalue j.p.d.f., the *univariate eigenvalue density* defined by

$$p(\lambda; n) := n \int_{\mathbb{R}^{n-1}} p_n(\lambda_1, \lambda_2, \dots, \lambda_n) d\lambda_2 \dots d\lambda_n,$$

which is normalized such that

$$\int_a^b p(\lambda) d\lambda$$

is the expected number of eigenvalues lying between a and b .

Example 6. (*The complex case for Orthogonal Polynomial*) On the complex plane one could write an ensemble with measure

$$dp_n^{(\beta)}(z_1, \dots, z_n) := \frac{1}{\mathcal{Z}_n^{(\beta)}} \prod_{j>k=1}^n |z_j - z_k|^\beta \prod_{j=1}^n e^{-\frac{\beta n}{2} Q(z_j)} dA(z_j). \quad (2.1)$$

This is the expression, if $\beta = 2$ for the measure of the random normal matrix ensemble. Setting $Q(z) = |z|^2$ this would be the so-called complex Ginibre ensemble. Consider, however, $Q(z) = q(|z|)$ for some $q : \mathbb{R} \mapsto \mathbb{R}$, i.e., Q is a radially symmetric potential. The theory says that the roots of such ensemble should be distributed in a disk or annulus. However, $\{z^j\}_j$ are the family of orthogonal polynomials associated with $w = e^{-\frac{\beta n}{2} Q(z)}$. Clearly its roots, which concentrate at zero, could not agree with the distribution of eigenvalues.

2.2 Invariant Ensembles

Invariant ensembles of random matrices can be thought as probability spaces which are invariant under conjugation of some appropriate matrix group - think of these ensembles as having probability distribution only depending on their eigenvalue distributions while their eigenvectors are independent and Haar (uniformly) distributed. The distribution of matrices should be uniform conditional on its eigenvalues. This is not completely unmotivated of course, conjugation invariance is, for example, a very natural physical assumption; it simply signifies that all coordinate systems are equivalent. To properly understand this concept, let's introduce

three instances of these ensembles: The *Unitary*, *Orthogonal* and *Symplectic* ensembles. Then we will make precise the definition of such an ensemble. Other ensembles many times can be of interest, but their theory may be less straightforward.

1. *Orthogonal Ensembles* consist of $n \times n$ real symmetric matrices $\hat{M} = \hat{M}^* = \hat{M}^T$ together with a probability distribution that is invariant under orthogonal conjugation $\hat{M} \mapsto \hat{O} \hat{M} \hat{O}^T$, $\hat{O}$ orthogonal, i.e., $\hat{O} \hat{O}^T = \hat{O}^T \hat{O} = \hat{I}$.
2. *Unitary Ensembles* consists of $n \times n$ Hermitian matrices $\hat{M} = \hat{M}^*$ together with a probability distribution that is invariant under unitary conjugation $\hat{M} \mapsto \hat{U} \hat{M} \hat{U}^*$, $\hat{U}$ unitary, i.e., $\hat{U}^* \hat{U} = \hat{U} \hat{U}^* = \hat{I}$.
3. *Symplectic Ensembles* consists of $2m \times 2m$ Hermitian self-dual matrices $\hat{M} = \hat{M}^* = \hat{J} \hat{M}^T \hat{J}^T$ where $\hat{J} = \text{diag}(\sigma, \dots, \sigma)$ with

$$\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

together with a probability distribution that is invariant under unitary-symplectic conjugation $\hat{M} \mapsto \hat{U} \hat{M} \hat{U}^*$, with $\hat{U}$ unitary and symplectic, i.e. $\hat{U} \hat{U}^* = \hat{U}^* \hat{U} = \hat{I}$ and $\hat{U} \hat{J} \hat{U}^T = \hat{J}$.

Examples of such invariant ensembles are the Gaussian Orthogonal, Unitary and Symplectic ensembles - *GOE*, *GUE*, *GSE*: these are matrices that take independent and identically distributed (i.i.d.) real, complex and quaternionic Gaussian entries (minus the symmetries). The Gaussian ensembles are infamous and deserve a lot more discussing, however, there is another class of ensembles more important to this work, the *Ginibre Ensembles*. The three types of Ginibre ensembles are, again: Ginibre Orthogonal, Unitary and Symplectic. These are matrices with i.i.d. standard real, complex and quaternionic Gaussian entries. In this case, the eigenvalues are distributed in the complex plane. We will come back to this examples, for now, let's make the definition precise.

Definition 3. (*Invariance*) Given a group $\mathcal{G}$ and a group action $\mathcal{A} : \mathcal{G} \times \mathcal{S} \rightarrow \mathcal{S}$, a random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is said to be **$\mathcal{A}$ -invariant** if the probability measure $d\mathcal{P}(\hat{X}) := \mathcal{P}(\hat{X})d\hat{X}$ satisfies

$$d\mathcal{P}(\mathcal{A}(g, \hat{X})) = d\mathcal{P}(\hat{X})$$

for every $g \in \mathcal{G}$.

The term *invariant matrix ensemble* usually refers to invariance under the mapping $\hat{X} \mapsto g \hat{X} g^{-1}$ for g drawn from the appropriate compact group $\mathcal{G}$; when a matrix ensemble is considered to be irreducible and its underlying set of matrix is $\mathcal{S} \subseteq \mathbb{M}_{n,n}(\mathbb{R}), \mathbb{M}_{n,n}(\mathbb{C}), \mathbb{M}_{n,n}(\mathbb{H})$, the group $\mathcal{G}$ must be one of $O(n), U(n), Sp(2n)$ - the orthogonal, unitary and symplectic space of matrices. Then, we say the ensemble is *Orthogonal*, *Unitary* or *Symplectic*.

Example 7. A random matrix ensemble is said to be a Classical Matrix Ensemble if its eigenvalue j.p.d.f. is of the form

$$\mathfrak{p}_n^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \frac{1}{\mathcal{N}_{n,\beta}^{(w)}} \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^\beta, \quad \beta = 1, 2, 4,$$

where $\mathcal{N}_{n,\beta}^{(w)}$ is the normalization constant, $\Delta_n(\lambda)$ is the Vandermonde determinant and the weight function has, among others, one of the following forms

$$w(\lambda) = \begin{cases} e^{-\lambda^2} & \text{Gaussian,} \\ \lambda^a e^{-\lambda} \chi_{\{\lambda > 0\}} & \text{Laguerre,} \\ \lambda^a (1-\lambda)^b \chi_{\{0 < \lambda < 1\}} & \text{Jacobi.} \end{cases}$$

where χ is the indicator function and $a, b > -1$. Especially, these are invariant ensembles under the orthogonal, unitary and symplectic conjugation (for $\beta = 1, 2, 4$).

We should be able to show that the probability measure $d\mathcal{P}(\hat{X}) = \mathcal{P}(\hat{X})d\hat{X}$ for the Ginibre and Classical Ensembles have the foretold orthogonal, unitary and symplectic invariance under the adjoint action of the appropriate group $\mathcal{G} \in \{O(n), U(n), Sp(2n)\}$. This is to say that we should establish that both the p.d.f. $\mathcal{P}(\hat{X})$ and the Lebesgue measure $d\hat{X}$ are invariant under this action. This is not trivial and also not the focus of this text. However, believing this to be an interesting concept that elucidates the definition we move forward using as guide, again, the Thesis by [Rahman \(2023\)](#). The p.d.f. $\mathcal{P}(\hat{X})$ is straightforward as it only depends on the eigenvalues which remain unchanged by any of the actions. The invariance of the Lebesgue measure is more involved and depends on whether or not the associated matrix set $\mathcal{S}$ consists of purely self-adjoint matrices. For the Classical Ensembles we can use the following proposition.

Proposition 1. ([RAHMAN, 2023](#)) Given an $n \times n$ real ($\beta = 1$), complex ($\beta = 2$) or quaternionic ($\beta = 4$) self-adjoint matrix variable $\hat{X}$, diagonalize it as $\hat{X} = \hat{U}\hat{\Lambda}\hat{U}^*$ where $\hat{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ is a diagonal matrix of eigenvalues, and $\hat{U}$ is corresponding matrix of eigenvectors; the spectral theorem dictates that $\hat{U}$ is an element of $O(n)$ in the real case, $U(n)$ in the complex case, and $Sp(2n)$ in the quaternionic case. To make the mapping $\hat{X} \mapsto \hat{U}\hat{\Lambda}\hat{U}^*$ bijective, we insist that the eigenvalues are listed in increasing order and the first row of $\hat{U}$ is non-negative real. The canonical Lebesgue measure

$$d\hat{X} = \prod_{i=1}^n dX_{ii} \prod_{1 \leq j < k \leq n} \prod_{s=1}^{\beta} dX_{jk}^s$$

on the set of adjoint matrices decomposes as

$$d\hat{X} = |\Delta_n|^\beta d\hat{\Lambda} d\mathcal{P}_{Haar}(\hat{U})$$

where $d\hat{\Lambda} = \prod_{i=1}^n d\lambda_i$ is the Lebesgue measure on the Weyl Chamber with $\lambda_1 < \dots < \lambda_n \subset \mathbb{R}^n$ and

$$d\mathcal{P}_{Haar}(\hat{U}) = \bigwedge_{1 \leq l < k \leq n} \bigwedge_{s=1}^{\beta} \left(\hat{U}^* [dU_{ij}]_{i,j=1}^n \right)_{lk}^{(s)}$$

is the unnormalized Haar measure on the quotient space $O(n)/O(1)^n$ ($\beta = 1$), $U(n)/U(1)^n$ ($\beta = 2$), or $Sp(2n)/Sp(2)^n$ ($\beta = 4$), given by the exterior product of the independent real components of the entries of the matrix of differentials $\hat{U}^* [dU_{ij}]_{i,j=1}^n$.

At this point, it is possible to check (albeit with some very non-trivial computations) that the Lebesgue measure is indeed invariant in these cases for the appropriate group action. This is a matter of checking that the Lebesgue measure on the Weyl Chamber is invariant, which is relatively straightforward, and that, less trivially, the unnormalized Haar measure on the quotient space is also invariant.

The situation concerning the $n \times n$ Ginibre Ensemble is more complicated. In the 1965 work, Ginibre extended the proposition above and its arguments starting with the diagonalization formula $\hat{G} = \hat{P}\hat{\Lambda}\hat{P}^{-1}$ - note that, with $\hat{\Lambda}$ diagonal of eigenvalues and $\hat{P}$ of eigenvectors, this formula is valid with probability one since the subset of matrices with repeated eigenvalues has measure zero. However, $\hat{P}$, diagonalizing matrix, is not necessarily orthogonal, unitary, nor symplectic. Nonetheless, QR decomposition can be used to write $\hat{P} = \hat{U}\hat{T}$ where $\hat{U}$ is as it was before and $\hat{T}$ is an upper triangular matrix with real positive diagonal entries. Then,

$$d\hat{G} = J(\lambda) d\hat{\Lambda} d\mathcal{P}_{Tri}(\hat{T}) d\mathcal{P}_{Haar}(\hat{U})$$

where $d\hat{\Lambda}$ is the Lebesgue measure on the space housing the eigenvalues, $d\mathcal{P}_{Haar}(\hat{U})$ is the Haar measure defined before, $J(\lambda)$ is a Jacobian consisting of products of Vandermonde determinants involving the eigenvalues, and $d\mathcal{P}_{Tri}(\hat{T})$ is some measure dependent on the entries of $\hat{T}$. From here, it can be shown that the Lebesgue measure on $\hat{G}$ is invariant under the desired conjugation action.

2.2.1 Orthogonal Polynomial Ensembles

The subclass of invariant matrix ensembles this section refers to are those self-adjoint matrices $\hat{X}$ with p.d.f.s $\mathcal{P}(\hat{X})$ such that one has the decomposition

$$\mathcal{P}(\hat{X}) = \prod_{i=1}^n w(\lambda_i),$$

where $\{\lambda_i\}$ are the indistinguishable eigenvalues of $\hat{X}$ and $w(\lambda)$ is some continuous weight function that is real valued on $\mathbb{R}$ and decays fast enough as $|\lambda| \rightarrow \infty$. This is the same as removing the constrain on the weight function of the classical ensembles. With all of this, we can make the definition more formal.

Definition 4. Let $\beta = 1, 2, 4$, fix $n \in \mathbb{N}$ and let $w(\lambda)$ be a continuous, non-negative, real-valued function such that $w(\lambda) = o(\lambda^{-\beta(n-1)-1})$. Then, the j.p.d.f.

$$p_n^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \frac{1}{\mathcal{N}_{n,\beta}^{(w)}} \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^\beta$$

describes the n -sized **orthogonal polynomial ensemble (OPE)** when $\beta = 2$, **skew-orthogonal polynomial ensemble of real type ($\mathbb{R}$ -SOPE)** when $\beta = 1$, and the **skew-orthogonal polynomial ensemble of quaternion type ($\mathbb{H}$ -SOPE)** when $\beta = 4$.

If we stare at this formula strongly and for long, the terms become less mysterious. One may notice that this expression is simply given by the normalized product of some weight function $w(\lambda) = e^{-\beta NV(\lambda)}$ and the Jacobian of the transformation $\hat{M} \mapsto \hat{G} \hat{\Lambda} \hat{G}^*$ to be given by the Vandermonde determinant (to some exponent β). This is an interesting observation and it is in accordance to ???. More than that, there is an observation to be made here in terms of the behavior of the eigenvalues. Notice that the Jacobian takes arbitrarily small values if there are equally arbitrarily close eigenvalues in the set. At first instance this means that our set is most likely simple, i.e. without repeated eigenvalues. Moreover it also means that the eigenvalues show some repulsion with each-other. Why then one might not expect the eigenvalues to scatter in the line? This is the role of the weight function. More than characterizing the ensemble, in the formula it has the role of ensuring integrability, i.e., holding the eigenvalues in a compact set. If this feels like some physical repelling system, fear not, this interpretation will be expanded in [section 5.1](#).

Example 8. The univariate eigenvalue density for the Gaussian Ensembles is expressed by

$$p_n(\lambda_1, \lambda_2, \dots, \lambda_n) = \frac{1}{\mathcal{Z}_{n,\beta}} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n \lambda_i^2 \right\} \prod_{i < j} |\lambda_i - \lambda_j|^\beta.$$

Where naturally we can identify

$$\exp \left\{ -\frac{1}{2} \sum_{i=1}^n \lambda_i^2 \right\} =: \prod_{i=1}^n w(\lambda_i).$$

This is the furthest we will take the theory for $\beta = 1, 4$. Although those cases are of interest, the structure for $\beta = 2$ allow us to work with in a simpler way. More than that, our main ensemble, to be treated in [Chapter 6](#), has $\beta = 2$. In any way, we mentioned the eigenvalues to be simple but did not provide any proof. For Hermitian matrices, the following proposition give us such result.

Proposition 2. The set of Hermitian matrices with simple spectra is open and dense in $\mathcal{M}_n$ (n -dimensional Hermitian matrices) and has full measure, that is, generally, hermitian

matrices have simple eigenvalues.

Proof. Let the Hermitian matrix $\hat{M}$ have the spectral decomposition

$$\hat{M} = \hat{U} \hat{\Lambda} \hat{U}^*.$$

There exists $\epsilon_1, \epsilon_2, \dots, \epsilon_n$ real, with $\sum_{i=1}^n |\epsilon_i| \rightarrow 0$, such that

$$\hat{M}_\epsilon = \hat{U} \begin{bmatrix} \lambda_1 + \epsilon_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n + \epsilon_n \end{bmatrix} \hat{U}^* = \hat{M}_\epsilon^*$$

has a simple spectrum. As $\hat{M}_\epsilon \rightarrow \hat{M}$, matrices with simple spectra are dense in $\mathcal{M}_n$ the space of n -dimensional Hermitian matrices. $\square$

We also stated that the expression encompasses the Jacobian of the transformation $\hat{M} \mapsto \hat{G} \hat{\Lambda} \hat{G}^*$: Now its time to show why that's true. We are interested in the probability distribution of the eigenvalues $\lambda_1, \dots, \lambda_n$ of $\hat{X}$. Because the matrices are self-adjoint, there exists $\hat{U} \in U(n)$, the Unitary group, such that

$$\hat{U}^* \hat{X} \hat{U} = \hat{\Lambda} := \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n).$$

To arrive at the expression for the eigenvalue distribution we make the previously stated transformation

$$\hat{X} \mapsto (\hat{U}, \hat{\Lambda})$$

and then integrate the eigenvector element away, so that we are left with the part depending on the eigenvalues

$$\begin{aligned} \text{dp}_n^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta = 2) &= \int_{U(n)/U(1)^n} \mathcal{P}(X_{1,1}, \dots, X_{n,n}) d\hat{X}, \\ &= \int_{U(n)/U(1)^n} \mathcal{P}(X_{1,1}(\hat{\Lambda}, \hat{U}), \dots, X_{n,n}(\hat{\Lambda}, \hat{U}) | J(\hat{X} \mapsto \{\hat{\Lambda}, \hat{U}\})) d\hat{\Lambda} d\hat{U}. \end{aligned}$$

This integral can be evaluated using Weyls formula.

Theorem 3. (Weyl) Suppose f is an integrable class function on $\mathbb{S}_n$, i.e., $f(\hat{U} \hat{X} \hat{U}^*) = f(\hat{X})$ for all $\hat{U} \in U(n)$. Then we have

$$\int_{\text{He}_n} f(\hat{X}) d\hat{X} = c_n \int_{\mathbb{R}^n} f(\lambda_1, \lambda_2, \dots, \lambda_n) \prod_{i < j} (\lambda_j - \lambda_i)^\beta d\lambda_1 \cdots d\lambda_n,$$

where $\mathbb{S}_n$ are the self-adjoint matrices of order n and

$$c_n = \frac{\pi^{n(n-1)/2}}{\prod_{j=1}^n j!}.$$

More generally,

$$dM = c_n \prod_{i < j} (\lambda_i - \lambda_j)^\beta d\lambda_1 \cdots d\lambda_n d\hat{U},$$

where $d\hat{U}$ is the Haar measure on $U(n)$

The condition on the weight function guarantees that the j.p.d.f. has a well-defined normalization constant. With Equation 3 we can finally state, as mentioned in the definition,

$$\begin{aligned} p_n^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta = 2) &= Vol(\mathcal{G}(n)/\mathcal{G}(1)^n) \mathcal{P}(X_{1,1}(\hat{\Lambda}), \dots, X_{n,n}(\hat{\Lambda})) |\Delta_n(\lambda)|^\beta \\ &= Vol(\mathcal{G}(n)/\mathcal{G}(1)^n) \mathcal{P}(X_{1,1}(\hat{\Lambda}), \dots, X_{n,n}(\hat{\Lambda})) \prod_{i < j} (\lambda_i - \lambda_j)^2 \\ &=: Vol(\mathcal{G}(n)/\mathcal{G}(1)^n) \prod_{i=1}^n w(\lambda_i) \prod_{i < j} (\lambda_i - \lambda_j)^2 \\ &=: \frac{1}{\mathcal{N}_{n,2}^{(w)}} \prod_{i=1}^n w(\lambda_i) \prod_{i < j} (\lambda_i - \lambda_j)^2. \end{aligned}$$

Moreover, if one wishes to integrate the structures given by the Jacobian against the product of weight functions, it is beneficial to use monic polynomials $\{q_j(\lambda)\}_{j=0}^\infty$ that are skew-orthogonal with respect to the weight $w(\lambda)$. This is the reason for the name of the polynomial ensembles. To properly show how to use this structure, we will manipulate the expression of p_n a little. Notice that

$$\prod_{i < j} |\lambda_i - \lambda_j|^2 = \det \left(q_{j-1}^{(2)}(\lambda_i) \right)_{n \times n}^2.$$

Using the fact that $(\det(A))^2 = \det(\sum_{j=1}^n A_{ji} A_{jk})$ we can rewrite

$$\prod_{i < j} |\lambda_i - \lambda_j|^2 = \det \left(\sum_{j=1}^n q_{j-1}^{(2)}(\lambda_i) q_{j-1}^{(2)}(\lambda_k) \right)_{n \times n}.$$

Then, start by putting the weight function inside the determinant in the following manner: Take $w(\lambda_i) = \sqrt{w(\lambda_i)} \sqrt{w(\lambda_i)}$ and multiply the i -th column of the matrix by one of the terms. The other term we multiply by the i -th line of the matrix. Now, we should get something like

$$\begin{aligned} p_n^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta = 2) &= \frac{1}{\mathcal{N}_{n,2}^{(w)}} \det \left(\sqrt{w(\lambda_i) w(\lambda_k)} \sum_{j=1}^n q_{j-1}^{(2)}(\lambda_i) q_{j-1}^{(2)}(\lambda_k) \right)_{i,k=1}^n, \\ &:= \frac{1}{\mathcal{N}_{n,2}^{(w)}} \det \left[K_n^{(2)}(\lambda_i, \lambda_k) \right]_{i,k=1}^n. \end{aligned}$$

This is the so-called *Determinantal Structure* of the $\beta = 2$ orthogonal polynomial ensembles. This will be recapitulated in Chapter 3.

Example 9. (Gaussian Measure) Let's make it explicit the probability distribution on Gaussian Ensembles. We define $\hat{M} = n^{-\frac{1}{2}} \hat{M}' \in \mathcal{M}_{n,n}(\mathcal{F})$ to be of the Gaussian Ensemble if the entries of $\hat{M}'$ are independent and normally distributed as follows:

$$M'_{i,j} \sim \begin{cases} \mathcal{N}_{\mathcal{F}}(0, 1/2) & \text{for } i \neq j, \\ \mathcal{N}_{\mathcal{F}}(0, 1) & \text{for } i = j, \end{cases} \quad \text{if } i \leq j.$$

where $\mathcal{F}$ is such as the real line, complex plane or quaternion space. Note that this turns $\hat{M}$ into a Scaled Wigner Matrix. We know by the spectral theorem and Weyl's formula that we can write

$$\begin{aligned} p_n(\lambda_1, \lambda_2, \dots, \lambda_n) &= \int_{\mathbb{V}_n} \mathcal{P}(\hat{M}) d\hat{M} \\ &= \int_{\mathbb{V}_n} \mathcal{P}(\hat{\Lambda}, \hat{O}) \prod_{i < j} |\lambda_j - \lambda_i|^2 d\hat{O} d\hat{\Lambda} \\ &= \mathcal{P}(\hat{\Lambda}) \prod_{i < j} |\lambda_j - \lambda_i|^2 \int_{\mathbb{V}_n} d\hat{O} \\ &= \mathcal{P}(\hat{\Lambda}) \prod_{i < j} |\lambda_j - \lambda_i|^2 \text{Vol}(\mathbb{V}_n) \end{aligned}$$

which lead us to write because of the Gaussian w form

$$p_n(\lambda_1, \lambda_2, \dots, \lambda_n) = \frac{1}{\mathcal{Z}_{n,\beta=2}} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n \lambda_i^2 \right\} \prod_{i < j} |\lambda_i - \lambda_j|^2.$$

with $\mathcal{Z}_{n,\beta}$ the partition function, or normalizer of the expression. We could also do the same calculation for the GUE and GSE ensembles, which would result in the following general expression, with $\beta = 1, 4$,

$$\begin{aligned} p_n(\lambda_1, \lambda_2, \dots, \lambda_n) &= \frac{1}{n! \mathcal{Z}_{n,\beta}} \exp \left\{ -\left(\sum_{i=1}^n \frac{\lambda_i^2}{2} - \sum_{i < j} \log |\lambda_i - \lambda_j|^\beta \right) \right\}, \\ &= \frac{1}{\mathcal{Z}_{n,\beta}^{(ord)}} e^{-\beta_n \mathcal{H}_n(\vec{\lambda})}, \end{aligned}$$

a very familiar expression for a physicist, as it is a Gibbs measure.

Let's retrace this steps with mathematically proper definitions. We start by properly defining the orthogonality condition on the polynomials.

Definition 5. With $w(\lambda)$ a suitable function, define the following inner products on polynomial space:

$$\langle f, g \rangle^{(2)} := \int_{\mathbb{R}} f(x) g(x) w(x) dx$$

For $\beta = 2$, let $\{q_j^{(\beta)}\}_{j=0}^\infty$ be a family of monic polynomials such that each $q_j^{(\beta)}(\lambda)$ is of degree j . Impose also that exists finite positive constants $\{r_j^{(\beta)}\}_{j=0}^\infty$ s.t.

$$\langle q_j^{(2)}, q_k^{(2)} \rangle^{(2)} = r_j^{(2)} \chi_{j=k}$$

for any $j, k > 0$. Then, we say that the family $\{q_j^{(2)}(\lambda)\}_{j=0}^\infty$ is **orthogonal in respect to** $w(\lambda)$.

The result we found in ?? is an infamous theorem that is better expressed in the following way.

Theorem 4. Let $\{q_j^{(\beta)}(x)\}_{j=0}^\infty, \{r_j^{(\beta)}\}_{j=0}^\infty$, be as in the definition above, and let $n \in \mathbb{N}$. The kernel corresponding respectively to the $\beta = 2$ case is

$$K_n^{(2)}(x, y) := \sqrt{w(x)w(y)} \sum_{k=0}^{N-1} \frac{1}{r_k^{(2)}} q_k^{(2)}(x) q_k^{(2)}(y).$$

Then, the eigenvalue j.p.d.f. of Definition 4 is given by

$$p_n^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta = 2) = \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^n.$$

The significance of this theorem is that, for $\beta = 2$, as for other values of beta in an analogous way, the kernels $K_n^{(\beta)}(x, y)$ have *Reproducing properties* such as

$$\int_{\mathbb{R}} \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^{k+1} d\lambda_{k+1} = (n-k) \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k.$$

Thus, one may integrate the j.p.d.f. over $d\lambda_{k+1} \cdots d\lambda_n$ to obtain the k -point correlation function

$$\begin{aligned} p_k^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_k; n; \beta = 2) &:= \frac{n!}{(n-k)!} \int_{\mathbb{R}^{n-k}} p_n^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) d\lambda_{k+1} \cdots d\lambda_n \\ &= \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k, \quad \beta = 2. \end{aligned}$$

This implies that the univariate eigenvalue density $p^{(w)} = p_1^{(w)}(\lambda)$ is given by the formula

$$p^{(w)}(\lambda; \beta = 2) = K_n^{(2)}(\lambda, \lambda).$$

Most properties of a determinantal point process - such as the eigenvalues of a random matrix from a unitary ensemble - follow from properties of the Kernel. Even more interestingly, for $\beta = 2$ we can use the *Christoffel-Darboux* formula:

Proposition 3. (*Christoffel-Darboux*) The $\beta = 2$ kernel $K_n^{(2)}(x, y)$ defined before simplifies

as

$$K_n^{(2)}(x, y) = \frac{\sqrt{w(x)w(y)} q_n^{(2)}(x)q_{n-1}^{(2)}(y) - q_{n-1}^{(2)}(x)q_n^{(2)}(y)}{r_{n-1}^{(2)}(x - y)}$$

where $r_j^{(2)}$ is a constant and $q_j^{(2)}$ is a monic polynomial are such that

$$\langle q_j^{(2)}, q_k^{(2)} \rangle^{(2)} = r_j^{(2)} \chi_{j=k}.$$

We will come back to this results on [Chapter 3](#) and [Chapter 4](#) to expand on it and in [Chapter 6](#) to properly calculate some asymptotics.

Note 1. Taking the limit $y \rightarrow x$ using L'Hopital's rule shows that for finite n the density has the form

$$\mathbf{p}^{(w)}(\lambda) = K_n^{(2)}(\lambda, \lambda) = \frac{w(\lambda)}{r_{n-1}^{(2)}} \left(q_{n-1}^{(2)}(\lambda) \partial_\lambda q_n^{(2)}(\lambda) - q_n^{(2)}(\lambda) \partial_\lambda q_{n-1}^{(2)}(\lambda) \right).$$

POINT PROCESSES

There are many instances where one might want to describe the behavior of a set of random points in time or in space. This could come from an investigation on system of diffusing particles or occurrences in time of some type of phenomena. Point processes are models to describe such random objects: The number of occurrence of events in a given time-frame or, alternatively, the number of points in regions of some space. For this work, we mainly focus on the latter application: density of points in regions of space. Those can be associated later with measure for eigenvalues of random matrices and the theory developed on those processes, especially the so-called determinantal point processes, will help us studying distributions for these matrices eigenvalues and associated quantities.

The definitions given in this section and some ideas follow very closely the well structured works in [Serfozo \(1990\)](#), [Johansson \(2006\)](#) and [Girotti \(2014\)](#).

3.1 Point Processes

Suppose there are m points at random locations in some region Λ , a complete separable metric space, given by some configuration $\chi_m = \{x_1, x_2, \dots, x_m\}$. We can define a quantity $\xi(A)$ that measures the amount of points ($\#\{\chi_m \cap A\}$) for any sub-region $A \subset \Lambda$ in the following way:

$$\xi(A) = \sum_{k=1}^m \mathbb{1}_A(x_k).$$

Moreover, let $\mathcal{I}$ denote all intervals or rectangles in Λ , $\mathcal{E}$ denote the class of Borel sets and $\mathcal{B}$ the class of bounded Borel sets. One can notice that this function ξ defines a *random counting measure* in the usual sense, i.e., a mapping $f : \mathcal{E} \rightarrow \{0, 1, \dots, \infty\}$ such that

$$f(\emptyset) = 0, \quad f(B) < \infty, \text{ for } B \in \mathcal{B},$$

and that, for disjoint $B_1, B_2, \dots$ in $\mathcal{B}$,

$$f\left(\bigcup_k B_k\right) = \sum_k f(B_k).$$

We also say that ξ is boundedly finite. Let $\mathcal{N}(\Lambda)$ denote the space of all counting measures ξ on Λ . If $B \in \mathcal{B}$, then $\xi(B)$ is finite and we can write

$$\xi|_B(A) = \sum_{i=1}^{\xi(B)} \mathbb{1}_A(x_i),$$

for some $\chi = \{x_1, x_2, \dots, x_{\xi(B)}\}$, $x_i \in B \forall 1 \leq i \leq \xi(B)$. Finally, we can explicitly define point processes.

Definition 6. (Point Processes) Let Λ be a complete separable metric space. A **point process** $\mathbb{P}$ on Λ is a probability measure on the space of all configuration of points χ or counting measures $\mathcal{N}(\Lambda)$. We say $\mathbb{P}$ to be a n -point process if $\mathbb{P}(\#\chi = n) = 1$ or, equivalently, $\mathbb{P}(\xi(\Lambda) = n) = 1$. We say that the counting measure ξ is simple if $\xi(\{x\}) \leq 1$ for all $x \in \Lambda$, then, the point process is simple if $\mathbb{P}(\xi \text{ is simple}) = 1$.

Example 10. The Poisson Point Process $\mathbb{P}^{(P)}$ is defined on $\mathbb{R}$ by some parameter Λ . This (temporal) point process $\mathbb{P}^{(P)}$ takes value k at some time t (in the interval $(0, t)$) with probability

$$\mathbb{P}^{(P)}[\xi^{(P)}(t) = k] = \frac{\Lambda^k e^{-\Lambda}}{k!}.$$

Where Λ is a parameter and equal to the expected value of $\xi^{(P)}(t)$. If it has the form $\Lambda = \lambda t$, then this process is homogeneous and λ is the rate or intensity of the process. Furthermore, for events in disjoint intervals on the line, we have independency, i.e.,

$$\mathbb{P}^{(P)}[\xi^{(P)}(t_0, t_1) = n_1, \dots, \xi^{(P)}(t_{k-1}, t_k) = n_k] = \prod_{i=1}^k \frac{(\lambda(t_i - t_{i-1}))^{n_i} e^{-\lambda(t_i - t_{i-1})}}{n_i!}.$$

An important note is that ξ is the same for any ordering of the subscripts on $x_1, x_2, \dots, x_n$. This lead us the the conclusion that if $u_n(x_1, x_2, \dots, x_n)$ is a probability function on $\mathbb{R}^n$ such that it is invariant under permutations, i.e.

$$u_n(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = u_n(x_1, x_2, \dots, x_n), \quad \forall \sigma \in S_n,$$

then u_n defines naturally a point process.

Example 11. (The j.p.d.f. of eigenvalues for GUE) Consider the measure for the Hermitian

matrices $\hat{H}$,

$$\mathcal{P}^{(G)}(\hat{H}) = \frac{1}{\mathcal{Z}_{n,\beta}^{(G)}} e^{-\text{Tr}(\hat{H}^2)} = \frac{1}{\mathcal{Z}_{n,\beta}^{(G)}} \prod_{j>k}^n |\lambda_j - \lambda_k|^\beta \prod_{j=1}^n e^{-\lambda_j^2} = p_n(\lambda_1, \lambda_2, \dots, \lambda_n).$$

This is a probability functions on the space of configuration of eigenvalues in $\mathbb{R}$ if $\mathcal{Z}_{n,\beta}^{(G)}$ is a finite normalization function. Moreover, it is simple, as configurations with repeated eigenvalues must have probability 0. Note also that the function p_n is invariant under permutations: This is of course the origin of the name Gaussian Unitary, Orthogonal and Symplectic Ensembles mentioned previously.

The mapping $\mathcal{E} \ni A \mapsto \mathbb{E}[\#(\chi \cap A)]$, that assigns to the the Borel set A the expected value of the number of points in itself under the configuration χ , is a measure on $\mathbb{R}$.

Definition 7. The *mean measure*, or *first moment*, of a point process on Λ is defined by

$$p(A) = \mathbb{E}[\xi(A)] := \mathbb{E}[\#(\chi \cap A)], \quad A \in \mathcal{E}.$$

This p is a measure on Λ in the standard sense that $p : \rightarrow [0, \infty]$ satisfies $p(\emptyset) = 0$ and, for disjoint $\Lambda_1, \Lambda_2, \dots$ in $\mathcal{E}$,

$$p\left(\bigcup_k \Lambda_k\right) = \sum_k p(\Lambda_k).$$

Assuming there exists a density $p_1 : \mathcal{E} \ni A \rightarrow \mathbb{R}_+$ with respect to the Lebesgue measure (**1-point correlation function**), there is also a way to define the mean measure as the integral of such density:

$$p(A) = \int_A p_1(x) dx.$$

This $p_1(x)$ can also be called the *rate* or *intensity* of ξ and is the infinitesimal probability of a point at x , i.e. $\mathbb{P}(\xi(dx) = 1) = p_1(x)dx$. In this case p is written as $p(dx) = p_1(x)dx$. This mean measure shows up in expressions of expectations for functionals of ξ .

Definition 8. More generally, the *k-th moment* p_k of ξ , written

$$p_k(A_1 \times A_2 \times \dots \times A_k) := \mathbb{E}[\xi_1(A) \xi_2(A) \dots \xi_k(A)], \quad A_1, A_2, \dots, A_k \in \mathcal{E},$$

can be expressed in respect to the **k-th correlation function** $p_k : A^k \rightarrow \mathbb{R}_+$ in the following way

$$p_k(A_1 \times A_2 \times \dots \times A_k) = \mathbb{E}\left[\prod_{j=1}^k (\#\chi \cap A_j)\right] = \int_{A_1} \dots \int_{A_k} p_k(x_1, \dots, x_k) dx_1 \dots dx_k.$$

This k -th correlation variable stand for the expected number of k -tuples $\chi_k = \{x_1, x_2, \dots, x_k\} \in \chi^k$ such that $x_j \in A_j \forall j$. For example, if $u_n(x_1, x_2, \dots, x_n)$ is a probability density function on

Λ^n , invariant under permutation of coordinates, then we can build an n -point process on Λ with correlation functions

$$p_k(x_1, x_2, \dots, x_k) := \frac{n!}{(n-k)!} \int_{\Lambda^{n-k}} u_n(x_1, x_2, \dots, x_n) dx_{k+1} \cdots dx_n.$$

Example 12. Let $\mathcal{H}_n$ be the space of all $n \times n$ Hermitian matrices. This space can be identified naturally with $\mathbb{R}^{n^2}$. If $\mathcal{P}_n$ is a probability measure on $\mathcal{H}_n$ and $\{\lambda_1(M), \dots, \lambda_n(M)\}$ denotes the set of eigenvalues of $M \in \mathcal{H}_n$, then

$$M \mapsto \sum_{j=1}^n \mathbb{1}_M(\lambda_j)$$

maps $\mathcal{P}_n$ to a finite point process $\mathbb{P}$ on $\mathbb{R}$. If dM is the Lebesgue measure on $\mathcal{H}_n$, then

$$d\mathcal{P}_n(M) = \frac{1}{\mathcal{Z}_n} e^{-\text{Tr} M^2} dM$$

is a Gaussian probability measure on $\mathcal{H}_n$, called the GUE. It can be shown that for any symmetric, continuous function f on $\mathbb{R}^n$ with compact support it holds that

$$\int_{\mathcal{H}_n} f(\lambda_1(M), \dots, \lambda_n(M)) d\mathcal{P}_n(M) = \int_{\mathbb{R}^n} f(x_1, x_2, \dots, x_n) p_n(x_1, x_2, \dots, x_n) d^n x$$

where

$$p_n(x_1, x_2, \dots, x_n) = \frac{1}{\mathcal{Z}_n n!} \prod_{1 \leq i < j \leq n} (x_i - x_j)^2 \prod_{j=1}^n e^{-x_j^2}$$

is the induced eigenvalue measure on $\mathbb{R}^n$. Hence the point process $\mathbb{P}$ on $\mathbb{R}$ defined by this induced measure has correlation functions

$$p_k(x_1, x_2, \dots, x_k) = \frac{n!}{(n-k)!} \int_{\mathbb{R}^{n-k}} \frac{1}{\mathcal{Z}_n n!} \prod_{1 \leq i < j \leq n} (x_i - x_j)^2 \prod_{j=1}^n e^{-x_j^2} dx_{k+1} \cdots dx_n,$$

as given by ??.

For simple points processes on $\mathbb{R}$ there is a direct way to understand correlation functions. Let $A_i = [x_i, x_i + \Delta x_i]$, $1 \leq i \leq k$, be disjoint intervals. If the Δx_i are small enough, then we should expect either one or zero particles in each interval A_i . Hence, the product $\xi(A_1) \cdots \xi(A_k)$ is typically either 1 or 0 depending on whether we have one particle in each interval. From ??, we should expect

$$p_k(x_1, x_2, \dots, x_k) = \lim_{\Delta x_i \rightarrow 0} \frac{\mathbb{P}(\text{one particle in each } A_i)}{\Delta x_1 \Delta x_2 \cdots \Delta x_k}.$$

This means that $p_1(x)$ is the density of particles at x , but since we have many particles, the events of finding particles at their respective points are not disjoint events, even for distinct points. It follows from the definition that if we have a simple point process, then $p_k(x_1, x_2, \dots, x_k)$ is exactly the probability of finding particles at $x_1, x_2, \dots, x_k$.

3.2 Determinantal Point Processes

Repulsive random processes are, in general, intractable, that is, they are usually difficult or even impossible to, for example, compute marginals and conditional probabilities. In these cases, their theoretical analysis is a major challenge. However, there is a class of processes, even repulsive ones, that are an rare exception to this rule: *Determinantal Point Processes*; or Fermionic point processes in the physics literature, are random processes that have the property of being both repulsive and tractable (TREMBLAY, 2025). This property comes from the fact that their correlation function can be expressed as a certain determinantal form. Let's make this precise.

Definition 9. Consider a point process $\mathbb{P}$ on a complete separable metric space Λ , with reference measure $\mathfrak{p}_n$, all of whose correlation functions $\mathfrak{p}_k$ exist. If there is a function $K : \Lambda \times \Lambda \rightarrow \mathbb{C}$ such that

$$\mathfrak{p}_k(x_1, x_2, \dots, x_k) = \det(K(x_i, x_j))_{i,j=1}^k$$

for all $x_1, x_2, \dots, x_k \in \Lambda$, $k \geq 1$, then we say that $\mathbb{P}$ is a **Determinantal Point Process**. We call K the **correlation kernel** of the process.

Example 13. (Eigenvalue for orthogonal polynomial ensembles) We have shown in [subsection 2.2.1](#) that

$$\mathfrak{p}_k^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_k; n; \beta = 2) = \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k,$$

for a subclass of the invariant ensembles called orthogonal polynomial ensembles. This is the main connection for the theory of random matrices and DDPs - and one that we take major advantage of. For $\beta = 2$, we should have

$$\mathfrak{p}_n^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; 2) \propto \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^2 = \det \left(w(\lambda_i) q_{j-1}^{(2)}(\lambda_i) \right)_{n \times n}^2$$

where $w(\lambda_i)$ is a weight function and $|\Delta_n(\lambda)|$ is the Vandermonde determinant. Then, we can write

$$\begin{aligned} \mathfrak{p}_n^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; 2) &\propto \det \left(\sqrt{w(\lambda_i) w(\lambda_k)} \sum_{j=1}^n q_{j-1}^{(2)}(\lambda_i) q_{j-1}^{(2)}(\lambda_k) \right)_{i,k=1}^n, \\ &\propto \det \left[K_n^{(2)}(\lambda_i, \lambda_k) \right]_{i,k=1}^n. \end{aligned}$$

The orthogonal polynomials $\{q_j^{(2)}(\lambda)\}_{j=0}^\infty$ that compose the kernel are known to satisfy the three-term recurrence (the coefficients depend on the choice of weight)

$$q_j^{(2)}(\lambda) = (\lambda + a_j) q_{j-1}^{(2)}(\lambda) - \frac{r_{j-1}^{(2)}}{r_{j-2}^{(2)}} q_{j-2}^{(2)}(\lambda).$$

All important information is contained in the correlation kernel for these processes and, therefore, all quantities of interest can be expressed in terms of the same K . The kernel K can be seen as an integral kernel of an operator K on $L^2(\Lambda, \lambda)$,

$$Kf(x) = \int_{\Lambda} K(x, y)f(y)d\lambda(y)$$

as long as the right and side is well defined. Important examples of DDPs are constructed by using the following result.

Theorem 5. (*Construction for DDP, (GIROTTI, 2014)*) Consider a kernel $K : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}_+$ with the following properties:

- *Trace-class:* $\text{Tr}(K) = \int_{\mathbb{R}} K(x, x)dx = n < +\infty$;
- *Positivity:* $\det[K(x_i, x_j)]_{i,j=1}^n$ is non-negative for every $x_1, x_2, \dots, x_n \in \mathbb{R}$;
- *Reproducing property:* $\forall x, y \in \mathbb{R}$

$$K(x, y) = \int_{\mathbb{R}} K(x, s)K(s, y)ds.$$

Then,

$$u_n(x_1, x_2, \dots, x_n) := \frac{1}{n!} \det[K(x_i, x_j)]_{i,j=1}^n$$

is a probability measure on $\mathbb{R}^n$, invariant under coordinated permutations. The associated point process is a determinantal point process with K as a correlation kernel.

Example 14. Consider the Airy Kernel,

$$A(x, y) = \int_0^\infty Ai(x+t)Ai(y+t)dt$$

where Ai is the Airy function, a positive convex and exponentially decreasing function, defined as

$$Ai(x) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right)dt.$$

This kernel is Hermitian and has

$$\int_t^\infty A(x, x)dx < \infty, \quad \forall t \in \mathbb{R}$$

due to orthogonality of the Airy function. It can be shown that $A(x, y)_{(t, \infty)}$ is the kernel of a trace class operator and there is a point process $\mathbb{P}$ on $\mathbb{R}$, the Airy Kernel Determinantal Point Process, which is determinantal with kernel $A(x, y)$. We also have

$$F_{TW}(t) := \mathcal{P}[\lambda^*(\xi) \leq t] = \det(I - A)_{L^2(t, \infty)}.$$

The distribution function $F_{TW}^{(2)}(t)$ for the last particle in the Airy kernel point process is a natural scaling limit of certain finite determinantal point processes. We call it the

Tracy-Widow distribution. Our case is to set $\beta = 2$, but there is also distributions referring to parameters $\beta = 1, 4$ - they can be found in other (non-determinantal) point processes such as the GOE and GSE. Note the asymmetry related to the transition of regimen and the spread increasing with the decrease of β - this can be interpreted as an increase in “temperature” of the particle system, to make sense in [Chapter 5](#).

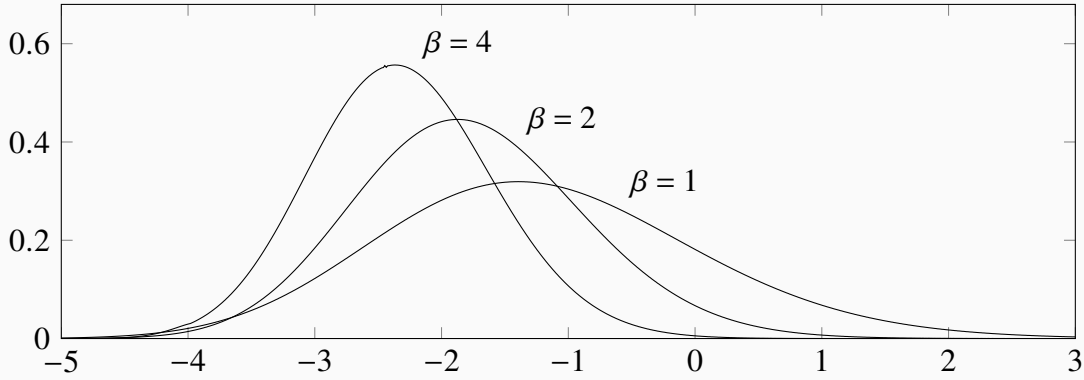


Figure 9 – Plots of $F_{TW}^{(\beta)}(t)$ for $\beta = 1, 2, 4$.

3.3 Limit and Transformations of DPP

Let $f \in C^1(\mathbb{R})$ be a bijection, with continuously differentiable inverse $f^{-1} = g \in C^1(\mathbb{R})$. Then, f maps configurations on $\mathbb{R}$ to configurations on $\mathbb{R}$. The push-forward of a point process $\mathbb{P}$ under such mapping is $f(\mathbb{P})$ and it is again a point process.

Proposition 4. ([GIROTTI, 2014](#)) *If K is the kernel of a DDP $\mathbb{P}$, then $f(\mathbb{P})$ is also a DDP with kernel*

$$\hat{K}(x, y) = \sqrt{g'(x)g'(y)}K(g(x), g(y))$$

where $f^{-1} = g$.

In particular, if we consider a linear function $f(x) = \alpha(x - x^*)$, for $\alpha > 0$ and $x^* \in \mathbb{R}$, then the transformed point process $f(\mathbb{P})$ is a scaled and translated version of the original point process $\mathbb{P}$. The new correlation kernel is

$$\hat{K}(x, y) = \frac{1}{\alpha} K\left(x^* + \frac{x}{\alpha}, x^* + \frac{y}{\alpha}\right).$$

Example 15. (*Karlin-McGregor Theorem & Dyson Bridges* ([KUIJLAARS, 2014](#)))

Consider n independent copies $X_1(t), \dots, X_n(t)$ of a one-dimensional strong Markov Process with continuous sample paths conditioned so that $X_j(0) = a_j$, where $a_1 < a_2 < \dots < a_n$ are certain given values. Let $p_t(x, y)$ be some transition probability function of the Markov process. Moreover, let $E_1, E_2, \dots, E_n$ be Borel sets so that $\sup E_j < \inf E_{j+1}$

for $j = 1, \dots, n-1$. Then

$$\int_{E_1} \cdots \int_{E_n} \det[p_t(a_i, x_j)]_{i,j=1}^n dx_1 \dots dx_n$$

is equal to the probability that the paths have not intersected in the time interval $[0, t]$ and $X_j(t) \in E_j$ for $j = 1, \dots, n$. This is the Karlin-McGregor theorem. Take now $p_t(a, x) = 1/\sqrt{(2\pi t)} \exp\{-(x-a)^2/2t\}$, a Gaussian transition, and consider that we condition our path to not only start at positions $a_1, a_2, \dots, a_n$ but finish at positions $b_1, b_2, \dots, b_n$ at some time $T > t$. If we let $a_j \rightarrow a$ and $b_j \rightarrow b$, the new j.p.d.f. can be non-trivially calculated as

$$\begin{aligned} P(x_1, \dots, x_n) &= \frac{1}{\mathcal{Z}_n} \det[p_t(a_i, x_j)]_{i,j=1}^n \det[p_{T-t}(x_i, b_j)]_{i,j=1}^n \\ &= \frac{1}{\mathcal{Z}_n} \det[x_j^{i-1}]_{i,j=1}^n \prod_{j=1}^n e^{-\frac{x_j^2}{2t}} \det[x_j^{i-1}]_{i,j=1}^n \prod_{j=1}^n e^{-\frac{x_j^2}{2(T-t)}} \\ &= \frac{1}{\mathcal{Z}_n} \det[x_j^{i-1}]_{i,j=1}^n \det[x_j^{i-1}]_{i,j=1}^n \prod_{j=1}^n e^{-\frac{Tx_j^2}{2t(T-t)}} \\ &= \frac{1}{\mathcal{Z}_n} \prod_{1 \leq i < j \leq n} (x_j - x_i)^2 \prod_{j=1}^n e^{-\frac{Tx_j^2}{2t(T-t)}} \end{aligned}$$

Two things are worth noting on this point. The first is that this can be reduced to the j.p.d.f. of the Gaussian Unitary Ensemble as in Example 4! The second thing is that it has a scaling in the weight function based on t . This means that for every t the distribution obeys the same law with a different scaling - it keeps being an determinant process as specified in Proposition 4. One can simulate such Brownian motions as below for $T = 1$.

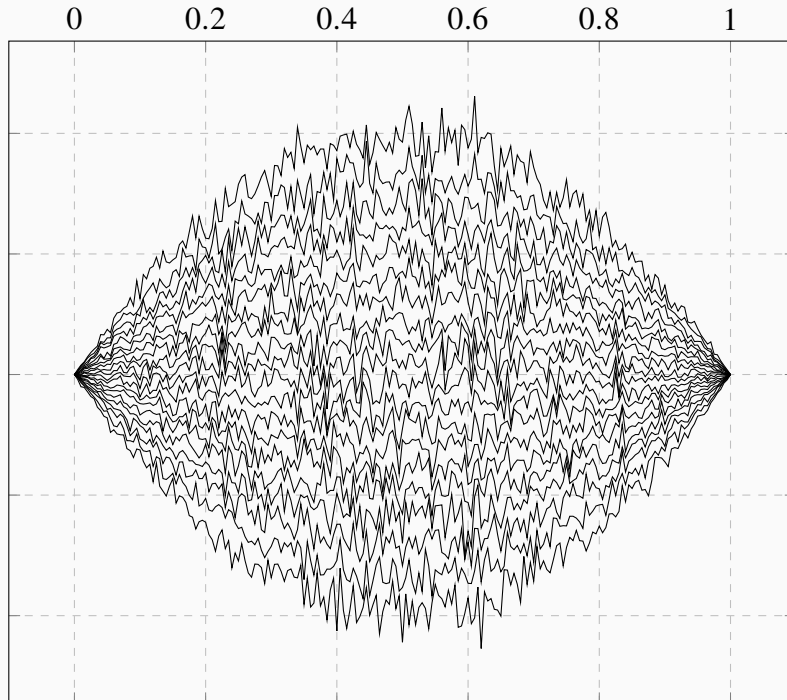


Figure 10 – 20 Non-intersecting Brownian paths for $0 < t < 1$ s.t. they start and end at $y = 0$.

Notice on the images below that the distribution of points is kept the same minus the re-scaling of $x/((2t(1-t))\sqrt{N})$.

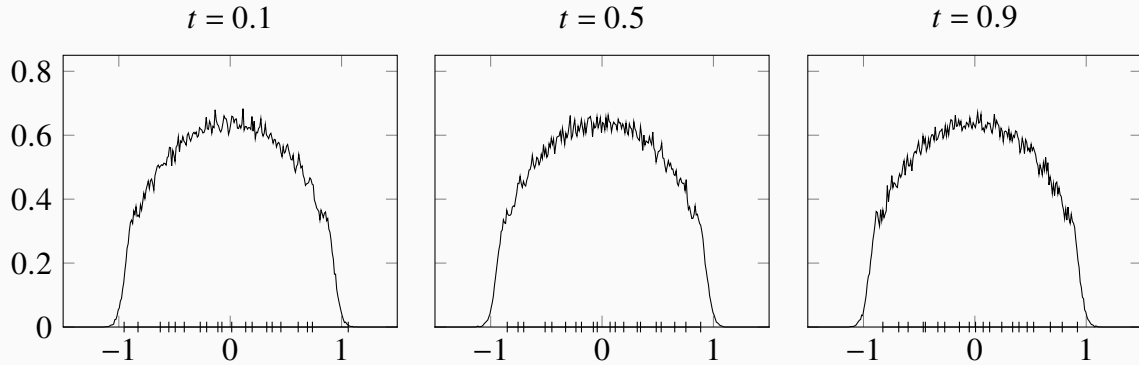


Figure 11 – Three conditions simulated from the 20 Dyson Bridges taken from $t = 0.1, 0.5, 0.9$. For each one the experimental density (line plot) and one sampled position (dashes in axis) were plotted. The three plots are re-scaled by $((2t(1-t))\sqrt{N})$.

Others transformations are also possible. For example, consider what happens when one fixes $c \in \mathbb{R}$ and, given that $c \in \chi$, removes c from the configuration. The resulting process $\chi \setminus \{c\}$ is a new point process.

Proposition 5. (GIROTTI, 2014) *Let K be the correlation kernel of a DDP and let c be a point such that $0 < K(c, c) < +\infty$. Then, the point process obtained by removing c is determinantal with kernel*

$$\hat{K}(x, y) = K(x, y) - \frac{K(x, c)K(c, x)}{K(c, c)}.$$

Even more importantly we will take limits of Kernels. We now want to consider sequences of finite determinantal point processes $\mathbb{P}_n$ with correlation kernel K_n and understand how the limit behaves.

Proposition 6. (GIROTTI, 2014) *Let $\mathbb{P}$ and $\mathbb{P}_n$ be determinantal point processes with kernels K and K_n respectively. Let K_n converge point-wise to K*

$$\lim_{n \rightarrow \infty} K_n(x, y) = K(x, y)$$

uniformly in x, y over compact subsets of $\mathbb{R}$. Then, the point processes $\mathbb{P}_n$ converge to $\mathbb{P}$ weakly.

A very important case studied here is of the limit of an scaled and centered determinantal point process. Suppose a sequence of kernels K_n and fixed point c of interest. We first perform a centering and re-scaling of the form

$$x \mapsto Cn^\alpha(x - c)$$

for some suitable value of $C, \gamma > 0$. Then the scaled kernels have a limit

$$\lim_{n \rightarrow \infty} \frac{1}{Cn^\alpha} K_n \left(c + \frac{x}{Cn^\alpha}, c + \frac{y}{Cn^\alpha} \right) = K(x, y)$$

Therefore, the scaling limit K is a kernel that corresponds to a determinantal point process with an infinite number of points. Interestingly, in many situations, *the same scaling limit K may appear*. This phenomenon is known as **universality** in RMT. Examples of such kernels are the sine kernel

$$K_{\sin} = \frac{\sin \pi(x - y)}{x - y}$$

that appears in bulk statistics of distributions of eigenvalues. Furthermore, we also have the important Airy kernel that describes edge behavior and is defined by

$$K_{Ai} = \frac{Ai(x)Ai'(y) - Ai'(x)Ai(y)}{x - y}$$

where Ai is the Airy function. Both these kernels can be identified as local laws on the bulk and edge of invariant ensembles, as posed in Theorem 6 below.

Theorem 6. *If*

$$\lim_{n \rightarrow \infty} \frac{1}{n} K_n(x, x) = \phi_V(x), \quad x \in \mathbb{R},$$

then the asymptotic distribution of the points in the point process (eigenvalues of $\hat{M}_n$) is given by the measure with density ϕ_V :

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n f(x_j) = \int f(x) \phi_V(x) dx.$$

Moreover, suppose ϕ_V to be supported on one interval $[a, b]$, then for $x^ \in (a, b)$ with $\phi_V(x^*) > 0$, one has*

$$\lim_{n \rightarrow \infty} \frac{1}{\phi_V(x^*)n} K_n \left(x^* + \frac{x}{\phi_V(x^*)n}, x^* + \frac{y}{\phi_V(x^*)n} \right) = \frac{\sin \pi(x - y)}{\pi(x - y)}.$$

Finally, if near the endpoint b

$$\phi_V(x) = \frac{c}{\pi} (b - x)^{\frac{1}{2}} (1 + o(1)), \quad x \rightarrow b,$$

then

$$\lim_{n \rightarrow \infty} \frac{1}{(cn)^{\frac{2}{3}}} K_n \left(b + \frac{x}{(cn)^{\frac{2}{3}}}, b + \frac{y}{(cn)^{\frac{2}{3}}} \right) = \frac{Ai(x)Ai'(y) - Ai'(x)Ai(y)}{x - y},$$

where Ai is the Airy function.

ASYMPTOTICS

There is a sense in the previous Chapters that we want to study some distributions when our matrices are large enough - be that for feasibility of computations or approximation of infinite dimension operators. We also have mentioned that our main interest here is the Invariant Ensembles, or specially in Orthogonal Polynomial ones. Because of this, studying large matrices is equivalent, in some sense, to studying the regime of large n for orthogonal polynomials. For some of those polynomials, a integral representation is available, allowing us to use asymptotic methods on integrals to solve for the large n behavior. This is the main discussion of this chapter.

Techniques and structuring here are based on the book by [Olver *et al.* \(2010\)](#) and [Alpert \(1999\)](#) for quadrature and on the lecture notes by [Silva \(2021\)](#) for asymptotics on integrals.

4.1 Quadrature Technique

Before doing asymptotics on integrals we discuss a technique that will become important when approximating large summations: quadratures, i.e., approximating integrals by summations and vice-versa. One can see that asymptotics on integrals can then also perform a role on discrete summations. We first start briefly introducing a useful tool in the development of the formula we wish to use: the Bernoulli numbers B_n , polynomials $B_n(x)$ and periodic polynomials $\tilde{B}_n(x)$. The Bernoulli numbers B_n are a sequence of rational numbers that frequently appears in the Taylor series expansion of trigonometric-type functions. These number can also be seen as special values obtained by the Bernoulli polynomials $B_n(x)$. This is because these polynomials can be written explicitly as

$$B_k(x) = \sum_{n=0}^k \binom{k}{n} B_n x^{k-n},$$

then, we must have that $B_k(0) = B_k$. More interestingly for us is the fact that these polynomials satisfy the differential equation

$$B'_n(x) = nB_{n-1}(x), \quad n = 1, 2, \dots$$

Finally, the periodic Bernoulli polynomial is simply the Bernoulli polynomials evaluated at the fractional part of the argument, i.e., $\tilde{B}_n(x) = B_n(x - \lfloor x \rfloor)$. Now, let B_n and $B_n(x)$ denote the n th Bernoulli number and polynomial, respectively, and $\tilde{B}_n(x)$ the n th Bernoulli periodic function $B_n(x - \lfloor x \rfloor)$. Assume that a, m and n are integers such that $n > a$, $m > 0$, and $f^{(2m)}(x)$ is absolutely integrable over $[a, n]$. Then,

Theorem 7. (*Euler-Maclaurin Formula, (OLVER et al., 2010)*) Following the context given above,

$$\begin{aligned} \sum_{j=a}^n f(j) &= \int_a^n f(x)dx + \frac{1}{2}(f(a) + f(n)) + \sum_{s=1}^{m-1} \frac{B_{2s}}{(2s)!} \left(f^{(2s-1)}(n) - f^{(2s-1)}(a) \right) \\ &\quad + \int_a^n \frac{B_{2m} - \tilde{B}_{2m}(x)}{(2m)!} f^{(2m)}(x)dx. \end{aligned}$$

This equation is called the *Euler-Maclaurin* formula. It can be extremely useful when estimating the asymptotic of the summation on the function f when its integral can be more easily evaluate. In essence, this is a quadrature technique; a numerical integration: the approximation of an integral $\int f(x)dx$ of some function f by a discrete summation $\sum w_i f(x_i)$ over a set of points x_i with some weight function w_i . For example, in the classical Riemann sums, one approximates the integral $\int f(x)dx$ over some interval $[a, b]$ with the limit of the summation $\sum f(x_i^*)\Delta x_i$, where $x_i^* \in [x_{i-1}, x_i]$ and $\Delta x_i = x_i - x_{i-1}$. Different methods can be generated by choosing x_i^* differently.

The trapezoidal rule, for example, is a Riemann sum where x_i^* is chosen to be the mean point $x_i^* = (x_{i-1} + x_i)/2$. In this way, we would estimate the area using trapezoids, as the name suggests. Then, one could write

$$\begin{aligned} \int_a^n f(x)dx &\approx \sum_{j=0}^n f(x_j^*) = \sum_{j=0}^n \frac{f(x_j + x_{j-1})}{2}, \\ &= \sum_{j=1}^{n-1} f(x_j^*) + \frac{1}{2}(f(n) + f(0)) = \sum_{j=0}^{n-1} f(x_j^*) + \frac{1}{2}(f(n) - f(n)). \end{aligned}$$

Of course, this has some error term. Suppose that for some (x_i, x_{i-1}) we have the same k -derivatives at both extreme points for every $k \geq 1$, i.e., the function is constant. Then, there is no error in the approximation. If however, the two extreme points differ only on the k th derivative, there is some curvature defined by this difference where the error comes from. Therefore, there is indication of some error function given in respect to higher order derivatives.

Proof. Following the computations in Alpert (1999), take a function $f \in C^p(\mathbb{R})$ for $p \geq 1$. To derive the Euler-Maclaurin for the interval $[a, b]$, let $h = (b - a)/n$ and $c = a + jh$, then we can express

$$\int_a^b f(x)dx = \sum_{j=0}^{n-1} \int_{a+jh}^{a+(j+1)h} f(x)dx$$

where we can use the property ?? and integration by parts to rewrite each integral as

$$\begin{aligned} \int_c^{c+h} f(x)dx &= h \int_0^1 B_0(1-x) f(c+xh)dx \\ &= -h \frac{B_1(1-x)}{1!} f(c+xh) \Big|_0^1 + h^2 \int_0^1 \frac{B_1(1-x)}{1!} f'(c+xh)dx, \\ &\vdots \\ &= -\sum_{r=0}^{p-1} \frac{h^{r+1} B_{r+1}(1-x)}{(r+1)!} f^{(r)}(c+xh) \Big|_0^1 + h^{p+1} \int_0^1 \frac{h^{r+1} B_p(1-x)}{(p)!} f^{(p)}(c+xh)dx, \\ &= h \frac{f(c) + f(c+h)}{2} - \sum_{r=0}^{p-1} \frac{h^{r+1} B_{r+1}(1-x)}{(r+1)!} [f^{(r)}(c+h) - f^{(r)}(c)] \\ &\quad + h^{p+1} \int_0^1 \frac{B_p(1-x)}{(p)!} f^{(p)}(c+xh)dx. \end{aligned}$$

where one can start to notice the trapezoidal rule creeping out on the first term. Notice that it was necessary to use that $-B_1(0) = B_1(1) = 1/2$ and $B_n(0) = B_n(1) = B_n$ for $n \neq 1$. Returning to the summation, we can rearrange terms as

$$\begin{aligned} h \left[\frac{f(a)}{2} + f(a+h) + \cdots + f(b-h) + \frac{f(b)}{2} \right] &= \frac{1}{2}(f(b) - f(a)) + \sum_{j=0}^{n-1} f(a+jh), \\ &= \int_a^b f(x)dx + \sum_{r=0}^{p-1} \frac{h^{r+1} B_{r+1}(1-x)}{(r+1)!} [f^{(r)}(b) - f^{(r)}(a)] - h^{p+1} \int_0^1 \frac{B_p(1-x)}{(p)!} f^{(p)}(c+xh)dx. \end{aligned}$$

□

This is the Euler-Maclaurin formula, an extension of the trapezoidal rule including the higher order terms.

4.2 Asymptotic on Integrals

The techniques we will expand on here refers to a specific type of complex function $\phi(\lambda) : \mathbb{C} \rightarrow \mathbb{R}$ in the regimen where $\lambda \rightarrow \infty$. Those functions are of the form

$$\phi(\lambda) := \int_{\Gamma} g(s) e^{-\lambda f(s)} ds,$$

where Γ is a given contour in the plane and f, g are suitable functions. Suitable here means that we impose g and f to be continuous (or analytic); g is integrable; f have global minima

with zero derivative (maybe in more than one point) and strictly positive second derivative at this point(s) of minima. There is also a more technical regularity condition on f to ensure the approximations holds to still be made clear.

Example 16. (Airy Equation) Function such as in ?? are not uncommon. Consider the function that satisfies the equation $\psi''(x) = x\psi(x)$. Taking the Fourier transform we have

$$-\omega^2\Phi(x) = i\Phi'(x)$$

for which we can solve with $\Phi(x) = e^{-i\omega^3/3}$ and then take the inverse Fourier Transform to get

$$\phi(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i/3\omega^3 + ix\omega} d\omega.$$

The imaginary part of the argument is an odd function and therefore zero when integrated. The real part is expressed

$$\phi(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \cos\left(\frac{\omega^3}{3} + x\omega\right) d\omega.$$

For this integral, the techniques applied in this chapter are of great use to understand its behavior as $\omega \rightarrow \infty$.

Moreover, one can somehow directly motivate such conditions. Consider $\phi(\lambda) : \mathbb{R} \rightarrow \mathbb{R}$, f with a single global minima and $\Gamma \equiv \mathbb{R}$ - for simplicity; the argument's idea should hold for other choices given the appropriate considerations. If ϕ is as in ??, and satisfying the conditions given for such function, then take x^* the single minimal critical point for f . We first note that if that is true, then, for any $|\epsilon| > 0$, the following limit should hold:

$$\lim_{\lambda \rightarrow \infty} \frac{e^{-\lambda(f(x^*))}}{e^{-\lambda(f(x^*-\epsilon))}} = \lim_{\lambda \rightarrow \infty} e^{-\lambda(f(x^*)-f(x^*-\epsilon))} = \lim_{\lambda \rightarrow \infty} e^{\lambda|c|} \rightarrow +\infty.$$

That is, the main contribution for the integral comes from a single point. The main idea then is to separate the integral in two regions: A first interval $(x^* - c/\sqrt{\lambda}, x^* + c/\sqrt{\lambda})$, $c > 0$, including this critical point and another interval that in theory should contribute only in error for the integral asymptotic. We start by approximating f around its global maximum by Taylor's theorem

$$f(x) = f(x^*) - \frac{1}{2}|f''(x^*)|(x-x^*)^2 + \epsilon_2(x-x^*)$$

and then perform a change of variables $\frac{x}{\sqrt{\lambda}} = s - x^*$, so that

$$\int_{x^* - \frac{c}{\sqrt{\lambda}}}^{x^* + \frac{c}{\sqrt{\lambda}}} g(s) e^{-\lambda f(s)} d(s) = \frac{e^{-\lambda f(x^*)}}{\sqrt{\lambda}} \int_{-c}^c g\left(x^* + \frac{x}{\sqrt{\lambda}}\right) e^{\frac{1}{2}|f''(x^*)|x^2 - \epsilon_2\left(\frac{x^3}{\sqrt{\lambda}}\right)} dx.$$

Now, using the technical assumption of regularity on the function f , one can argue that the error term ϵ_2 goes to zero sufficiently fast, leaving a Gaussian Integral. More that that, one need

to argue that the integral outside this small neighborhood of x^* is exponentially smaller than $e^{-\lambda f(x^*)}$. This should follow by an argument where the asymptotic for the integral

$$\int_{x^* + \frac{c}{\sqrt{\lambda}}}^{\infty} g(s) e^{-\lambda f(s)} d(s) \leq e^{-\lambda f(x^* + c/\sqrt{\lambda})} \|g\|_{L^1}$$

is exponentially less than the previous region.

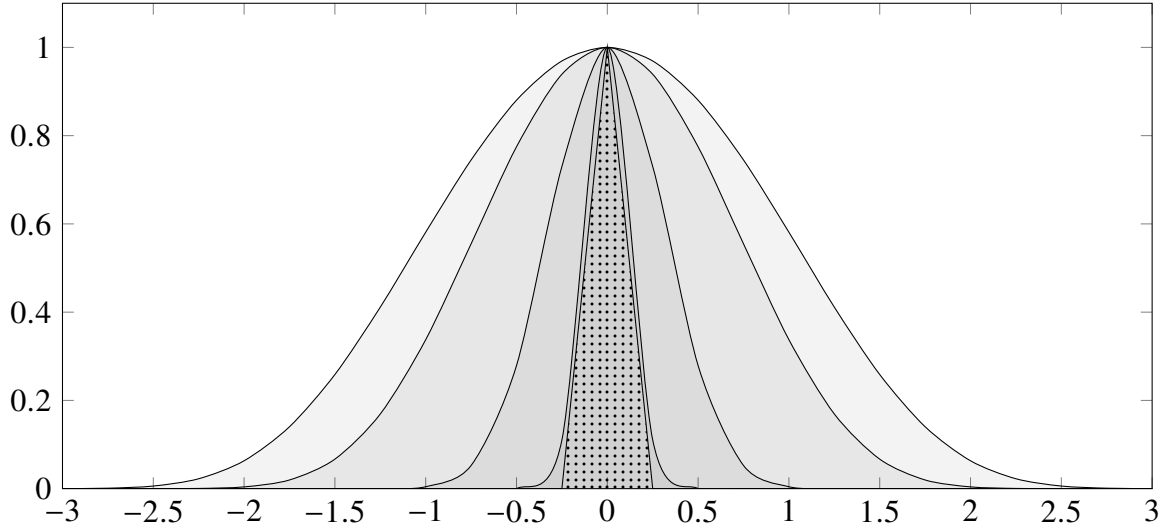


Figure 12 – In the Plot it is represented the plot for $e^{-\lambda(\cosh(x)-1)}$ and an increasing $\lambda = 1, 2, 10, 70$ - darker represent larger values of the parameter. The dotted patter represents $e^{-\lambda x^2}$ for large lambda.

With this, we only note a few details of interest. Firstly, more precise error estimates can be obtained if one uses higher order terms of the Taylor Series expansion of f . For maximizer x^* which have null second derivative (a double critical point), the Laplace Approximation can be used expanding to the third order terms of the Taylor Series $f(x) = f(x^*) = \frac{1}{6}f'''(x^*)(x - x^*)^3 + \epsilon_3(x - x^*)$ and making the change of variables $t/\sqrt[3]{n} = x - x^*$ instead. This argument and result could also be used on curves on the plane. This is motivated from the fact that every (smooth) curve is locally a straight line. This will lead us to the method of *Steepest Descent*. These next subsections are in the spirit of proving a bit more formally the expansion given by the *Laplace Method* and introducing the mentioned *Steepest Descent*.

4.2.1 Watson's Lemma

We start this topics in the real line by looking at the functions $F : \mathbb{R} \rightarrow \mathbb{R}$ of the form

$$F(\lambda) = \int_0^T f(t) e^{-\lambda t} dt, \quad (4.1)$$

which one may think of as the Laplace transform of a function f supported on $[0, T]$. Note that F fits well with the models we described previously, and so one should expect the main contribution to come from the minima of the exponent function. Let's formalize the conditions for such expectation and the consequent result.

Theorem 8. (Watson's Lemma) Take $f \in L^1(0, T)$ and suppose that for some $\delta > 0$ it is of the form

$$f(t) = t^\alpha f_0(t), \quad 0 < t \leq \delta,$$

where $\alpha \in (-1, \infty)$ and f_0 is continuously differentiable on $[0, \delta]$ with $f_0(0) \neq 0$. Then the function F in (4.1) satisfies

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1 + O(\lambda^{-1}))$$

when $\lambda \rightarrow \infty$. Also, Γ is the usual Gamma function.

We must first mention that a weak estimation of the integral could be given by

$$\int_0^T f(t) e^{-\lambda t} dt \leq \|f\|_{L^1(0, T)}$$

where we only use the fact that $f \in L^1(0, T)$. However, we have more information on the behavior of the function in the section $0 < t < \delta$. Because of that fact, it is expected that in this set we could refine the approximation.

Proof. We start by decomposing the function in two parts

$$F(\lambda) = \int_0^\delta f(t) e^{-\lambda t} dt + \int_\delta^T f(t) e^{-\lambda t} dt =: (1) + (2).$$

For (2), we can use the same weak estimate as commented before. This is because we have no further information on the function in this sector of the line. This leave us with

$$F(\lambda) \leq \int_0^\delta f(t) e^{-\lambda t} dt + \|f\|_{L^1(\delta, T)} e^{-\delta\lambda}.$$

Now, we remind of the condition of continuous differentiability and the hypothesis on $f(t)$ to get $f(t) = t^\alpha (f_0(0) + r(t))$ for some differentiable error function $r(t)$ satisfying $|r(t)| \leq \|f'_0\|_{L^\infty(0, \delta)} t$ with $\|f'_0\|_{L^\infty(0, \delta)} = \inf \{c \geq 0 \mid \mu|_{(0, \delta)}(\{|f'_0| > c\}) = 0\}$ as usual. Then, write (1) as

$$\begin{aligned} (1) &= \int_0^\delta t^\alpha (f_0(0) + r(t)) e^{-\lambda t} dt = f_0(0) \int_0^\delta t^\alpha e^{-\lambda t} dt + \int_0^\delta t^\alpha r(t) e^{-\lambda t} dt, \\ &=: f_0(0) \cdot (1.1) + (1.2). \end{aligned}$$

For these two integrals we can try to simplify the problem by rewriting

$$\begin{aligned} (1.1) &= \int_0^\infty t^\alpha e^{-\lambda t} dt - \int_\delta^\infty t^\alpha e^{-\lambda t} dt = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - \int_\delta^\infty t^\alpha e^{-\lambda t} dt, \\ |(1.2)| &\leq \|f'_0\|_{L^\infty(0, \delta)} \int_0^\delta t^{\alpha+1} e^{-\lambda t} dt, \\ &= \|f'_0\|_{L^\infty(0, \delta)} \left(\int_0^\infty t^{\alpha+1} e^{-\lambda t} dt - \int_\delta^\infty t^{\alpha+1} e^{-\lambda t} dt \right), \\ &= \|f'_0\|_{L^\infty(0, \delta)} \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - \int_\delta^\infty t^{\alpha+1} e^{-\lambda t} dt \right). \end{aligned}$$

We are now only missing one ingredient to get the result, the order of

$$G(\lambda, \beta) := \int_{\delta}^{\infty} t^{\beta} e^{-\lambda t} dt, \quad \lambda > 1, \quad \beta > -1.$$

Using Cauchy-Schwartz we can estimate

$$\begin{aligned} |G(\lambda, \beta)|^2 &\leq \left(\int_{\delta}^{\infty} e^{-\lambda t} dt \right) \left(\int_{\delta}^{\infty} t^{2\beta} e^{-\lambda t} dt \right), \\ &= \left(\frac{e^{-\lambda \delta}}{\lambda} \right) \left(\int_{\delta}^{\infty} t^{2\beta} e^{-\lambda t} dt \right), \\ &\stackrel{(u=\lambda t)}{=} \left(\frac{e^{-\lambda \delta}}{\lambda} \right) \left(\int_{\delta}^{\infty} e^{-u} u^{2\beta} \frac{1}{\lambda^{2\beta+1}} du \right), \\ &= \frac{e^{-\lambda \delta}}{\lambda^{2\beta+2}} M_{\beta} \end{aligned}$$

where M_{β} is some constant independent of λ . That leave us with

$$G(\lambda, \beta) = O\left(\frac{e^{-\lambda \delta/2}}{\lambda^{\beta+1}}\right) = O\left(e^{-\epsilon \lambda}\right)$$

where $\epsilon > 0$ depends on β and δ . Finally we can rewrite

$$(1.1) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - G(\lambda, \alpha) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + O\left(e^{-\epsilon \lambda}\right),$$

$$|(1.2)| \leq \|f'\|_{L^{\infty}(0, \delta)} \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - G(\lambda, \alpha+1) \right) = \|f'\|_{L^{\infty}(0, \delta)} \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} + O\left(e^{-\epsilon \lambda}\right) \right) = O\left(\lambda^{-\alpha-2}\right),$$

and

$$F(\lambda) \leq O\left(\lambda^{-\alpha-2}\right) + f_0(0) \left(\frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + O\left(e^{-\epsilon \lambda}\right) \right) + \|f\|_{L^1(\delta, T)} e^{-\delta \lambda}.$$

By the dominance of the exponential error, we get the result

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}} (1 + O(\lambda^{-1})).$$

□

In the estimation above, the Gamma function and the factor of λ^{-1} is the result of integrating the linear exponent of $e^{-\lambda t}$. If we, however, change this exponent to be of the general type $-\lambda t^d$ we can still have a similar result on the behavior of the function.

Proposition 7. For $T, S > 0$, define the function

$$G(\lambda) := \int_{-S}^T g(t) e^{-\lambda t^2} dt$$

and assume $g(x)$ absolutely integrable in $[-S, T]$, of class C^1 in a fixed neighborhood of

the origin. Then the function satisfies

$$G(\lambda) = \frac{g(0)\Gamma(1/2)}{\lambda^{1/2}} + O\left(\frac{1}{\lambda^{3/2}}\right)$$

when $\lambda \rightarrow \infty$ and where Γ is the Gamma function.

Proof. The proof is a reduction of the problem to Watson's Lemma. Assume, $T > S$ and write

$$G(\lambda) = \int_{-S}^S g(t) e^{-\lambda t^2} dt + \int_S^T g(t) e^{-\lambda t^2} dt.$$

The second integral we know to be of order

$$\int_S^T g(t) e^{-\lambda t^2} dt \leq \|g\|_{\infty} e^{-\lambda S^2},$$

exponential on λ , so we drop it. For the first integral we rewrite

$$\begin{aligned} \int_{-S}^S g(t) e^{-\lambda t^2} dt &= \int_0^S (g(t) + g(-t)) e^{-\lambda t^2} dt, \\ &\stackrel{(u=t^2)}{=} \frac{1}{2} \int_0^{S^2} \left(\frac{g(\sqrt{u}) + g(-\sqrt{u})}{\sqrt{u}} \right) e^{-\lambda u} du \end{aligned}$$

where now we can use Watson's Lemma to finish the proof. $\square$

Example 17. Take the function called Gamma given by the expression

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} dt.$$

We are interested in what happens to the function when we take the variable x to $+\infty$. To see what happens, let's fit it into Proposition 11. Rewrite ?? as

$$\int_0^{\infty} e^{-t+x \log(t) - \log(t)} dt.$$

This lead us to identify ϕ as $-\log(t)$, which would give us $\phi'(t) < 0$, that is, we would have it's minimum at $t = +\infty$. However $e^{\log(t)} = t^1$ is not integrable near the origin. Something need to change about the identification. Let's introduce a variable to rescale the integral: $xs = t$, s.t.,

$$\Gamma(x) = x^x \int_0^{\infty} e^{-\lambda(s-\log(s))} e^{-s} ds.$$

Where we have used $\lambda = x - 1$. Thus we identify $\phi(s) = s - \log(s)$ and $g(s) = e^{-s}$. in fact, now ϕ has a global minima at $s = 1$ and $\phi''(1) = 1 \neq 0$. Hence

$$\begin{aligned} \Gamma(x) &= \frac{\sqrt{2\pi} x^x e^{-x}}{(x-1)^{(-1/2)}} (1 + O(x^{-1})), \\ &= \sqrt{2\pi} x^{x-1/2} e^{-x} (1 + O(x^{-1})), \\ &= (x - 1/2) \log(x) - x + \frac{1}{2} \log(2\pi), \end{aligned}$$

which is called the Stirling formula.

4.2.2 Laplace Method

A seemly natural extension of the type of function we were dealing with are functions of the type

$$\Phi(\lambda) := \int_a^b g(t) e^{-\lambda\phi(t)} dt$$

for $a, b \in \mathbb{R} \cup \{-\infty, \infty\}$ and $a < b$ and a well behaved function g . For now, ϕ is a real-valued function. To begin the study let's make some assumptions. The first one is that a is finite, that means that we can set $a = 0$ without loss of generality. The second assumption is that ϕ is a increasing function on $[0, b]$, so that $\phi(0) < \phi(t)$ for any t in the interval. With that we note that the coefficient $e^{-\lambda\phi(t)} / e^{-\lambda\phi(0)}$ decreases exponentially, that is, the integral is accumulated as the neighborhood of 0, i.e.

$$\Phi(\lambda) \approx \int_0^\delta g(t) e^{-\lambda\phi(t)} dt.$$

As we know $\phi(t)$ to be increasing, it's first non zero derivative must be of even order, so that when we expanded the function by Taylor,

$$\phi(t) \approx \phi(0) + \phi_{(2k+1)}(0)t^{2k+1},$$

for some $k \geq 0$ and $|t| < \delta$ where $\phi_{(2k+1)} := \phi_{(2k+1)} / (2k+1)!$. As ϕ_{2k+1} , the $(2k+1)$ th-derivative, must be positive we also have that, as $\lambda \rightarrow \infty$,

$$\begin{aligned} \Phi(\lambda) &\approx e^{-\lambda\phi(0)} \int_0^\delta g(t) e^{-\lambda\phi_{(2k+1)}(0)t^{2k+1}} dt, \\ &\stackrel{(u=t(\lambda\phi_{(2k+1)}(0))^{1/(2k+1)})}{\approx} e^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{(2k+1)}(0))^{\frac{1}{2k+1}}} \int_0^\infty e^{-u^{2k+1}} du, \\ &\stackrel{(s=u^{2k+1})}{\approx} e^{-\lambda\phi(0)} \frac{g(0)}{(2k+1)(\lambda\phi_{(2k+1)}(0))^{\frac{1}{2k+1}}} \int_0^\infty e^{-s} ds. \end{aligned}$$

where the integral can be evaluated as a Gamma function. More than anything we need to focus on an important assumption that was made: that ϕ has a global minima at one of the endpoints of integration. If that were not the case, for example if $a < 0 < b$ and $\phi(t)$ had a global minima at $t = 0$ we would now have to consider the expansion

$$\phi(t) \approx \phi(0) + \phi_{2k}(0)t^{2k}$$

which would lead us to the expression

$$\Phi(\lambda) \approx e^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{2k}(0))^{\frac{1}{2k}}} \int_{-\infty}^\infty e^{-u^{2k}} du.$$

This differentiates of the previously result more importantly by the limits of integration. This is closely related to physical boundaries of the problem. Before enunciating the proper Laplace Method we state the Implicit Function Theorem, which will become important in a series of results that follows.

Theorem 9. (*Implicit Function Theorem*) Assume that S is an open subset of $\mathbb{R}^{n+k}$ and that $F : S \rightarrow \mathbb{R}^k$ is a function of class C^1 . Assume also that (a, b) is a point in S such that

$$F(a, b) = 0, \quad \text{and} \quad D_y F(a, b) \neq 0.$$

Then

1. There exist $r_0, r_1 > 0$ such that for every $x \in \mathbb{R}^n$ such that $|x - a| < r_0$, there exists a unique $y \in \mathbb{R}^k$ such that $|y - b| < r_1$

$$F(x, y) = 0.$$

That is, defines a function $y = f(x)$ for $x \in \mathbb{R}^n$ near a and y close to b .

2. Moreover, the function $f : B(r_0, a) \rightarrow B(r_1, b) \subset \mathbb{R}^k$ from part (1) is of class C^1 , and its derivatives may be determined by differentiating the identity

$$F(x, f(x)) = 0$$

and solving to find the partial derivatives of f .

Now we turn to our main interest.

Theorem 10. (*Laplace's Method, endpoint contributions*) Let Φ be as in ??, with a finite and $g \in L^1(a, b)$. Assume that ϕ has a unique global minimum at $t = a$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form

$$g(t) = (t - a)^\alpha g_0(t)$$

for $|t - a| < \delta$ and g_0 continuously differentiable in $[a, a + \delta]$ and $g_0(a) \neq 0$, whereas ϕ is twice continuously differentiable on $[a, a + \delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = \frac{1}{\lambda^{\alpha+1}} \frac{g_0(a) \Gamma(\alpha+1)}{\phi'(a)^{\alpha+1}} e^{-\lambda \phi(a)} \left(1 + O(\lambda^{-1})\right)$$

as $\lambda \rightarrow \infty$.

Proof. Start by assuming, without loss of generality, that $a = 0$ and dividing the integral in two intervals, one in the region where we have information about the behavior of g and another in the rest of the interval,

$$\Phi(\lambda) = \int_0^b g(t) e^{-\lambda \phi(t)} dt = \int_0^\delta g(t) e^{-\lambda \phi(t)} dt + \int_\delta^b g(t) e^{-\lambda \phi(t)} dt = (1) + (2).$$

Knowing that ϕ has a global minima at $t = 0$ we can assume, changing δ as needed, that

$\phi(t) > \phi(\delta)$ for any $t > \delta$. That give us a simple limitation for (2):

$$(2) = \int_{\delta}^b g(t) e^{-\lambda\phi(t)} dt \leq e^{-\lambda\phi(\delta)} \int_{\delta}^b g(t) dt \leq e^{-\lambda\phi(\delta)} \|g\|_{L^1[0,b]}.$$

We now get back to (1) and start by studying the difference give by $\phi(t) - \phi(0) = s$ in a similar way that we would think of a change of variable from t to s . Define the multivariate function

$$H(t, s) := \phi(t) - \phi(0) - s$$

such that $H(0, 0) = 0$ and $\partial_t H(0, 0) \neq 0$. This means we are in the condition to use Theorem 9 to say that there exists a unique bijective and continuously differentiable function $t(s) := t$ such that

$$\begin{cases} \phi(t(s)) - \phi(0) - s = 0, \\ t'(0) = \frac{\partial_s H(0, 0)}{\partial_t H(0, 0)} = \frac{1}{\phi'(0)}. \end{cases}$$

If we set $T = t^{-1}(\delta)$ we would then have for (1)

$$\int_0^{\delta} g(t) e^{-\lambda\phi(t)} dt = \int_0^T g(t(s)) e^{-\lambda\phi(t(s))} t'(s) ds.$$

Now, to apply the assumption on g we write $g(t(s)) = t^{\alpha}(s)g_0(t(s))$. As t is C^2 and vanishes at the origin, using its Taylor expansion, we have

$$t(s) = st'(0)(1 + r(s)) = \frac{s}{\phi'(0)}(1 + r(s)).$$

Joining both results,

$$\begin{aligned} \int_0^T g(t(s)) e^{-\lambda\phi(t(s))} t'(s) ds &= e^{\lambda\phi(0)} \int_0^T g(t(s)) e^{-\lambda(\phi(t(s)) - \phi(0))} t'(s) ds \\ &= e^{\lambda\phi(0)} \int_0^T t^{\alpha}(s) g_0(t(s)) e^{-\lambda s} t'(s) ds \\ &= \frac{e^{\lambda\phi(0)}}{\phi'(0)^{\alpha}} \int_0^T s^{\alpha} (1 + r(s))^{\alpha} g_0(t(s)) e^{-\lambda s} t'(s) ds \end{aligned}$$

where, if we set $f(s) := s^{\alpha} (1 + r(s))^{\alpha} g_0(t(s))$ we can apply Watson's Lemma to finish the result. $\square$

Theorem 11. (*Laplace's Method, interior contributions*) Let Φ be as in ??, with a finite and $g \in L^1(a, b)$. Assume that ϕ has a unique global minimum at $t = c \in (a, b)$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form

$$g(t) = (t - a)^{\alpha} g_0(t)$$

for $|t - a| < \delta$ and g_0 continuously differentiable in $[c - \delta, c + \delta]$ and $g_0(a) \neq 0$, whereas ϕ is three times continuously differentiable on $[c - \delta, c + \delta]$ with $\phi'(a) > 0$. Then the function

Φ satisfies an estimate of the form

$$\Phi(\lambda) = e^{-\lambda\phi(c)} \left(\frac{\sqrt{2\pi}g(c)}{\sqrt{|\phi''(c)|}\lambda^{1/2}} + O(\lambda^{-3/2}) \right)$$

as $\lambda \rightarrow \infty$.

Proof. Although the proof may be similar there is an important distinction in the solution. Suppose we wanted to try the same approach. We would first assume, w.l.g., that $c = 0$. Then, we could write

$$\begin{aligned} \Phi(\lambda) &= \int_a^b g(t) e^{-\lambda\phi(t)} dt \\ &= \int_a^{-\delta} g(t) e^{-\lambda\phi(t)} dt + \int_{-\delta}^{\delta} g(t) e^{-\lambda\phi(t)} dt + \int_{\delta}^b g(t) e^{-\lambda\phi(t)} dt \\ &=: (1) + (2) + (3). \end{aligned}$$

For (1) and (3) the result is as it was before

$$\begin{cases} (1) = \int_{\delta}^a g(-t) e^{-\lambda\phi(-t)} dt \leq e^{\lambda\phi(-\delta)} \|g\|_{L^1[a,b]}, \\ (3) = \int_{\delta}^b g(t) e^{-\lambda\phi(t)} dt \leq e^{\lambda\phi(\delta)} \|g\|_{L^1[a,b]}. \end{cases}$$

For (2) we would try to define a function

$$H(t, s) := \phi(t) - \phi(0) - s^2.$$

However we could not apply Theorem 9 as the partial derivative $\partial_t H = \phi'$ vanishes at $t = 0$. A nice fact about this indetermination is that it doesn't come from the fact that there are no solutions to $H(t, s) = 0$ but from the fact that there are two. To solve this we will use a technique called *blow up*. First, we introduce a variable $v := v(t)$ such that $vs = t$. Now, we redefine our function of interest to be

$$L(s, v) := \frac{\phi(sv) - \phi(0)}{s^2} - 1.$$

for $s \in (-\delta, \delta) - \{0\}$. We would think this function is also singular but note that, using the Taylor Expansion of ϕ we can write

$$\phi(t) = \frac{\phi''(0)}{2} t^2 + t^3 R(t)$$

for some error function R , which leave us with

$$L(s, v) = \frac{\phi''(0)}{2} v^2 + sv^3 R(sv) - 1,$$

and we can indeed apply the Implicit Function Theorem. In fact, we know $L \in C^2$ and as we want to solve for $L(s, v) = 0$, we need that the v_0 we take to satisfy $L(0, v_0) = 0$, ie,

$$v_0 = \sqrt{\frac{2}{\phi''(0)}}.$$

Now, we also know that

$$\partial_v L(0, v_0) = \phi''(0) v_0 \neq 0.$$

So we get a bijective and continuously differentiable function $v = v(s)$ such that

$$\begin{cases} L(s, v(s)) = 0 \\ v(0) = v_0 \end{cases}$$

But, $L(s, v(s))$ means exactly that

$$\phi(t) - \phi(0) = s^2$$

with $t = t(s) = sv(s)$. Getting back to our equation, we have that

$$e^{\lambda\phi(0)} \int_{-\delta}^{\delta} g(t) e^{-\lambda(\phi(t)-\phi(0))} dt = e^{\lambda\phi(0)} \int_{\alpha}^{\beta} g(sv(s)) (v(s) + sv'(s)) e^{-\lambda s^2} ds$$

where we can calculate $\alpha = -\sqrt{\phi(-\delta) - \phi(0)}$ and $\beta = \sqrt{\phi(\delta) - \phi(0)}$. One can now again finish the proof with Proposition 7. $\square$

Example 18. (NICA; ORTMANN, 2025) For $x \in \mathbb{R}$ fixed, i.e. $x = o(1)$, we have the following asymptotic as $n \rightarrow \infty$:

$$\text{He}_n(x) = \sqrt{2} \left(\frac{n}{2}\right)^{n/2} e^{x^2/4} \cos\left(\frac{n\pi}{2} - x\sqrt{n}\right) (1 + o(1))$$

To see that, consider the integral representation of the Hermite Polynomial and re-write

$$\begin{aligned} \text{He}_n(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x+it)^n e^{-t^2/2} dt = \frac{2}{\sqrt{2\pi}} \Re \left\{ \int_0^{\infty} (x+it)^n e^{-t^2/2} dt \right\} \\ &= \frac{2}{\sqrt{2\pi}} \Re \left\{ \int_{\Re(z)=x, \text{Im}(z)>0} e^{n \ln(z) + \frac{1}{2}(z-x)^2} \frac{dz}{i} \right\} \\ &= \frac{2}{\sqrt{2\pi}} \Re \left\{ \int_{\Re(z)=0, \text{Im}(z)>0} e^{n \ln(z) + \frac{1}{2}(z-x)^2} \frac{dz}{i} \right\} \end{aligned}$$

where you start with a line integral in $\mathbb{C}$ and move the contour over to the positive imaginary axis - justified by the Cauchy's Theorem and the exponential decay of the integrand near infinity. Now, we focus on $n \ln(z) + \frac{1}{2}(z-x)^2$ which has maximum at $z^* = i\sqrt{n} + o(\sqrt{n})$. The idea is to change variables by a factor of $\sqrt{z}$ so that the location of the maximum contribution stays $O(1)$ as n gets large. Specifically, we do

$$z = i\sqrt{n}s, \quad \frac{dz}{i} = \sqrt{n}ds, \quad \ln(z) = \ln(\sqrt{n}) + \ln(s) + i\frac{\pi}{2}$$

which give us

$$\frac{1}{2}(z-x)^2 = -\frac{1}{2}ns^2 + \frac{1}{2}x^2 - i\sqrt{n}sx$$

so that we remain with

$$\begin{aligned} \text{He}_n(x) &= \frac{2}{\sqrt{2\pi}} \Re \left\{ \int_0^{\infty} e^{n \ln(\sqrt{n})} e^{n \ln(s)} e^{ni\frac{\pi}{2}} e^{\frac{1}{2}x^2} e^{-i\sqrt{n}sx} \sqrt{n}ds \right\} \\ &= \frac{2}{\sqrt{2\pi}} \sqrt{n} e^{n \ln(\sqrt{n})} e^{\frac{1}{2}x^2} \int_0^{\infty} e^{n(\ln(s) - \frac{1}{2}s^2)} \cos\left(n\frac{\pi}{2} - \sqrt{n}sx\right) ds, \end{aligned}$$

where we can apply Laplace's approximation discussed before. Noting that $\phi(s) = \ln(s) - \frac{1}{2}s^2$ has $\phi'(s) = \frac{1}{s} - s$ and $\phi''(s) = -\frac{1}{s^2} - 1$, with global maximum at $s^* = 1$ with $\phi(1) = \frac{1}{2}$ and $\phi''(1) = -2$, yielding

$$\text{He}_n(x) = \frac{2}{\sqrt{2\pi}} \sqrt{n} e^{n \ln(\sqrt{n})} e^{\frac{1}{2}x^2} \left(e^{n\phi(1)} \sqrt{\frac{2\pi}{n|\phi''(1)|}} \cos\left(n\frac{\pi}{2} - \sqrt{n}x\right) (1 + o(1)) \right),$$

which simplifies to the result.

4.2.3 Steepest Descent

Let's now finally explore integrals over the complex plane with complex valued functions. We retake the integral

$$\phi(\lambda) := \int_{\Gamma} g(s) e^{-\lambda f(s)} ds,$$

where Γ is a contour in the complex plane and f, g are given complex-valued, analytic functions. Suppose that Γ is parametrized by $\gamma : [a, b] \rightarrow \mathbb{C}$ and let us decompose the functions we have into their real and imaginary parts such as

$$f(z) = u(z) + iv(z), \quad g(z) = p(z) + iq(z), \quad \gamma(t) = \alpha(t) + i\beta(t)$$

with all functions in the decomposition being real valued. Of course we now have

$$e^{-\lambda f(z)} = e^{-\lambda u(z)} e^{-i\lambda v(z)},$$

and when λ grows large, we have the exponentially growing/decaying term $e^{-\lambda u(z)}$ and a oscillating term coming from $e^{-i\lambda v(z)}$ which is hard to control. The idea to get around this fact is to use paths in which this term is constant. If we have such a case, we would write

$$\begin{aligned} \Phi(\lambda) &= e^{-i\lambda v_0} \int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t)) e^{-\lambda u(\gamma(t))} dt \\ &\quad + i e^{-i\lambda v_0} \int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t)) e^{-\lambda u(\gamma(t))} dt \end{aligned}$$

where each integral can now be solved by Laplace's method.

Theorem 12. (*Steepest Descent*) Consider Φ as in ?? with a smooth contour Γ . Assume that the exponent $f = u + iv$ is an analytic function on a neighborhood of Γ that satisfies

$$v(z) \equiv v_0, \quad z \in \Gamma$$

In addition, suppose that u has a unique minimum at an interior point of Γ , say on $z_0 \in \Gamma$, with $f''(z_0) \neq 0$, and also that g is analytic on a neighborhood of Γ and satisfies

$$\int_{\Gamma} |g(z)| |dz| < \infty.$$

Then Φ has a leading asymptotic as $\lambda \rightarrow +\infty$ given by

$$\Phi(\lambda) = e^{-\lambda f(z_0)} \left(\frac{\sqrt{2\pi} g(z_0) e^{i\theta(z_0)}}{\lambda^{1/2} \sqrt{|f''(z_0)|}} + O(\lambda^{-3/2}) \right), \quad \lambda \rightarrow +\infty,$$

where $\theta(z_0)$ is the angle of the oriented tangent of Γ at z_0 .

Proof. Fix a parametrization $\gamma : [a, b] \rightarrow \mathbb{C}$ of Γ with $\gamma(c) = z_0, a < c < b$, and decompose Φ as before in ???. By the assumptions on the functions f, g we can apply Laplace's method on both equations, and yields, as $\lambda \rightarrow +\infty$,

$$\int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t)) e^{-\lambda u(\gamma(t))} dt = \sqrt{2\pi} e^{-\lambda u(z_0)} \left(\frac{p(z_0)\alpha'(c) - q(z_0)\beta'(c)}{\sqrt{|(u \circ \gamma)''(c)|} \lambda^{1/2}} + O(\lambda^{3/2}) \right)$$

and

$$\int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t)) e^{-\lambda u(\gamma(t))} dt = \sqrt{2\pi} e^{-\lambda u(z_0)} \left(\frac{p(z_0)\beta'(c) - q(z_0)\alpha'(c)}{\sqrt{|(u \circ \gamma)''(c)|} \lambda^{1/2}} + O(\lambda^{3/2}) \right)$$

Using that $v_0 + u(z_0) = f(z_0)$ and also that,

$$(p(z_0)\alpha'(c) - q(z_0)\beta'(c)) + i(p(z_0)\beta'(c) - q(z_0)\alpha'(c)) = g(z_0)\gamma'(c).$$

Combining the estimative for both parts we get

$$\Phi(\lambda) = \sqrt{2\pi} e^{-\lambda f(z_0)} \left(\frac{g(z_0)\gamma'(c)}{\sqrt{|(u \circ \gamma)''(c)|} \lambda^{1/2}} + O(\lambda^{3/2}) \right)$$

and it suffices now to get an expression for the second derivative in the denominator. Knowing the imaginary part is constant we can write

$$\frac{d}{dt} u(\gamma(t)) = \frac{d}{dt} f(\gamma(t)) = \gamma'(t) f'(\gamma(t))$$

and

$$\frac{d^2}{dt^2} u(\gamma(t)) = \frac{d^2}{dt^2} f(\gamma(t)) = (\gamma'(t))^2 f''(\gamma(t)) + \gamma''(t) f'(\gamma(t)).$$

Hence,

$$\frac{\gamma'(c)}{\sqrt{|(u \circ \gamma)''(c)|}} = \frac{1}{\sqrt{|f''(z_0)|}} \frac{\gamma'(c)}{|\gamma'(c)|}$$

which concludes the proof. $\square$

We now need to discuss why this assumption of constant value of $\text{Im}(f(z))$ is not an assumption as strong as one might assume at first. Suppose we want to integrate in a random contour Γ of the plane. Of course, our assumption that the functions were analytic permit us to deform the contour keeping the value of the integral. We are going to use this to find an contour

$\tilde{\Gamma}$ such that sections of the contour are such that $\text{Im}(f(z))$ is constant. We already know that being analytic means satisfying the *Cauchy-Riemann* equations,

$$u_x(z) = v_y(z), \quad \text{and} \quad u_y(z) = -v_x(z).$$

An immediate result from these equation is that the gradients of u and v must be normal. Indeed, if $\langle \cdot, \cdot \rangle$ is the inner product on $\mathbb{R}^2$,

$$\langle \nabla u(z), \nabla v(z) \rangle = u_x(z)v_x(z) + u_y(z)v_y(z) = u_x(z)v_x(z) - v_x(z)u_x(z) = 0.$$

As we're following the curves where $\text{Im}(f(z))$ is constant, we're also following the path of Steepest Descent of the real part of the integral. As we also are looking to use Laplace's method on the integrals, we may want to have our paths going through the critical points of the function. This fully motivates the method described.

Example 19. *There are three asymptotics regimes for $\text{He}_n(x)$ when $x \rightarrow \infty$ depending on the value of $\lim_{N \rightarrow \infty} x/2\sqrt{N}$ for $N = n + 1$, each of which has qualitatively different behavior*

- *Small x values $-2\sqrt{N} < x < 2\sqrt{N}$: in this regime the Hermite Polynomials will be oscillating, and have a similar qualitative behavior as in Example 18, with modification to phase and amplitude. These polynomials will have some connections to the arcsine law and semi-circle law.*
- *Large x values $|x| > 2\sqrt{N}$: In this regime the polynomial doesn't have more oscillations and is monotone increasing. As x grows larger, this approximation behaves $\approx x^N e^{-N^2/2x^2}$.*
- *Edge x values $x = \pm 2\sqrt{N} + O(N^{-\frac{1}{6}})$: in this regime the polynomials will be approximated by the Airy Function. This interpolates between the oscillatory small x regime and the monotone large x regimes.*

We write the integral representation given by:

$$\begin{aligned} \text{He}_n(x) &:= \frac{n!}{2\pi i} \int_{\Gamma} z^{-n-1} e^{xz - \frac{z^2}{2}} dz \\ &= \frac{n!}{2\pi i} \int_{\Gamma} e^{xz - \frac{z^2}{2} - (n+1)\ln(z)} dz \end{aligned}$$

where Γ is a contour that goes around the origin coming and going from minus infinity such as $\Gamma = \{\lambda - i\delta : \lambda < 0\} \cup \{\delta e^{i\theta} : -\pi/2 < \theta < \pi/2\} \cup \{\lambda + i\delta : \lambda < 0\}$. This is equivalent to using $\Gamma = \{e^{i\theta} : 0 \leq \theta < 2\pi\}$. To determine He_n , set $N = n + 1$ and consider the scaling

$x \mapsto N^a \alpha$ and $z = N^b \beta$

$$\begin{aligned} \text{He}_n(N^a \alpha) &= \frac{(N-1)!}{2\pi i} \int_{\Gamma} e^{(N^a \alpha)(N^b \beta) - \frac{1}{2}(N^b \beta)^2 - N \ln(N^b \beta)} d\beta \\ &\stackrel{(a=b=1/2)}{=} \frac{(N-1)! N^{\frac{1}{2}(1-N)}}{2\pi i} \int_{\Gamma} e^{-N(-\alpha \beta + \frac{1}{2}\beta^2 + \ln(\beta))} d\beta \end{aligned}$$

So that we have

$$\text{He}_n(\sqrt{N}\alpha) = C_n^0 \int_{\Gamma} e^{-N\phi_{\alpha}(\beta)} d\beta$$

with

$$\phi_{\alpha}(\beta) = \frac{1}{2}\beta^2 - \alpha\beta + \ln(\beta).$$

As we want to calculate the asymptotic by steepest descent, we need to determine the critical points of such a function and then the steepest descent paths passing through this points. We can easily determine the critical points as $\beta_c^{\pm} = \frac{\alpha \pm \sqrt{\alpha^2 - 4}}{2}$ - solutions to the quadratic formula given by solving the critical point of the function (the zeros of its derivatives). Now, the behavior of the critical points will depend on the interval of the real parameter α .

$$\beta_c^{\pm} = \frac{\alpha \pm \sqrt{\alpha^2 - 4}}{2} = \begin{cases} e^{\pm\gamma}, & \alpha = 2 \cosh(\gamma) > 2, \quad \gamma > 0; \\ e^{\pm i\gamma}, & \alpha = 2 \cos(\gamma) \in [0, 2), \quad \gamma \in (0, \pi/2]; \\ 1, & \alpha = 2. \end{cases}$$

each of the cases refers to one of the regimes we described before.

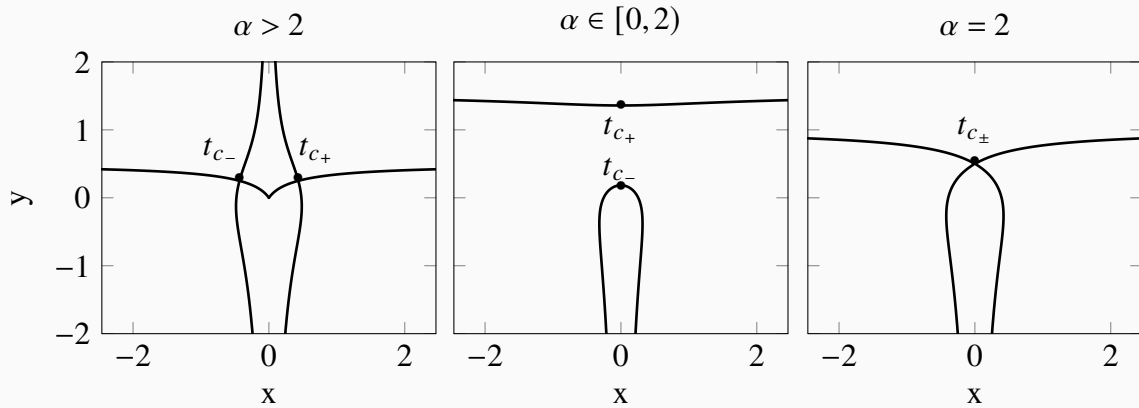


Figure 13 – The lines represent the solution to $\text{Im}[f(z) - f(t_c)] = 0$ for the three possible cases of the Hermite Polynomial. These are the curves we use to perform the asymptotic analysis on the integral. Note the case $\alpha = 2$; the coalescence of critical point gives origin to the Airy function.

EQUILIBRIUM

We have seen that the study of eigenvalues of random matrices is of interest and can be seen as a Point Process. For some specific models, one can further find a more rigid structure related to Determinantal Point Processes. These allow us to see eigenvalue distribution in terms of some correlation Kernel much easier to deal with than the initial formulation. Now, we will see that eigenvalue dynamics for invariant models can be thought of as another DDP: The so called *Coulomb Gases*. These can be thought of as a physical gas of repelling ions restricted to some conductor under some confining potential. Although it may be daunting to have yet another model to work with, these gases lead us directly to some nice interpretations and results. Namely, one can think of *thermodynamic equilibrium* for a large enough number of particles and this makes complete sense in a physical setting. In our matrices, these equilibrium distributions refers to the univariate eigenvalue distribution, a very interesting characterization of a RMT model. On this chapter we will expand on these ideas.

The section on Equilibrium Measures was based on the works of [Saff and Totik \(2024\)](#) and [Deift \(1999\)](#). We will define Coulomb Gases and try to motivate its importance in RMT. Afterwards we will explore the concept of equilibrium measures for these particle systems and state some important results relevant to computations that follows.

5.1 Coulomb Gases

We start by describing a familiar distribution or physicists: The Boltzmann-Gibbs distribution. This is a probability distribution that represents the probability of a given state for a system of particles as a function of the system state's *energy* ($\mathcal{H}$) and the *temperature* ($1/(\beta n^2)$). This is the distribution that maximizes the entropy: lower energy states have higher probability of being occupied than states with higher energy.

Suppose a set of n particles at positions $x_1, x_2, \dots, x_n \in \Sigma$, a *conductor* of dimension d ,

or some configuration $\xi_n = (x_1, x_2, \dots, x_n)$, embedded in some space $\mathbb{R}^{dn}$.

Definition 10. If p_n is a probability measure on $(\mathbb{R}^d)^n$, we name it **Coulomb gas** if it has the form of a Boltzmann-Gibbs probability measure, namely if

$$dp_n(\xi_n) = \frac{e^{-\beta n^2 \mathcal{H}_n(\xi_n)}}{\mathcal{Z}_n^\beta} dx_1 dx_2 \dots dx_n, \quad (5.1)$$

where

$$\mathcal{H}_n(\xi_n) = \frac{1}{n} \sum_{i=1}^n Q(x_i) + \frac{1}{2n^2} \sum_{i \neq j}^n g(x_i - x_j)$$

is usually called the **Hamiltonian**, or more concretely, the **configurational energy** of the system. The function $Q: \mathbb{R}^d \cong \Sigma \rightarrow \mathbb{R}$ is the external potential and $\mathcal{Z}_n^\beta$ is the **partition function**, or normalization function, given by

$$\mathcal{Z}_n^\beta = C_{n,\beta} \int_{\mathbb{R}^{dn}} \prod_{j=1}^n dx_j e^{-\beta n^2 \mathcal{H}_n(\xi_n)}.$$

The measure p_n models an Coulomb-interacting gas of charged indistinguishable particles under some external field Q . If one wishes dp_n to be a probability measure, then one must ensure that Q is such that the normalizing constant $\mathcal{Z}_n^\beta$ is finite for all n and that the measure support $\text{supp}(dp_n(\chi_n))$ is compact. This is all to say that the potential should be confining: As the number of particles increases, it must counter the repulsion effect and keep limited the region where particles move around. The function $g: \Sigma \rightarrow (-\infty, \infty]$ is the *Coulomb Interacting Kernel*, solution to the Poisson equation given by $-\nabla g(\vec{x}) = c_n \delta_0$ and is the reason in the first place why the particles may diverge.

Recall the expression for invariant ensembles in Definition cDefinition: polynomial ensemble, note that, rearranging the terms in Equation 5.1 and considering a logarithm interaction kernel we get

$$dp_n(x_1, x_2, \dots, x_n) = \frac{1}{\mathcal{Z}_n^\beta} \prod_{1 \leq j < k \leq n} |x_j - x_k|^\beta \prod_{j=1}^n e^{-\beta n Q(x_j)} dx_1 dx_2 \dots dx_n$$

This is to say that for invariant ensembles, their eigenvalues behave as a Coulomb gas. This opens the way for interpretations on the distribution of eigenvalues but also to computations using simulation methods for particle dynamics.

Example 20. As explored in [Chafaï and Ferré \(2018\)](#), a few classical algorithms of simulation are very successful in replicating the measures for Coulomb gas systems. Equivalently, if one is interested in sampling eigenvalues for $\beta = 2$ invariant ensembles, one can sample from its configuration equilibrium $p_Q(\lambda)$ in respect to some potential V using simulation of the particles. Consider for example the non-trivial case explored in [Balogh](#),

Grava and Merzi (2016), where, taken $d = 2$ and $n = 2$ we will have $W(x) = g(x) = \log |x|$ and impose the potential

$$Q(z) = |z|^{2a} - \operatorname{Re} t z^a.$$

This potential has disjoint intervals for some combinations of the parameters t and a - it presents a transition in $t_c \approx \sqrt{(1/a)}$. Analytically, it was just recently solved. However we can easily sample from this measure using the simulation as shown below. Data was taken from Pimenta and Silva (2024).

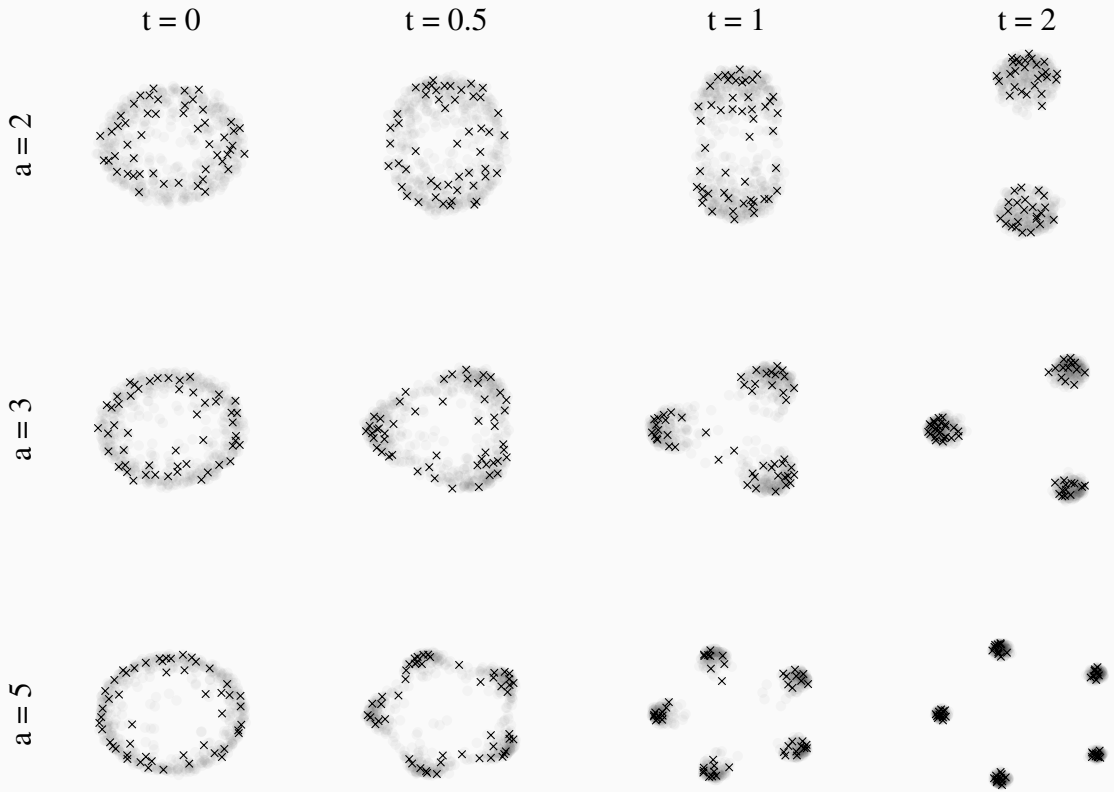


Figure 16 – Shown in the shadow is the mean configurations - in some an approximation of the limit distribution of points, and in the scatter is a random $n = 50$ sample from the distribution. We show configurations for combinations of the parameters $t = 0, 0.5, 1, 2$ from left to right and $a = 2, 3, 5$ from top to bottom.

5.2 Equilibrium Measure

The description of the eigenvalues as a Coulomb gas and as a Point Process lead us to the idea of minimizing some type of energy of the system by finding equilibrium configurations $\xi_n = \{x_1, x_2, \dots, x_n\}$. Especially interesting is the case when n tends to a large enough number as it simplifies some structures and the results.

To clarify the point, suppose we are dealing with the density

$$p_n(\xi_n) = \frac{e^{-\beta n^2 \mathcal{H}_n(\chi_n)}}{Z_{n,\beta}}, \quad \mathcal{H}_n(\chi_n) = \frac{2}{n} \sum_{i=1}^n Q(x_i) + \frac{1}{n^2} \sum_{i < j}^n \ln \frac{1}{|x_i - x_j|}.$$

For any n -configuration $\xi_n = \{x_1, x_2, \dots, x_n\}$, let μ_x be the normalized counting measure on $\mathbb{C}$,

$$\mu_x = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, \quad \text{such that} \quad \int_{\mathbb{R}} d\mu_x = 1.$$

Then

$$\frac{1}{n^2} \sum_{i < j} \ln |x_i - x_j|^{-1} + \frac{2}{n} \sum_{i=1}^n Q(x_i)$$

may be written in the form

$$\int \int \ln |t - s|^{-1} d\mu_x(t) d\mu_x(s) + 2 \int Q(x_i) d\mu_x(s).$$

This lead us to the energy minimization problem

$$\begin{aligned} E^{(Q)} &= \inf_{\mu \in \mathcal{M}^1(\mathbb{R})} \left[\int \int \ln |t - s|^{-1} d\mu_x(t) d\mu_x(s) + 2 \int Q(s) d\mu_x(s) \right] \\ &:= \inf_{\mu \in \mathcal{M}^1(\mathbb{R})} [I_Q(\mu_x)] \end{aligned}$$

where

$$\mathcal{M}^1(\mathbb{R}) = \left\{ \mu \in \text{Borel measures on } \mathbb{R} \mid \int d\mu = 1 \right\}.$$

The quantity $E^{(Q)}$ is the *equilibrium energy* of the system and $I_Q(\mu_x)$ is the *energy integral*. However, we must step backwards so that we can advance.

The classical case for such minimization problems corresponds to the cases when the conductor Σ is compact and the external field Q is zero. Under some mild conditions on the external field, however, the equilibrium measure μ_Q - the measure that minimizes such problem - has a unique solution which is a measure of compact support. For it to be compact, Q the confining potential, must increase sufficiently fast around infinity. Let $\Sigma \subseteq \mathbb{C}$ be a compact subset of the complex plane and $\mathcal{M}(\Sigma)$ be the collection of all positive Borel measure of compact support in Σ . Let $\mu \in \mathcal{M}(\Sigma)$ be such a measure in the complex plane $\mathbb{C}$.

Definition 11. The associated **logarithmic potential** is defined by

$$U^\mu(z) := \int \ln \frac{1}{|z - t|} d\mu(t).$$

Also, $I(\mu)$, called the **logarithmic energy** of $\mu \in \mathcal{M}(\Sigma)$ is defined

$$I(\mu) := \int \int \ln \frac{1}{|z - t|} d\mu(z) d\mu(t),$$

the **energy** V of Σ is

$$V := \inf \{ I(\mu) \mid \mu \in \mathcal{M}(\Sigma) \}$$

and the measure μ_Σ^V , if V is finite, which defines the energy is the **equilibrium measure**.

This logarithmic potential can be interpreted as the equivalent interaction potential of the system of n identical particles distributed with μ where the repelling force between two particles is proportional to the distance between them. There are a few useful properties one can extract about this quantity.

Proposition 8. (SAFF; TOTIK, 2024) *The logarithmic potential are super-harmonic on the whole plane, that is, they have the following properties.*

1. $U^\mu(z) > -\infty$ for all z ,
2. U^μ is lower semi-continuous,
3. $U^\mu(z) \geq \frac{1}{2\pi} \int_{-\pi}^{\pi} U^\mu(z)(x + r e^{i\theta}) d\theta$ for all z and all $r \geq 0$.

Proof. Firstly, property (1) follows directly by the properties of μ , especially the compact support. The property (2) can be proven noticing that

$$U^\mu(z) = \lim_{M \rightarrow \infty} \int \min\left(M, \ln \frac{1}{|z-t|}\right) d\mu(t).$$

The functions on the right-hand side are continuous and increasing with M , such limit of an increasing sequence of functions must be lower semi-continuous. To prove (3), note that the kernel $\ln(1/|z-t|)$ is super-harmonic in $\mathbb{C}$ for each t . If some F is analytic in a domain D and $\rho > 0$, then $|F(z)|^\rho$ is sub-harmonic in D ; furthermore, $\ln|F(z)|$ is sub-harmonic in D provided F is not identically zero. Thus,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\theta} - t|} d\theta \leq \ln \frac{1}{|z-t|}, \quad z, t \in \mathbb{C}.$$

Using the Fubini-Tonelli theorem and the inequality above

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} U^\mu(z + r e^{i\theta}) d\theta &= \int \frac{1}{2\pi} \int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\theta} - t|} d\theta d\mu(t), \\ &\leq \int \ln \frac{1}{|z-t|} d\mu(t) = U^\mu(z). \end{aligned}$$

□

Also, for technical reasons we also state the following Lemma regarding the logarithmic energy that will become useful in a later result although we do not to prove it.

Lemma 1. (SAFF; TOTIK, 2024) *Let $\mu = \mu_1 - \mu_2$ be a signed Borel measure with compact support and total mass $\mu(\mathbb{C}) = 0$. Suppose further that each of the positive measures μ_1 and μ_2 has finite logarithmic energy. Then the logarithmic energy of μ*

$$I(\mu) = \int \int \ln \frac{1}{|z-t|} d\mu(z) d\mu(t),$$

is non-negative and it is zero if and only if $\mu = 0$.

The energy V can be finite or $+\infty$, and in the finite case there is a unique measure $\mu = \mu_\Sigma \in \mathcal{M}(\Sigma)$ for which the minimum defining energy V is reachable. Such a measure μ_Σ is called the *equilibrium distribution* or *equilibrium measure* of the compact set Σ . For such a function we have that its logarithm potential is always smaller or equal to the energy I_Σ .

Definition 12. Define also the logarithmic capacity

$$\text{cap}(\Sigma) := e^{-V}.$$

The capacity for an arbitrary Borel set E is defined by the supreme of the capacity of all the compact subsets $K \subseteq E$. Naturally, the capacity of a subset of a Borel set of capacity zero is said to be zero. The union of zero-capacity Borel sets can also be shown to be a zero-capacity Borel set. A property is said to hold *quasi-everywhere* (q.e.) on a set E if the set of non conforming points is of capacity zero. Then, we can say that

$$U_\Sigma^\mu(z) = V, \quad \text{q.e. } z \in \Sigma.$$

We should at this point reintroduce the external potential Q . The main difference now is that we can loose the compacity restriction on the sets Σ and impose that there is no positive mass around infinity. There is a natural way to introduce this idea. Let $\Sigma \subset \mathbb{C}$ be a closed set and $w : \Sigma \rightarrow [0, \infty)$. We call such a function a *weight function* on Σ .

Definition 13. A weight function w is said to be **admissible** if it satisfies the following three conditions

1. w is upper semi-continuous;
2. $\Sigma_0 := \{z \in \Sigma | w(z) > 0\}$ has positive capacity;
3. if Σ is unbounded, then $|z|w(z) \rightarrow 0$ as $|z| \rightarrow \infty$, $z \in \Sigma$.

Example 21. (Classical weights) The classical weights, for some real parameters $a, b > -1$, are

$$w(\lambda) = \begin{cases} e^{-\lambda^2} & \text{Gaussian,} \\ \lambda^a e^{-\lambda} \chi_{\{\lambda > 0\}} & \text{Laguerre,} \\ \lambda^a (1 - \lambda)^b \chi_{\{0 < \lambda < 1\}} & \text{Jacobi.} \end{cases}$$

These are all admissible weights by Definition 13. They are positive continuous functions. The Gaussian and Laguerre have Σ_0 with positive capacity as $I(w)$ is finite and $|z|w(z) \rightarrow 0$ as $|z| \rightarrow \infty$, $z \in \Sigma$. Jacobi is direct as $I(w)$ is evaluated in a compact interval for a finite function.

Assume our weight function to have the form

$$w(z) := e^{-nQ(z)}.$$

Then, for the weight to be admissible, $Q : \Sigma \rightarrow (\infty, \infty]$ must be lower semi-continuous, $Q(z) < \infty$ on a set of positive capacity and if Σ is unbounded, then

$$\lim_{|z| \rightarrow \infty, z \in \Sigma} \{Q(z) - \ln |z|\} = \infty.$$

Let $\mathcal{M}(\Sigma)$ be the collection of all positive unit Borel measures μ with support in Σ . Our definition for the weighted energy integral

$$\begin{aligned} I_w(\mu) &= \int \int \ln |t-s|^{-1} d\mu(t) d\mu(s) + 2 \int Q(s) d\mu(s) \\ &= \int U^\mu(s) d\mu(s) + 2 \int Q(s) d\mu(s) \end{aligned}$$

is recovered if both integrals exist and are finite. We can now move on to the matter of existence of an *equilibrium measure* μ^Q for a given external field Q given by some polynomial

$$Q(x) = t_{2m}x^{2m} + t_{2m-1}x^{2m-1} + \cdots + t_0, \quad t_{2m} > 0, V(x) \geq 0. \quad (5.2)$$

Theorem 13. (DEIFT, 1999) *Let Q be as in Equation 5.2. Then, there exists a unique probability measure $\mu = \mu^w$ such that*

$$E^w = \inf_{\mu \in \mathcal{M}^1} I^w(\mu) = I^w(\mu^w).$$

Moreover, μ^w has compact support.

Example 22. *For example, take the invariant ensemble correspondent to taking the Coulomb gas restricted to the line on some two dimensional conductor and under the potential*

$$Q(x) = tx^{2m}.$$

One can determine that for such system there is an unique μ_Q^ that minimizes the logarithmic energy. This measure is defined by*

$$\text{supp } \mu_Q^*(x) = [-a, a], \quad \mu_Q^*(x) = \frac{mt}{\pi} \sqrt{a^2 - x^2} h(x),$$

where

$$a := \left(mt \prod_{l=1}^m \frac{2l-1}{2l} \right) \quad \text{and} \quad h(x) := x^{2m-2} + \sum_{j=1}^{m-1} x^{2m-2-2j} a^{2j} \prod_{l=1}^j \frac{2l-1}{2l}.$$

Note that this simplifies to the Gaussian case if $Q(x) = x^2/2$ ($t, m = 1$). In such case we already shown that the equilibrium measure is defined by the Semicircle distribution

$$\text{supp } \mu_Q^*(x) = [-\sqrt{2}, \sqrt{2}], \quad \mu_Q^*(x) = \frac{1}{\pi} \sqrt{2 - x^2}.$$

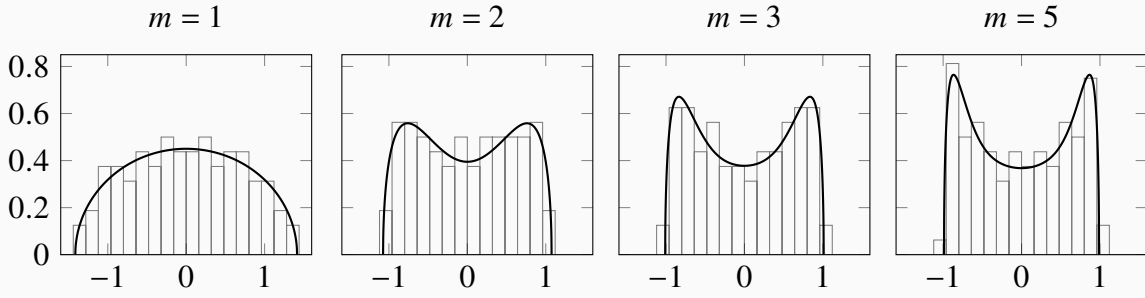


Figure 17 – Four conditions for the Monic potential, the equilibrium measure is expressed by the line plot. For all cases we used $t = 1$ and set $m = 1, 2, 3, 5$ from left to right. The bin-plot is the result of the simulation of the equivalent Coulomb Gas taken from [Pimenta and Silva \(2024\)](#).

5.2.1 Euler-Lagrange

Associated with the variational problem introduced on [section 5.2](#) defined by

$$\begin{aligned}
 E^w &= \inf_{\mu \in \mathcal{M}^1} I^w(\mu), \\
 &= \inf_{\mu \in \mathcal{M}^1} \left[\int \int \ln |t - s|^{-1} d\mu(t) d\mu(s) + 2 \int Q(s) d\mu(s) \right], \\
 &= I^w(\mu^w),
 \end{aligned}$$

one could define the so-called *Euler-Lagrange* type variational equations. They are important as they provide a tool, for some cases, to solve these equations for the equilibrium measure μ^w .

Theorem 14. (*Variational Equations, (DEIFT, 1999)*) The equilibrium measure μ^w for the given variational problem in [section 5.2](#) satisfies the following conditions.

There exists a real constant l such that

1.

$$\int \left[2 \int \ln |x - y|^{-1} d\mu^w(y) + 2Q(x) \right] d\tilde{\mu}(x) \geq l$$

for all $\tilde{\mu} \in \mathcal{M}^1(\mathbb{C})$ of compact support with $I^w(\tilde{\mu}) < \infty$.

2.

$$2 \int \ln |x - y|^{-1} d\mu^w(y) + 2Q(x) = l$$

for μ^w -almost everywhere.

Conversely, if $\mu \in \mathcal{M}^1(\mathbb{C})$ has compact support, satisfies conditions (1) and (2) above, and $I^w(\mu) < \infty$, then $\mu = \mu^w$.

Proof. ($\implies$) We recall that μ^w has compact support by [Theorem 13](#) and $I^w(\mu^w) = E^w < \infty$. Suppose $\mu^* \in \mathcal{M}^1(\mathbb{C})$ has compact support and $I^w(\mu^*) < \infty$. Then, defining some convex

combination

$$\mu_t = t\mu^* + (1-t)\mu^w = \mu^w + t(\mu^* - \mu^w) \in \mathcal{M}^1(\mathbb{C}), \quad 0 \leq t \leq 1,$$

we should have

$$\begin{aligned} I^w(\mu_t) &= \int \int \ln|t-s|^{-1} d\mu^w(t) d\mu^w(s) + 2 \int Q(s) d\mu^w(s) \\ &\quad + 2t \int \int \ln|t-s|^{-1} d\mu^w(t) d(\mu^* - \mu^w)(s) \\ &\quad + 2t \int Q(s) d(\mu^* - \mu^w)(s) \\ &\quad + t^2 \int \int \ln|t-s|^{-1} d(\mu^* - \mu^w) d(\mu^* - \mu^w)(s). \end{aligned}$$

This is all well since μ^w, μ^* have compact support and $I^w(\mu^w), I^w(\mu^*) < \infty$. As $I^w(\mu_t) \geq I^w(\mu^w)$ for $0 \leq t \leq 1$ by assumption of minimality and Lemma 1, it is necessary that

$$\int \left[2 \int \ln|x-y|^{-1} d\mu^w(y) + 2Q(x) \right] d(\mu^* - \mu^w)(x) \geq 0$$

and therefore

$$\int \left[2 \int \ln|x-y|^{-1} d\mu^w(y) + 2Q(x) \right] d\mu^*(x) \geq l \quad (5.3)$$

where

$$\int \left[2 \int \ln|x-y|^{-1} d\mu^w(y) + 2Q(x) \right] d\mu^w(x) := l$$

proving the first assertion in (1). Now, let

$$B = \left\{ x \mid 2 \int \ln|x-y|^{-1} d\mu^w(y) + 2Q(x) < l \right\}.$$

Then, B must be a bounded set of $\mathbb{C}$. Suppose $\mu^*(B) > 0$ and set

$$\mu_B^* = \frac{\chi_B}{\mu^*(B)} \mu^*$$

where χ_B is the characteristic function of the set B . Then $\mu_B^* \in \mathcal{M}^1(\mathbb{C})$ has compact support and $I^w(\mu_B^*) < \infty$. inserting $d\mu_B^*$ in Equation 5.3, we find

$$l \leq \int \left[2 \int \ln|x-y|^{-1} d\mu^w(y) + 2Q(x) \right] d \frac{\chi_B}{\mu^*(B)} \mu^*(x) < l$$

a contradiction. Therefore,

$$\mu^*(B) = \mu^* \left(\left\{ x \mid 2 \int \ln|x-y|^{-1} d\mu^w(y) + 2Q(x) < l \right\} \right) = 0$$

for all measures $\mu^* \in \mathcal{M}^1(\mathbb{C})$ with support compact and $I^w(\mu^*) < \infty$. in particular, it is true for $\mu^* = \mu^w$. Since,

$$\begin{aligned} 0 &= \int \left[2 \int \ln|x-y|^{-1} d\mu^w(y) + 2Q(x) - l \right] d\mu^w(x) \\ &= \int_{\mathbb{C}-B} \left[2 \int \ln|x-y|^{-1} d\mu^w(y) + 2Q(x) - l \right] d\mu^w(x) \end{aligned}$$

and (2) follows.

($\Leftarrow$) If, now, $\mu \in \mathcal{M}^1$ satisfies (1) and (2), $I^w(\mu) < \infty$, and suppose μ has compact support. Then write $\mu^w = \mu + (\mu^w - \mu)$. We have, as before

$$\begin{aligned} I^w(\mu) &\geq I^w(\mu^w) \\ &= I^w(\mu) + \int \left[2 \int \ln|x-y|^{-1} d\mu(y) + 2Q(x) - l \right] d(\mu^w - \mu)(x) \\ &\quad + \int \int \ln|t-s|^{-1} d(\mu^w - \mu)(t) d(\mu^w - \mu)(s). \end{aligned}$$

Using (2),

$$\int \left[2 \int \ln|x-y|^{-1} d\mu(y) + 2Q(x) - l \right] d\mu^w(x) - l \geq l - l = 0$$

but also the third term is strictly positive by Lemma 1 unless it is identically null. If, however, $\mu^w - \mu \neq 0$, then we showed that

$$I^w(\mu) \geq I^w(\mu^w) > I^w(\mu),$$

a contradiction. Thus, if μ satisfies (1), (2), then it must be μ^w . $\square$

Moreover, if there is the knowledge that

$$d\mu^w = \phi(x)dx \tag{5.4}$$

for some continuous function ϕ of compact support, then

$$2 \int \ln|x-y|^{-1} d\mu(y) + 2Q(x)$$

is continuous and the previous theorem takes the following more simple form

Theorem 15. (Strong Variational Equations, (DEIFT, 1999)) Suppose that $d\mu^w = \phi(x)dx$ for some continuous function ϕ of compact support. Then the conditions of Theorem 13 can be replaced by

1.

$$2 \int \ln|x-y|^{-1} d\mu^w(y) + 2Q(x) \geq l$$

for all $z \in \mathbb{C}$.

2.

$$2 \int \ln|x-y|^{-1} d\mu^w(y) + 2Q(x) = l$$

on $\{z : \phi(z) > 0\}$.

We mentioned in Equation 5.4 the possibility of knowing the density ϕ of the equilibrium measure μ^w in relation to the Lebesgue measure. We will now determine ϕ . As we did many

times, assume that μ is finite and of compact support in the complex plane. For notation, let A denote the two-dimensional Lebesgue measure on $\mathbb{C}$ and Δ denote the Laplacian operator

$$\Delta F := \frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2}.$$

We have the theorem below.

Theorem 16. ([SAFF; TOTIK, 2024](#)) *If in a region R the potential U^μ has continuous second partial derivatives, then μ is absolutely continuous in R with respect to two-dimensional Lebesgue measure A , and on R we have the formula*

$$d\mu = -\frac{1}{2\pi} \Delta U^\mu dA$$

5.3 Solving for the Free Energy

Remember the expression for the Gibbs-Boltzmann measure of Coulomb gases

$$dp_n(\xi_n) = \frac{1}{\mathcal{Z}_n^\beta} e^{-\beta n^2 \mathcal{H}_n(\xi_n)} \prod_{i=1}^n dx_i.$$

As discussed, this expression refers to a thermodynamic fluid of particles with positions $\xi_n = x_1, x_2, \dots, x_n$ on a conductor Σ , in equilibrium at some inverse temperature β and submitted to a confining potential Q and a repulsive logarithmic term of interaction. Because of the term βn^2 , taking the asymptotic $n \rightarrow \infty$ means at the same time zero temperature and the thermodynamic limit. Physically, understanding the equilibrium positions would be equivalent to minimizing the *Free Energy* $F = -(1/\beta) \ln \mathcal{Z}_n^\beta$. Let's describe the idea behind such minimization. More details can be found in [Livan, Novaes and Vivo \(2018\)](#).

Lets first take the normalized counting measure, already introduced in [Chapter 3](#),

$$n(x) = \frac{1}{n} \sum_{i=1}^n \delta(x - x_i).$$

Note that using this function we can write the unity as

$$1 = \int d[n(x)] \delta \left[n(x) - \frac{1}{n} \sum_{i=1}^n \delta(x - x_i) \right],$$

and with this we can also rewrite our partition function to be written as

$$\mathcal{Z}_n^\beta = C_{n,\beta} \int d[n(x)] \int_{\mathbb{R}^{dn}} \prod_{j=1}^n dx_j e^{-\beta n^2 \mathcal{H}_n(\xi_n)} \delta \left[n(x) - \frac{1}{n} \sum_{i=1}^n \delta(x - x_i) \right].$$

We don't want to deal with summations, in this context, integrals can be much easier to deal with. Then, we also want to convert the expression for the Hamiltonian. The confining term is easier, we can write

$$\frac{1}{n} \sum_{i=1}^n Q(x) = \frac{1}{n} \int_{\mathbb{R}^d} n(x) Q(x) dx.$$

For the interaction term, one must first note that

$$\frac{1}{2n^2} \sum_{i \neq j} \ln |x_i - x_j| = \frac{1}{2n^2} \left[\sum_{i,j} \ln |x_i - x_j| - \sum_i \Delta x_i \right]$$

where we call Δx_i is the cutoff distance and refers to the term $i = j$ in the discrete case and two neighboring charges approximating in the continuous case. This should ensure that the functional will still produce a finite result. Now we can write

$$\frac{1}{2n^2} \sum_{i \neq j} \ln |x_i - x_j| = \frac{1}{2n^2} n^2 \int \int_{\mathbb{R}^{2d}} dx dy n(x) n(y) \ln |x - y| - \frac{1}{2n^2} n \int_{\mathbb{R}^d} dx n(x) \ln \Delta(x).$$

and then

$$\begin{aligned} \mathcal{Z}_n^\beta &= C_{N,\beta} \int d[\mathbf{n}(x)] e^{-\beta n^2 \mathcal{H}_n(\mathbf{n}(x))} \int_{\mathbb{R}^{dn}} \prod_{j=1}^n dx_j \delta \left[\mathbf{n}(x) - \frac{1}{n} \sum_{i=1}^n \delta(x - x_i) \right] \\ &=: C_{N,\beta} \int d[\mathbf{n}(x)] e^{-\beta n^2 \mathcal{H}_n(\mathbf{n}(x))} I_n[\mathbf{n}(x)], \end{aligned}$$

where

$$\mathcal{H}_n(\mathbf{n}(x)) = \int_{\mathbb{R}^d} n(x) Q(x) dx + \frac{1}{2} \int \int_{\mathbb{R}^{2d}} dx dy n(x) n(y) \ln |x - y| - \frac{1}{2n} \int_{\mathbb{R}^d} dx n(x) \ln \Delta(x).$$

There is some interpretations of me made about $I_n[\mathbf{n}(x)]$. Physically it is possible to think of it as a count of how many microstates ξ_n are compatible with the macrostate $\mathbf{n}(x)$. The logarithm of this quantity is proportional to the entropy of such system. Rewrite this function as

$$\begin{aligned} I_n[\mathbf{n}(x)] &= \int d[\mathbf{n}(x)] \int_{\mathbb{R}^{dn}} \prod_{j=1}^n dx_j \exp \left[i n \int dx n(x) \hat{\mathbf{n}}(x) - i \int dx \hat{\mathbf{n}}(x) \sum_{i=1}^n \delta(x - x_i) \right] \\ &= \int d[\mathbf{n}(x)] \exp \left[i n \int_{\mathbb{R}^{dn}} dx n(x) \hat{\mathbf{n}}(x) \right] \left[\int_{\mathbb{R}^d} dy e^{i \int_{\mathbb{R}^d} dx \hat{\mathbf{n}}(x) \delta(x - x_i)} \right]^n \\ &= \int d[\hat{\mathbf{n}}(x)] e^{nS[\hat{\mathbf{n}}(x)|\mathbf{n}(x)]}, \end{aligned} \tag{5.5}$$

where

$$S[\hat{\mathbf{n}}(x)|\mathbf{n}(x)] = i \int_{\mathbb{R}^{dn}} dx n(x) \hat{\mathbf{n}}(x) + \log \int_{\mathbb{R}^d} dy e^{-i \hat{\mathbf{n}}(x)}.$$

The most important thing to notice here is that Equation 5.5 is of the form required for the Laplace asymptotic technique discussed in Chapter 4. This means that t can be estimated, if S has a maxima with strictly negative second derivative, by the evaluation at its maximum point. One can find

$$\begin{aligned} 0 &= \frac{\partial S}{\partial \hat{\mathbf{n}}(x)} = i n(x) - i \frac{e^{-i \hat{\mathbf{n}}(x)}}{\int_{\mathbb{R}^d} dy e^{-i \hat{\mathbf{n}}(y)}} \\ \implies -i \hat{\mathbf{n}}(x) &= -\ln(n(x)) - \log \int_{\mathbb{R}^d} dy e^{-i \hat{\mathbf{n}}(y)}, \end{aligned}$$

for the critical counting function. Then, substituting,

$$I_n[\mathbf{n}(x)] \approx \exp \left[-n \int_{\mathbb{R}^{nd}} d\mathbf{x} \mathbf{n}(x) \ln(\mathbf{n}(x)) \right],$$

which has the form of the Shannon entropy of the density $\mathbf{n}(x)$. there is an classical argument that states that $\Delta(x) \approx c/(n\mathbf{n}(x))$, i.e., the higher the density of particles around a point x , the smaller the average distance between them. Thus,

$$\begin{aligned} \int_{\mathbb{R}^d} d\mathbf{x} \mathbf{n}(x) \ln \Delta(x) &\approx \int_{\mathbb{R}^d} d\mathbf{x} \mathbf{n}(x) \ln \left(\frac{c}{n\mathbf{n}(x)} \right) \\ &= \int_{\mathbb{R}^d} d\mathbf{x} \mathbf{n}(x) (\ln c - \ln n + \ln \mathbf{n}(x)) \\ &= \int_{\mathbb{R}^d} d\mathbf{x} \mathbf{n}(x) \ln c - \int_{\mathbb{R}^d} d\mathbf{x} \mathbf{n}(x) \ln n + \int_{\mathbb{R}^d} d\mathbf{x} \mathbf{n}(x) \ln \mathbf{n}(x). \end{aligned}$$

Return now to the partition function and use the relations found to write

$$\begin{aligned} \mathcal{Z}_n^\beta &\approx C_{N,\beta} \int d[\mathbf{n}(x)] \exp \left[-\beta n^2 \mathcal{H}_n(\mathbf{n}(x)) \right] \exp \left[-n \int_{\mathbb{R}^{nd}} d\mathbf{x} \mathbf{n}(x) \ln(\mathbf{n}(x)) \right], \\ &\approx C_{N,\beta} \int d[\mathbf{n}(x)] \exp \left[-\beta n^2 \mathfrak{F}_0[\mathbf{n}(x)] + \frac{\beta n}{2} \mathfrak{F}_1[\mathbf{n}(x)] + \frac{\beta}{2} n \ln(n) - \frac{\beta}{2} n \ln(c) \right] \exp \left[-n \mathfrak{F}_1[\mathbf{n}(x)] \right], \\ &\approx C_{N,\beta} \int d[\mathbf{n}(x)] \exp \left[-\beta n^2 \mathfrak{F}_0[\mathbf{n}(x)] + \frac{\beta}{2} n \ln(n) + n \left(\frac{\beta}{2} - 1 \right) \mathfrak{F}_1[\mathbf{n}(x)] - \frac{\beta}{2} n \ln(c) \right], \end{aligned}$$

where

$$\begin{aligned} \mathfrak{F}_0[\mathbf{n}(x)] &:= \int_{\mathbb{R}^d} \mathbf{n}(x) Q(x) dx + \frac{1}{2} \int \int_{\mathbb{R}^{2d}} d\mathbf{x} d\mathbf{y} \mathbf{n}(x) \mathbf{n}(y) \ln |x - y|, \\ \mathfrak{F}_1[\mathbf{n}(x)] &:= \int_{\mathbb{R}^d} d\mathbf{x} \mathbf{n}(x) \ln \mathbf{n}(x). \end{aligned}$$

Note that, now, one can again perform a saddle point estimation for the asymptotic in a critical point $\mathbf{n}^*(x)$ to get

$$\mathcal{Z}_n^\beta \approx \exp \left[-\beta n^2 \mathfrak{F}_0[\mathbf{n}^*(x)] + \frac{\beta}{2} n \ln(n) + n \left(\frac{\beta}{2} - 1 \right) \mathfrak{F}_1[\mathbf{n}^*(x)] - \frac{\beta}{2} n \ln(c) \right]$$

and finally for the Free Energy

$$\ln(\mathcal{Z}_n^\beta) = -\beta n^2 \mathfrak{F}_0[\mathbf{n}^*(x)] + \frac{\beta}{2} n \ln(n) + n \left(\frac{\beta}{2} - 1 \right) \mathfrak{F}_1[\mathbf{n}^*(x)] - \frac{\beta}{2} n \ln(c) + O(n).$$

This expression for the Free Energy is the main goal for [Chapter 6](#).

PLANAR ENSEMBLES

We are going to discuss in this chapter the ideas developed in the paper [Byun, Kang and Seo \(2023\)](#), where asymptotics were developed for a large class of random matrix ensembles using relatively straightforward asymptotic techniques; taking advantage of the orthogonal polynomials present in these models. In the end, we hope to clarify the origin for the novel limit-behavior expression for the *Partition Function* $\mathcal{Z}_N^{\beta=2}$ in the shape of the expression

$$\ln(\mathcal{Z}_{N,\beta=2}^Q) = -I_Q N^2 + \frac{1}{2} N \ln N + E_Q N + G \ln N + F^Q + o(1).$$

Partition functions perform the role of the normalization for the probability measure relative to the Coulomb gas being worked on - a system of interacting particles confined by some potential as described in [section 5.1](#). In this instance, we deal with complex plane gases. This relates in RMT, to the *random normal matrix* and *planar symplectic ensembles*, depending on the symmetry imposed. If one takes, for example, the quadratic potential $Q(x) = x^2/2$ using the $2 - \beta$ regime, one gets the *complex* and *planar* Ginibre ensembles, falling back into the classical models. For the results presented, another restriction is made; the confining potentials must be radially symmetric. For a large class of such potentials Q , the first term I_Q in the expression was known for a long time, given by the system's energy when distributed as the associated equilibrium measure. The coefficient $1/2$ of the order $N \ln N$, as well as the entropy term E_Q , were described in a recent work of [Leblè and Serfaty \(2017\)](#), also for rather large classes of Q . The remaining terms in this expansion are new. The term G appears to depend only on whether the limiting spectrum is an annulus or a disk which, surprisingly, seems to not have been observed in the previous literature.

Although some recent work, such as in [Hedenmalm and Wennman \(2020\)](#), may indicate a direction to expand some of these results to larger classes of potentials, this is, so far, still a necessary condition for the techniques presented henceforth. Once the model is established, the techniques are involved, but should be reasonable: We express the partition function in terms of a summation of norming constants, for which we have a integral representation. Then,

the integral can be estimated using the Laplace technique - separating the integral in a central main-contributing region and a bounded error-contributing complement region. Finally, the limit on the summation can be estimated with quadrature techniques, namely the Euler-Maclaurin formula, giving us the final results after computations.

6.1 Working Model

In this section we properly define the model and specifics of the working model to which we wish to get the asymptotics from. We have discussed in [section 5.1](#) how eigenvalue statistics can be interpret as some Coulomb gas. On the complex plane one could have for example the measure

$$dp_N^{(\beta)}(z_1, \dots, z_N) := \frac{1}{\mathcal{Z}_N^{(\beta)}} \prod_{j>k=1}^N |z_j - z_k|^\beta \prod_{j=1}^N e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j) \quad (6.1)$$

where $dA(z) := d^2z/\pi$ is the area measure. This measure would imply that the normalization constant - or in our context, the *Partition Function*, is given by

$$\mathcal{Z}_N^{(\beta)} := \int_{\mathbb{C}^N} \prod_{j>k=1}^N |z_j - z_k|^\beta \prod_{j=1}^N e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j).$$

Definition 14. Let $p_N^{(\beta)}$ be as in Equation 6.1. If $\beta = 2$, $p_N^{(2)}$ is the measure for the ensemble we call **random normal matrix**. If, for example, one sets $Q(z) = |z|^2$ this would be called **complex Ginibre ensemble**.

We could also define the measure for the configurational canonical Coulomb gas ensemble in the upper-half plane - which has an additional complex conjugation symmetry and is governed by the law

$$d\tilde{p}_N^{(\beta)}(z_1, \dots, z_N) := \frac{1}{\mathcal{Z}_N^{(\tilde{\beta}eta)}} \prod_{j>k=1}^N |z_j - z_k|^\beta |z_j - \bar{z}_k|^\beta \prod_{j=1}^N |z_j - \bar{z}_j|^\beta e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j) \quad (6.2)$$

and would imply the normalization constant

$$\mathcal{Z}_N^{(\tilde{\beta}eta)} := \int_{\mathbb{C}^N} \prod_{j>k=1}^N |z_j - z_k|^\beta |z_j - \bar{z}_k|^\beta \prod_{j=1}^N |z_j - \bar{z}_j|^\beta e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j).$$

Definition 15. Let $p_N^{(\beta)}$ be as in Equation 6.2. If $\beta = 2$, $p_N^{(2)}$ is the measure for the ensemble we call **planar symplectic ensemble**. If, for example, one sets $Q(z) = |z|^2$ this would be called **symplectic Ginibre ensemble**.

Virtually all results we'll state for the random normal matrix ensemble are valid in the context of the planar symplectic ensemble with only a few adaptations. However, in the name of

history telling and simplicity we will omit this measure: Adding this case to our chapter would cloud the point. For now on, we will only consider $\beta = 2$, this has some algebraic advantages that become clear by the Determinantal Point Processes structure (see [Chapter 3](#) for details) - because of this, we omit the β in the variables ($\mathcal{Z}_N^{(\beta)} := \mathcal{Z}_N$).

For the models of random matrices we are interested in, we must consider $\{P_n(z) = \sum_{j=0}^n a_j z^j\}_n^{(w)}$ a family of orthogonal polynomials in respect to w where P_m has order m . We're dealing with weight functions of the type

$$w(z) = e^{-NQ(z)}, \quad (6.3)$$

where $Q(z)$ is some complex potential that satisfies:

1. We have

$$Q(z) = q(|z|)$$

for some $q : \mathbb{R} \mapsto \mathbb{R}$, i.e., Q is a radially symmetric potential;

2. The potential Q grows sufficiently fast, that is,

$$\liminf_{|z| \rightarrow \infty} \frac{Q(z)}{2 \ln(|z|)} > 1;$$

3. Furthermore, assume Q to be smooth, sub-harmonic in $\mathbb{C}$, and strictly sub-harmonic in a neighborhood of the droplet.

This demands are essential for what follows and not all counter-intuitive. To require radial symmetric is to say that the integral in the plane can be somehow exchanged by a line integral, as all directions would be homogeneous. The growth requirement comes from the need for a confinement of the particles in the system. Remember that the interaction energy has logarithmic behavior. Smoothness makes sure we can expand the potential in a series without much worries about the continuity and existence of derivatives. The conditions on sub-harmonicity may not be as clear but are requirements that can be thought of as convexity. This will ensure uniqueness of the minima for the function and the connectedness of the support of the equilibrium measure. There is still a question of what are exactly the orthogonal polynomials for such weight function.

Lemma 2. *The set $\{z^j\}_j^{(w)}$ is the proper orthogonal family in relation to the weight function $w(z) = e^{-NQ(z)}$ given.*

Proof. We know by the assumption of orthogonality that

$$\langle P_n, P_m \rangle_w = \begin{cases} 0 & \text{if } m \neq n, \\ 1 & \text{if } m = n. \end{cases}$$

Let us take some arbitrary base of the polynomial space $\{z^j\}_j^{(w)}$. We can apply the Gram Schmidt scheme to get the proper orthogonality in respect to the weight w . If $P_n = x^n$, then the orthogonal element of the basis $\tilde{P}_n$ should be

$$\tilde{P}_n = P_n - \sum_{j=0}^{n-1} \frac{\langle P_n, P_j \rangle_w}{\langle P_j, P_j \rangle_w} P_j,$$

but notice that

$$\begin{aligned} \langle P_n, P_j \rangle_w &= \int_{\mathbb{C}} z^{n-j} e^{-NQ(z)} dA(z), \\ &= \int_0^\infty \int_0^{2\pi} r^{n-j} e^{i\theta(n-j)} e^{-Nq(r)} r dr d\theta, \\ &= \int_0^{2\pi} e^{i\theta(n-j)} d\theta \int_0^\infty r^{n-j+1} e^{-Nq(r)} dr, \\ &= 0, \quad \text{since } n \neq j. \end{aligned}$$

Therefore, we have that $\{z^j\}_j^{(w)}$ is the proper orthogonal family. More than that, if we impose the normality,

$$\begin{aligned} \int_{\mathbb{C}} a_n^2 z^{n-n} e^{-NQ(z)} dA(z) &= \int_{\mathbb{C}} a_n^2 e^{-NQ(z)} dA(z) = 1 \\ \implies a_n &= \sqrt{\frac{1}{\int_{\mathbb{C}} w(z) dA(z)}}. \end{aligned}$$

□

6.2 Asymptotics on Droplets

We will use some radially symmetric potential $Q(z) : \mathbb{C} \rightarrow \mathbb{R}$ which satisfies some nice growth and smoothness conditions, as specified in [section 6.1](#). Because of the radial symmetry, one can take some function $q : \mathbb{R} \rightarrow \mathbb{R}$ such that $q(|z|) = Q(z)$; If we say that the Q is subharmonic on the plane, i.e., $\Delta Q(z) \geq 0$ for all $z \in \mathbb{C}$, then

$$\Delta Q(z) \geq 0 = \frac{1}{r} (rq'(r))' \geq 0 \implies rq'(r) \text{ is increasing } \forall r \in \mathbb{R},$$

and strictly increasing in the droplet by the same reason. We begin by quickly considering the auxiliary result.

Lemma 3. *When dealing with a p.d.f. of invariant measure, one can write*

$$\mathcal{Z}_N = N! \prod_{k=0}^{N-1} h_k,$$

where h_k is the norming constant as in $\langle q_j^{(2)}, q_k^{(2)} \rangle^{(2)} = h_j^{(2)} \chi_{j=k}$.

Proof. We start stating the following equality which follows from Theorem 4 that

$$\det \left[K_N^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^N = p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_N; \beta),$$

$$\prod_{k=1}^N h_k^{-1} \det \left[\sqrt{w(x)w(y)} \sum_{k=0}^{N-1} q_k^{(2)}(x) q_k^{(2)}(y) \right]_{i,j=1}^N = \prod_{1 \leq i \leq j \leq N} (\lambda_j - \lambda_i)^2 \prod_{k=0}^N w(\lambda_k)$$

therefore

$$N! \det \left[\tilde{K}_N^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^N = N! \prod_{k=1}^N h_k \prod_{1 \leq i \leq j \leq N} (\lambda_j - \lambda_i)^2 \prod_{k=0}^N w(\lambda_k)$$

where

$$\mathcal{Z}_{N,2} = N! \prod_{k=1}^N h_k.$$

□

Thus, one can write the free energy as

$$\ln \mathcal{Z}_N = \ln N! + \sum_{j=0}^{N-1} \ln h_j, \quad \text{where} \quad h_k := \int_{\mathbb{C}} |z|^{2j} e^{-NQ(z)} dA(z) \quad (6.4)$$

are the norming constant for the monomial family of orthogonal polynomials in respect to w as in Equation 6.3. One could also write, for the planar symplectic ensemble

$$\ln \tilde{\mathcal{Z}}_N = \ln N! + \sum_{j=0}^{N-1} \ln 2\tilde{h}_{2j+1}, \quad \text{where} \quad \tilde{h}_k := \int_{\mathbb{C}} |z|^{2j} e^{-2NQ(z)} dA(z).$$

The expressions derived for the normal ensemble would be easily adapted for this model, and it is done in [Byun, Kang and Seo \(2023\)](#), however we will not follow that computation in name of clarity and storytelling.

In order to derive the asymptotic formulas for the free energy we need to derive the asymptotics for the orthogonal polynomials, specifically for the integral of its norming constants expressed in Equation 6.4. The first step in the asymptotics is then to prepare the integral for the Laplace method. Our goal is to identify a region that gives the main contribution, in order, for value of the integral - and consequently to identify a error-contributing region of the integration domain. Let's reformulate our expression to identify the usual integral form for the Laplace method, such as in [Chapter 4](#). We know $Q(z) = q(|z|)$ a radially symmetric potential such that, defining $z := r e^{i\theta}$, that is, $|z| = r$, we would have

$$\begin{aligned} h_j &= \int_{\Theta} \int_{\mathbb{R}_+} r^{2j} e^{-Nq(r)} \frac{r}{\pi} dr d\theta \\ &= \frac{2\pi}{\pi} \int_{\mathbb{R}_+} r^{2j+1} e^{-Nq(r)} dr \\ &= \int_{\mathbb{R}_+} 2r e^{-N\left(q(r) - \frac{2j}{N} \ln(r)\right)} dr. \end{aligned}$$

We first redefine our potential such that the integrand has a expression better suited to apply the Laplace method. Using

$$V_{\tau(j)} = q(r) - 2\tau(j) \ln(r),$$

where $\tau(j) := j/N \in [0, 1)$, then one can rewrite the integral as

$$h_j = \int_{\mathbb{R}_+} 2r e^{-NV_{\tau(j)}(r)} dr.$$

Now, we have set the stage for the Laplace technique. Note that now τ will perform the function of the previous index j and relates to which coefficient we're calculating. Taking a look at the first derivative of the newly defined potential we find that the critical point r_c must satisfy

$$V'_\tau(r) = q'(r) - \frac{2\tau}{r} \implies r_c q'(r_c) = 2\tau.$$

This is: We can find a critical point of the potential r_c , parameterized by the degree of the polynomial we are calculating, or more concretely, by τ . Of course, because we know the range of τ , it is worth defining the limital points

$$\begin{cases} r_0, & \text{the largest solution of } r_0 q'(r_0) = 0, \\ r_1, & \text{the smallest solution of } r_1 q'(r_1) = 2. \end{cases}$$

Our main goal here is to evaluate the integral using the surroundings of the critical point. If this point is unique, then we need no more than to calculate the integral in a single small interval around $r_c(\tau)$. This is, of course, true: We have mentioned that the properties on the potential directly implies that $r q'(r)$ is a strictly increasing function of $[r_0, r_1]$, then the solution must be unique. Now, as outside the droplet the growth is not strict, the definitions of r_0 and r_1 had to be as mentioned before: Respectively the largest and smallest solution to the proper instance of $r q'(r) = 2\tau$. An important distinction must be made between the cases when $r_0 = 0$ and $r_0 > 0$, corresponding to [subsection 6.2.2](#) and [subsection 6.2.1](#), respectively. We begin noticing that $r_\tau | r_\tau q'(r_\tau) = 2\tau$ is an increasing function on τ and that $r_0 \leq r_\tau \leq r_1 \forall \tau \in [0, 2]$. As we'll be performing the Laplace technique, Gaussian integrals will be a big part of the result, more details on why can be found in [Chapter 4](#). As such, we state the following proposition.

Proposition 9. *The Gaussian integrals below can be evaluated as follows.*

$$\int_{\mathbb{R}} e^{-2ar^2} dr = \sqrt{\frac{\pi}{2}} \frac{1}{a^{1/2}} \implies \int_{\mathbb{R}} e^{-2\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right)y^2} = \sqrt{\frac{\pi}{2}} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}},$$

$$\int_{\mathbb{R}} e^{-2ar^2} r^4 dr = \sqrt{\frac{\pi}{2}} \frac{3}{16} \frac{1}{a^{5/2}} \implies \int_{\mathbb{R}} e^{-2\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right)y^2} y^4 = \sqrt{\frac{\pi}{2}} \frac{3}{16} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{5}{2}},$$

$$\int_{\mathbb{R}} e^{-2ar^2} r^6 dr = \sqrt{\frac{\pi}{2}} \frac{15}{64} \frac{1}{a^{7/2}} \implies \int_{\mathbb{R}} e^{-2\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right)y^2} y^6 = \sqrt{\frac{\pi}{2}} \frac{15}{64} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\right)^{\frac{7}{2}}.$$

6.2.1 Asymptotics - Annulus

We start by considering the annulus regime, with $r_0 > 0$. We know the measure in Equation 6.1 to be the law for the distribution of the particles in this problem. When we say that $r_0 > 0$, we are saying that the region of space $[r_0, r_1]$ where this measure is non-zero for $N \rightarrow \infty$ has the annulus shape. If we think of the equivalent Coulomb gas, this means the particles will concentrate in this shape when we perform an asymptotic on N . This is the support for the limit measure.

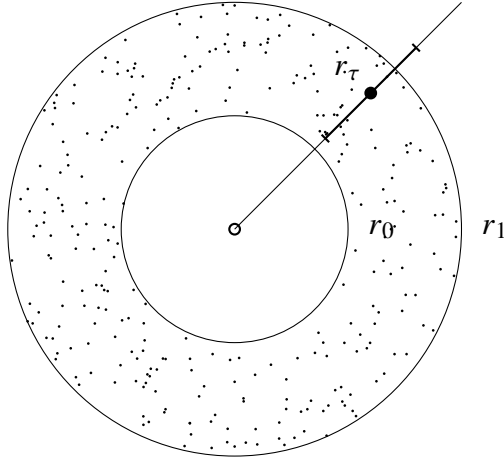


Figure 18 – The figure depicts an annulus referring to the equilibrium measure when $r_0 > 0$. As the complex plane integral is reduced to a line integral, we draw the domain change and the point of interest r_τ where V has a critical point. Around this point we depict the region around the minima, which in the method that should account for the main order of the integral we solve for.

Theorem 17. *For the annulus case, where $r_0 > 0$, we should have the following asymptotic for the free energy asymptotically on N given by*

$$\begin{aligned} \ln \mathcal{Z}_N = & -N^2 I_Q[\mu_Q] + \frac{N}{2} (\ln(2\pi) - E_Q[\mu_Q] - 2) + \frac{N}{2} \ln(N) \\ & + \frac{\ln(2\pi)}{2} + \frac{1}{2} \ln(N) + F_V[A_{[r_0, r_1]}] + \mathcal{O}(N^{-1}) \end{aligned}$$

where we set

$$\begin{aligned} F_V[A_{[r_0, r_1]}] := & -\frac{2}{3} \ln\left(\frac{r_1}{r_0}\right) + \frac{5}{32} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) \\ & - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr, \end{aligned}$$

$$I_Q[\mu_Q] := \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right],$$

and

$$E_Q[\mu_Q] := \frac{1}{2} \int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (s q'(s))' ds.$$

We start by studying the asymptotic for the norming constants h_j . Then, we try to derive the formulas for the summation on its logarithm $\ln(h_j)$. These results should be enough to prove the previous theorem. The following subsections are referring to these two results.

Theorem 18. *We have the following asymptotic on N for h_j , with $0 \leq j \leq N-1$,*

$$h_j = \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{N} \mathfrak{B}(r_{\tau(j)}) + O(N^{-2}) \right], \quad (6.5)$$

where we have set

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3} \frac{5}{24}.$$

The proof for such theorem passes through Lemmas 4 and 5. We begin as usual when applying asymptotic techniques in integrals; separating a central part where the integral value is concentrated and another part where we can control the order of contribution. This is possible because the integrand is proportional to some exponential, with negative exponent, with a unique critical point, around which, the value of the expression concentrates, as discussed in Chapter 4. Define $\delta_N = \ln(N)/\sqrt{N}$ - this will be our cut-size for the interest region and will be justified during the calculations - we have

$$h_j = \int_{|r-r_j| > \delta_N} 2r e^{-NV_{\tau(j)}(r)} dr + \int_{|r-r_j| < \delta_N} 2r e^{-NV_{\tau(j)}(r)} dr = (1) + (2). \quad (6.6)$$

As mentioned, the first integral must be somehow controlled and the second estimated as the main value for the result. We begin by showing that the first term has an error contribution.

Lemma 4. *If we choose M to be such that*

$$\int_{|r-r_j| > \delta_N} \left(e^{-q(r)} r^2 \right)^M r dr < \infty,$$

then, with (1) as in Equation 6.6 and $0 \leq j \leq N-1$, asymptotically on N ,

$$(1) := \int_{|r-r_j| > \delta_N} 2r e^{-NV_{\tau(j)}(r)} dr = e^{-NV_{\tau(j)}(r_{\tau(j)})} \epsilon_N$$

where $\epsilon_N = O(e^{-c(\ln(N))^2})$.

Proof. Let's first show that we can control the first part of the integral, for that, rewrite

$$\begin{aligned}
 (1) &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r e^{-N(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\
 &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r e^{-N(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} e^{-(M-M)(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\
 &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} e^{-(N-M)(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr.
 \end{aligned}$$

Now, we need some way to estimate the difference $(V_{\tau(j)}(r) - V_{\tau(j)}(r_{\tau(j)}))$. If we are able to say that this difference is at least $O(N^{-1})$, we could control the growth of the integral choosing M to be large enough such that

$$\int_{|r-r_j|>\delta_N} \left(e^{-q(r)} r^{2\tau(j)} \right)^M r dr$$

converges. Using the definition of the function V and the mean value theorem, we start by evaluating the difference $\Delta(j) = V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)})$ in the following manner

$$\begin{aligned}
 \Delta(j) &= q(r_{\tau(j+1)}) - q(r_{\tau(j)}) - 2(\tau(j+1) \ln(r_{\tau(j+1)}) - \tau(j) \ln(r_{\tau(j)})), \\
 &= q'(c_1)(r_{\tau(j+1)} - r_{\tau(j)}) - \frac{2}{N} \left(\ln(r_{\tau(j+1)}) + j \ln\left(\frac{r_{\tau(j+1)}}{r_{\tau(j)}}\right) \right), \\
 &= q'(c_1)r'(c_2)(\tau(j+1) - \tau(j)) - \frac{2}{N} \left(\ln(r_{\tau(j+1)}) + jO(N^{-1}) \right), \\
 &= q'(c_1)r'(c_2)O(N^{-1}) - O(N^{-1}), \\
 &= O(N^{-1}).
 \end{aligned}$$

The last equality holds as both functions are continuous and c_1, c_2 are values that are limited; in a compact disk on the plane. Then, we can add zero and use the estimation

$$V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j+1)}(r) + V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) = O(N^{-1})$$

to derive the result we are after. It is enough to estimate $V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)})$. For that, because $r q'(r)$ is increasing, for all $r > r_{\tau(j)} + \delta_N$ we can say that

$$\begin{aligned}
 V_{\tau(j+1)}(r) &\geq V_{\tau(j+1)}(r_{\tau(j)} + \delta_N), \\
 &= V_{\tau(j+1)}(r_{\tau(j+1)}) + \frac{1}{2} V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1)})^2 + O(\delta_N^3),
 \end{aligned}$$

and therefore can write

$$\begin{aligned}
 V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) &= \frac{1}{2} V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1)})^2 + O(\delta_N^3), \\
 &= \delta^2 c + O(N^{-1}) + O(\delta_N^3), \\
 &\geq \delta^2 c.
 \end{aligned}$$

Finally, using this one can say

$$V_{\tau(j+1)}(r) - V_{\tau(j+1)}(r_{\tau(j+1)}) \geq c\delta_N^2.$$

Getting this inequality back to the integral expression end up with

$$\begin{aligned} (1) &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} e^{-(N-M)(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\ &\leq e^{-NV_{\tau(j)}(r_{\tau(j)})} e^{-c\delta_N^2(N-M)} \int_{|r-r_j|>\delta_N} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \end{aligned}$$

where you can check that

$$\begin{aligned} &\int_{|r-r_j|>\delta_N} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr \\ &= \int_{|r-r_j|>\delta_N} 2r e^{-M((q(r)-2\tau(j)\ln(r))-(q(r_{\tau(j)})-2\tau(j)\ln(r_{\tau(j)})))} dr, \\ &= 2 \left(e^{-q(r_{\tau(j)})} r_{\tau(j)}^{-2\tau(j)} \right)^M \int_{|r-r_j|>\delta_N} \left(e^{-q(r)} r^{2\tau(j)} \right)^M r dr. \end{aligned}$$

By assumption, we chose M to be a big enough number such that the integral above converges.

If that is true,

$$(1) = \int_{|r-r_j|>\delta_N} 2r e^{-NV_{\tau(j)}(r)} dr \leq e^{-NV_{\tau(j)}(r_{\tau(j)})} \epsilon_N,$$

with $\epsilon_N = O(e^{-c(\ln(N))^2})$. □

We now move on to the value-contributing part of the integral. The other order term should be absolved in this one if Theorem 18 is to be right.

Lemma 5. *For the second part of the integral we have, for (2) as in Equation 6.6 and $0 \leq j \leq N-1$, asymptotically on N ,*

$$(2) := \int_{|r-r_j|<\delta_N} 2r e^{-NV_{\tau(j)}(r)} dr = \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{N} \mathfrak{B}(r_{\tau(j)}) + O(N^{-2}) \right] \quad (6.7)$$

where $\mathfrak{B}(r_{\tau(j)})$ is as in Theorem 18.

Proof. As we're dealing with a symmetric (in relation to the critical point) integral, we might as well set a centralized variable $x = r - r_{\tau}$. With this, the integral becomes

$$\begin{aligned} (2) &= \int_{-\delta_N}^{\delta_N} 2(x + r_{\tau(j)}) e^{-NV_{\tau(j)}(x+r_{\tau(j)})} dx, \\ &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{-\delta_N}^{\delta_N} 2(x + r_{\tau(j)}) e^{-N[V_{\tau(j)}(x+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})]} dx. \end{aligned}$$

There is also a question of scale. Our interval grows with $\sqrt{N}/\ln(N)$. By re-scaling the reference variable using $y = \sqrt{N}x$ we get

$$(2) = \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\ln(N)}^{\ln(N)} 2 \left(\frac{y}{\sqrt{N}} + r_{\tau(j)} \right) e^{-N[V_{\tau(j)}(\frac{y}{\sqrt{N}}+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})]} dy.$$

Now, the idea here is quite direct: to use the Taylor expansion for $V_{\tau(j)}\left(\frac{y}{\sqrt{N}} + r_{\tau(j)}\right)$ near the critical point $r_{\tau(j)}$. That is,

$$\begin{aligned}
 (2) &= \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\ln(N)}^{\ln(N)} 2\left(\frac{y}{\sqrt{N}} + r_{\tau(j)}\right) e^{-N\left(\frac{y^2}{2N} V_{\tau(j)}^{(2)}(r_{\tau(j)}) + \frac{y^3}{3!N^{\frac{3}{2}}} V_{\tau(j)}^{(3)}(r_{\tau(j)}) + \frac{y^4}{4!N^2} V_{\tau(j)}^{(4)}(r_{\tau(j)}) + O\left(\left(\frac{y}{\sqrt{N}}\right)^5\right)\right)} dy, \\
 &= \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{(2r_{\tau(j)})^{-1}\sqrt{N}} \int_{-\ln(N)}^{\ln(N)} \left(\frac{y}{r_{\tau(j)}\sqrt{N}} + 1\right) e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)}) - \frac{y^3}{3!\sqrt{N}} V_{\tau(j)}^{(3)}(r_{\tau(j)}) - \frac{y^4}{4!N} V_{\tau(j)}^{(4)}(r_{\tau(j)}) + O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)} dy, \\
 &= \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{(2r_{\tau(j)})^{-1}\sqrt{N}} \int_{-\ln(N)}^{\ln(N)} \left(\frac{y}{r_{\tau(j)}\sqrt{N}} + 1\right) e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} \left(e^{-\frac{y^3}{3!\sqrt{N}} V_{\tau(j)}^{(3)}(r_{\tau(j)}) - \frac{y^4}{4!N} V_{\tau(j)}^{(4)}(r_{\tau(j)}) + O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)} \right) dy.
 \end{aligned}$$

Now we take the Taylor Series for

$$\begin{aligned}
 e^{-\frac{y^3}{3!\sqrt{N}} V_{\tau(j)}^{(3)}(r_{\tau(j)}) - \frac{y^4}{4!N} V_{\tau(j)}^{(4)}(r_{\tau(j)}) + O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)} &= 1 - \left[\frac{y^3}{3!\sqrt{N}} V_{\tau(j)}^{(3)}(r_{\tau(j)}) + \frac{y^4}{4!N} V_{\tau(j)}^{(4)}(r_{\tau(j)}) + O\left(\frac{y^5}{N^{\frac{3}{2}}}\right) \right] \\
 &\quad - \frac{1}{2} \left[\frac{y^6}{N} \left(\frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6} \right)^2 + \frac{y^8}{N^2} \left(\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24} \right)^2 + \frac{y^7}{N^3} \left(\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{6} \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{24} \right) + O\left(\frac{y^8}{N^2}\right) \right].
 \end{aligned}$$

To get the asymptotic expansion (since odd terms vanish) we reorganize the terms in the integral again to obtain

$$\begin{aligned}
 (2) &= 2r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\ln(N)}^{\ln(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} \left[1 - \frac{y^4}{3!r_{\tau(j)}N} V_{\tau(j)}^{(3)}(r_{\tau(j)}) \right. \\
 &\quad \left. - \frac{y^4}{4!N} V_{\tau(j)}^{(4)}(r_{\tau(j)}) - \frac{y^6}{2 \cdot (3!)^2 N} V_{\tau(j)}^{(3)}(r_{\tau(j)})^2 + O\left(\frac{y^6}{N^2}\right) \right] dy, \quad (6.8)
 \end{aligned}$$

then, distributing the integral,

$$\begin{aligned}
 (2) &= 2r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left[\left(-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24N} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6r_{\tau(j)}N} \right) \int_{-\ln(N)}^{\ln(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} y^4 dy \right. \\
 &\quad \left. - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{72N} \int_{-\ln(N)}^{\ln(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} y^6 dy + \int_{-\ln(N)}^{\ln(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} dy + O\left(N^{-2}\right) \right].
 \end{aligned}$$

At this point, the result can be directly evaluated using the Gaussian integrals in Proposition 9.

So that we can finally substitute the values to get

$$\begin{aligned}
 (2) &= \sqrt{2\pi} r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left[\left(-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24N} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6r_{\tau(j)}N} \right) \frac{3}{16} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{5}{2}} \right. \\
 &\quad \left. - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{72N} \frac{15}{64} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{7}{2}} + \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} + O\left(N^{-2}\right) \right],
 \end{aligned}$$

and, rearranging the terms we should get our result

$$(2) = \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{N} \mathfrak{B}(r_{\tau(j)}) + O(N^{-2}) \right] \quad (6.9)$$

where we defined $\mathfrak{B}(r_{\tau(j)})$ as

$$\mathfrak{B}(r_{\tau(j)}) = -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3} \frac{5}{24}.$$

Notice that the error term of Lemma 4 is absorbed into the error order of Equation 6.9. Therefore we have the asymptotic expression for h_j given by Equation 6.5 and the proof for Theorem 18 follows directly. $\square$

We return to Equation 6.4 and turn to the calculation of each term of $\sum_{j=0}^{N-1} \ln h_j$ using the logarithm of the expansion calculated previously in Theorem 18

$$\sum_{j=0}^{N-1} -NV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \left(\ln(2\pi r_{\tau(j)}^2) - \ln N - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right) + \frac{1}{N} \mathfrak{B}(r_{\tau(j)}) + O(N^{-2}). \quad (6.10)$$

That is, we need to prove the following theorem.

Theorem 19. *Asymptotically on N ,*

$$\sum_{j=0}^{N-1} \ln h_j = -N^2 I_Q[\mu_Q] - \frac{N}{2} \ln N + N \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) + F_V[A_{[r_0, r_1]}] + O(N^{-1})$$

where $F_V[A_{[r_0, r_1]}]$, $I_Q[\mu_Q]$, $E_Q[\mu_Q]$ are as in Theorem 17.

The proof comes from gathering the terms in Equation 6.10 with its respective expressions derived in Lemmas 6 and 7. We go term by term. Lemma 6 refers to the asymptotic of

$$\sum_{j=0}^{N-1} -NV_{\tau(j)}(r_{\tau(j)}),$$

and Lemma 7 refers to the asymptotics of both terms

$$\sum_{j=0}^{N-1} \left(\ln(2\pi r_{\tau(j)}^2) - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right).$$

Then, the final expression becomes a question of balancing the results.

Lemma 6. *Asymptotically on N ,*

$$-N \sum_{j=0}^{N-1} V_{\tau(j)}(r_{\tau(j)}) = -N^2 I_Q[\mu_Q] + NU_{\mu_Q}(0) - \frac{1}{6} \ln\left(\frac{r_0}{r_1}\right) + \mathcal{O}(N^{-2})$$

where we have set $I_Q[\mu_Q]$ as in Theorem 17 and

$$-2U_{\mu_Q}(0) := \int_{r_0}^{r_1} \ln(s)(sq'(s))' ds = q(r_1) - q(r_0) + 2\ln(r_1).$$

Proof. Using the Euler-Maclaurin formula

$$\begin{aligned} \sum_{j=0}^{N-1} V_{\tau(j)}(r_{\tau(j)}) &= \int_0^N V_{\tau(t)}(r_{\tau(t)}) dt - \frac{1}{2} (V_{\tau(N)}(r_{\tau(N)}) - V_{\tau(0)}(r_{\tau(0)})) + \\ &\quad \frac{1}{12} [\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0}] + \mathcal{O}(N^{-3}). \end{aligned} \quad (6.11)$$

The first term in Equation 6.11 is solved doing a integral by parts. We remember the expression for the derivative of the potential $V_{\tau}'(r) = q'(r) - 2\tau/r$. Note that in the critical point $r_{\tau(j)}$, we should have $r_{\tau(j)}q'(r_{\tau(j)}) = 2\tau(j)$. This implies that, for $r(\tau) := r_{\tau}$, the differential can be written as

$$\begin{aligned} 2d\tau(j) &= d(r_{\tau}q'(r_{\tau})), \\ &= dr_{\tau}q'(r_{\tau}) + dr_{\tau}r_{\tau}q''(r_{\tau}), \\ &= dr_{\tau}(q'(r_{\tau}) + r_{\tau}q''(r_{\tau})). \end{aligned}$$

And, therefore,

$$\frac{dr(\tau)}{d\tau} = \frac{2}{r(\tau)V_{\tau}^{(2)}(r(\tau))} = \frac{2}{(rq'(r))'|_{r=r(\tau)}}.$$

We then are able to rewrite, setting $s = r_{\tau(t)}$,

$$\begin{aligned} \int_0^N V_{\tau(t)}(r_{\tau(t)}) dt &= \int_0^N (q(r_{\tau(t)}) - 2\tau(t) \ln(r_{\tau(t)})) dt, \\ &= \frac{1}{2} \int_{r_0}^{r_1} (q(s) - sq'(s) \ln(s)) s V^{(2)}(s) N ds, \\ &= N \left[\frac{1}{2} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \frac{1}{2} \int_{r_0}^{r_1} s^2 q'(s) \ln(s) V^{(2)}(s) ds \right], \end{aligned}$$

and finally using that $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$ we can apply the integration by parts on the

second term to get

$$\begin{aligned}
\frac{1}{2} \int_{r_0}^{r_1} s^2 q'(s) \ln(s) V^{(2)}(s) ds &= \frac{1}{2} \int_{r_0}^{r_1} s^2 q'(s) \ln(s) \frac{1}{s} (s q'(s))' ds, \\
&= \frac{1}{2} \int_{r_0}^{r_1} s q'(s) \ln(s) (s q'(s))' ds, \\
&= \frac{1}{4} \int_{r_0}^{r_1} ((s q'(s))^2)' \ln(s) ds, \\
&= \frac{1}{4} (s q'(s))^2 \ln(s) \Big|_{r_0}^{r_1} - \frac{1}{4} \int_{r_0}^{r_1} (s q'(s))^2 \frac{1}{s} ds \\
&= \ln(r_1) - \frac{1}{4} \int_{r_0}^{r_1} s q'(s)^2 ds, \\
&= \ln(r_1) - \frac{1}{2} q(r_1) + \frac{1}{4} \int_{r_0}^{r_1} q(s) (s q'(s))' ds.
\end{aligned}$$

Leaving us with

$$\begin{aligned}
\int_0^N V_{\tau(t)}(r_{\tau(t)}) dt &= N \left[\frac{1}{2} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) - \frac{1}{4} \int_{r_0}^{r_1} q(s) (s q'(s))' ds \right], \\
&= N \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right], \\
&=: NI_Q[\mu_Q].
\end{aligned} \tag{6.12}$$

We also need an expression for $V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)})$. This is direct noting that

$$V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)}) = V_1(r_1) - V_0(r_0) = q(r_1) - q(r_0) + 2 \ln(r_1) =: 2U_{\mu_Q}(0). \tag{6.13}$$

And finally we can the remaining term, we use the Leibniz rule and obtain

$$\begin{aligned}
\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} &= \partial_t (q(r_{\tau(t)}) - 2\tau(t) \ln(r_{\tau(t)}))|_{t=N}, \\
&= \partial_\tau (q(r_{\tau(t)}) - 2\tau(t) \ln(r_{\tau(t)}))|_{\tau=1} \partial_t (\tau(t))|_{t=N}, \\
&= \frac{1}{N} (\partial_\tau (q(r_\tau)) - 2(\ln(r_\tau) + \tau \partial_\tau (\ln(r_\tau))|_{\tau=1})), \\
&= \frac{1}{N} [\partial_t (q(r_{\tau(t)}) - 2\ln(r_{\tau(t)}))|_{\tau=1} - 2\ln(r_1)]
\end{aligned}$$

and

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{N} [\partial_t (q(r_{\tau(t)}) - 2\ln(r_{\tau(t)}))|_{\tau=0} - 2\ln(r_0)].$$

Since we can calculate, from $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$

$$\begin{aligned}
\partial_\tau q(r_\tau)|_{\tau=0} &= \left(\frac{2q'(r)}{(r q'(r))'} \right)_{r_0} = 0, \\
\partial_\tau q(r_\tau)|_{\tau=1} &= \left(\frac{2q'(r)}{(r q'(r))'} \right)_{r_1} = \frac{4}{r_1 (q'(r_1) + r_1 q''(r_1))}, \\
\partial_\tau (2\ln(r_\tau))|_{\tau=1} &= \left(\frac{4}{r (r q'(r))'} \right)_{r_1} = \frac{4}{r_1 (q'(r_1) + r_1 q''(r_1))},
\end{aligned}$$

we can also express the difference as

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{12N} [2(\ln(r_0) - \ln(r_1))] = \frac{1}{6N} \ln\left(\frac{r_0}{r_1}\right). \quad (6.14)$$

The proof for the summation $\sum_{j=0}^{N-1} V_{\tau(j)}(r_{\tau(j)})$ is complete if one gathers the terms in Equation 6.12, Equation 6.14 and Equation 6.13,

□

Lemma 7. Asymptotically on N ,

$$\sum_{j=0}^{N-1} \ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) = NE_Q[\mu_Q] - \frac{1}{2} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + O(N^{-1})$$

and

$$\sum_{j=0}^{N-1} \ln(2\pi r_{\tau(j)}^2) = N \ln(2\pi) + 2 \left(-NU_{\mu_Q}(0) - \frac{1}{2} \ln\left(\frac{r_1}{r_0}\right) + O(N^{-1}) \right)$$

where $U_{\mu_Q}(0)$ as in Lemma 6 and $E_Q[\mu_Q]$ as in Theorem 17.

Proof. Lets start now with the second summation, to which we apply Euler-Maclaurin again

$$\begin{aligned} \sum_{j=0}^{N-1} \ln\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right) &= \int_0^N \frac{1}{4} \ln V_{\tau(t)}^{(2)}(r_{\tau(t)}) dt - \frac{1}{2} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + \\ &\quad \frac{1}{12} \left[\partial_t \frac{1}{4} \ln V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=N} - \partial_t \frac{1}{4} \ln V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=0} \right] + O(N^{-3}). \end{aligned}$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we would have

$$\int_0^N \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)}) dt = \frac{N}{2} \int_{r_0}^{r_1} \ln\left(\frac{1}{4} V^{(2)}(s)\right) (sq'(s))' ds.$$

And, noticing that

$$\partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)}) = \partial_V \left(\ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right) \frac{1}{4} \partial_r \left(V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right) \partial_\tau(r_{\tau(t)}) = \frac{\left(\partial_r V^{(2)} \right)(r_{\tau(t)})}{r_{\tau(t)} \left(V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right)^2},$$

we have for the third term on the right hand side,

$$\begin{aligned} \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=N} - \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=0} &= \frac{1}{2N} \left(\frac{\left(\partial_r V^{(2)} \right)(r_1)}{r_1 \left(V_1^{(2)}(r_1) \right)^2} - \frac{\left(\partial_r V^{(2)} \right)(r_0)}{r_0 \left(V_0^{(2)}(r_0) \right)^2} \right) \\ &= O(N^{-1}). \end{aligned}$$

Therefore,

$$\begin{aligned} \sum_{j=0}^{N-1} \ln \frac{1}{4} V_{\tau(j)}^{(2)}(r_{\tau(j)}) &= \frac{N}{2} \int_{r_0}^{r_1} \ln \left(\frac{1}{4} V^{(2)}(s) \right) (sq'(s))' ds - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) + O(N^{-1}), \\ &:= NE_Q[\mu_Q] - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) + O(N^{-1}). \end{aligned}$$

Similarly, for the first summation, we apply Euler-Maclaurin again

$$\begin{aligned} \sum_{j=0}^{N-1} \ln(2\pi r_{\tau(j)}^2) &= \ln(2\pi) + 2 \left(\int_0^N \ln(r_{\tau(t)}) dt - \frac{1}{2} \ln \left(\frac{r_1}{r_0} \right) \right. \\ &\quad \left. + \frac{1}{12} [\partial_t \ln r_{\tau(t)}|_{\tau=1} - \partial_t \ln r_{\tau(t)}|_{\tau=0}] + O(N^{-3}) \right). \end{aligned}$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we would have

$$\int_0^N \ln r_{\tau(t)} dt = \frac{N}{2} \int_{r_0}^{r_1} \ln(s) (sq'(s))' ds.$$

And, noticing that

$$\partial_t \ln r_{\tau(t)} = \partial_r (\ln(r_{\tau(t)})) \partial_\tau (r_{\tau(t)}) = \frac{2}{V_{\tau(t)}^{(2)}(r_{\tau(t)}) (r_{\tau(t)})^2},$$

we have for the third term on the right hand side,

$$\partial_t \ln r_{\tau(t)}|_{\tau=1} - \partial_t \ln r_{\tau(t)}|_{\tau=0} = \frac{1}{2N} \left(\frac{2}{V^{(2)}(r_1) (r_1)^2} - \frac{2}{V^{(2)}(r_0) (r_0)^2} \right) = O(N^{-1}).$$

Then

$$\sum_{j=0}^{N-1} \ln(2\pi r_{\tau(j)}^2) = N \ln(2\pi) + 2 \left(\frac{N}{2} \int_{r_0}^{r_1} \ln(s) (sq'(s))' ds - \frac{1}{2} \ln \left(\frac{r_1}{r_0} \right) + O(N^{-1}) \right).$$

□

Proof. (Theorem 19) We only need to notice now that

$$\begin{aligned} \frac{1}{N} \sum_{j=0}^{N-1} \mathfrak{B}(r_{\tau(j)}) &= \frac{1}{4} \int_0^{r_1} \mathfrak{B}(s) (sq'(s))' ds + O(N^{-2}), \\ &= \frac{1}{4} \int_0^{r_1} \mathfrak{B}(s) s V^{(2)}(s) ds + O(N^{-2}). \end{aligned}$$

Using the definition for $\mathfrak{B}(r_{\tau(j)})$ and integrating by parts,

$$\begin{aligned} \frac{1}{N} \sum_{j=0}^{N-1} \mathfrak{B}(r_{\tau(j)}) &= \frac{1}{4} \int_{r_0}^{r_1} \left[-\frac{V^{(4)}(r)}{V^{(2)}(r)^2} \frac{1}{8} - \frac{V^{(3)}(r)}{V^{(2)}(r)^2} \frac{1}{2r} - \frac{V^{(3)}(r)^2}{V^{(2)}(r)^3} \frac{5}{24} \right] r V^{(2)}(r) dr + O(N^{-2}), \\ &= -\frac{1}{32} \int_{r_0}^{r_1} \frac{V^{(4)}(r)}{V^{(2)}(r)} r dr - \frac{1}{8} \int_{r_0}^{r_1} \frac{V^{(3)}(r)}{V^{(2)}(r)} dr - \frac{5}{96} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(N^{-2}), \\ &= -\frac{1}{8} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - \frac{1}{32} \int_{r_0}^{r_1} \frac{V^{(4)}(r)}{V^{(2)}(r)} r dr - \frac{5}{96} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(N^{-2}), \end{aligned}$$

where

$$\int_{r_0}^{r_1} \frac{V^{(4)}(r)}{V^{(2)}(r)} r dr = \frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} - \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) + \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr.$$

Therefore

$$\frac{1}{N} \sum_{j=0}^{N-1} \mathfrak{B}(r_{\tau(j)}) = -\frac{3}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + \mathcal{O}(N^{-2}), \quad (6.15)$$

and the theorem follows from

$$\begin{aligned} \sum_{j=0}^{N-1} \ln h_j &= -N^2 I_Q[\mu_Q] + N U_{\mu_Q}(0) - \frac{1}{6} \ln \left(\frac{r_0}{r_1} \right) - \frac{N}{2} \ln(N) - \frac{1}{2} \left(N E_Q[\mu_Q] - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) \right) \\ &\quad + \frac{1}{2} \left(N \ln(2\pi) + 2 \left(-N U_{\mu_Q}(0) - \frac{1}{2} \ln \left(\frac{r_1}{r_0} \right) \right) \right) - \frac{3}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) \\ &\quad - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + \mathcal{O}(N^{-1}) \end{aligned}$$

that reduces to

$$\begin{aligned} \sum_{j=0}^{N-1} \ln h_j &= -N^2 I_Q[\mu_Q] - \frac{N}{2} \ln(N) + N \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{2}{3} \ln \left(\frac{r_0}{r_1} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) \\ &\quad - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + \mathcal{O}(N^{-1}). \end{aligned}$$

□

We can finally prove Theorem 17.

Proof. (Theorem 17) From Theorem 18,

$$\begin{aligned} \sum_{j=0}^{N-1} \ln(h_j) &= -N^2 I_Q[\mu_Q] - \frac{N}{2} \ln(N) + N \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{2}{3} \ln \left(\frac{r_0}{r_1} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) \\ &\quad - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + \mathcal{O}(N^{-1}). \end{aligned}$$

We can get the asymptotic expansion for the Free energy using that

$$\ln(N!) = N \ln(N) - N + \frac{1}{2} \ln(N) + \frac{1}{2} \ln(2\pi) + \mathcal{O}(N^{-1})$$

that is,

$$\begin{aligned}
\ln \mathcal{Z}_N &= \ln N! + \sum_{j=0}^{N-1} \ln h_j, \\
&= N \ln(N) - N + \frac{1}{2} \ln(N) + \frac{1}{2} \ln(2\pi) + \left(-N^2 I_Q[\mu_Q] - \frac{N}{2} \ln(N) + N \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) \right. \\
&\quad \left. - \frac{2}{3} \ln \left(\frac{r_0}{r_1} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr \right) + O(N^{-1}), \\
&= -N^2 I_Q[\mu_Q] + \frac{N}{2} (\ln(2\pi) - E_Q[\mu_Q] - 2) + \frac{N}{2} \ln(N) + \frac{1}{2} \ln(N) + \frac{\ln(2\pi)}{2} \\
&\quad - \frac{2}{3} \ln \left(\frac{r_0}{r_1} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(N^{-1}), \\
&:= -N^2 I_Q[\mu_Q] + \frac{N}{2} (\ln(2\pi) - E_Q[\mu_Q] - 2) + \frac{N}{2} \ln(N) + \frac{\ln(2\pi)}{2} + \frac{1}{2} \ln(N) + F_V[A_{[r_0, r_1]}] + O(N^{-1}),
\end{aligned}$$

and the theorem follows. $\square$

6.2.2 Asymptotics - Disk

In this case, we consider $r_0 = 0$. As before, the measure in Equation 6.1 to be the law for distribution of the particles in this model. When we say $r_0 = 0$, we mean that the support for this configuration space of particles is a disk in the complex plane. The expression we get is different from taking the limit in the case $r_0 > 0$. (BYUN; KANG; SEO, 2023)

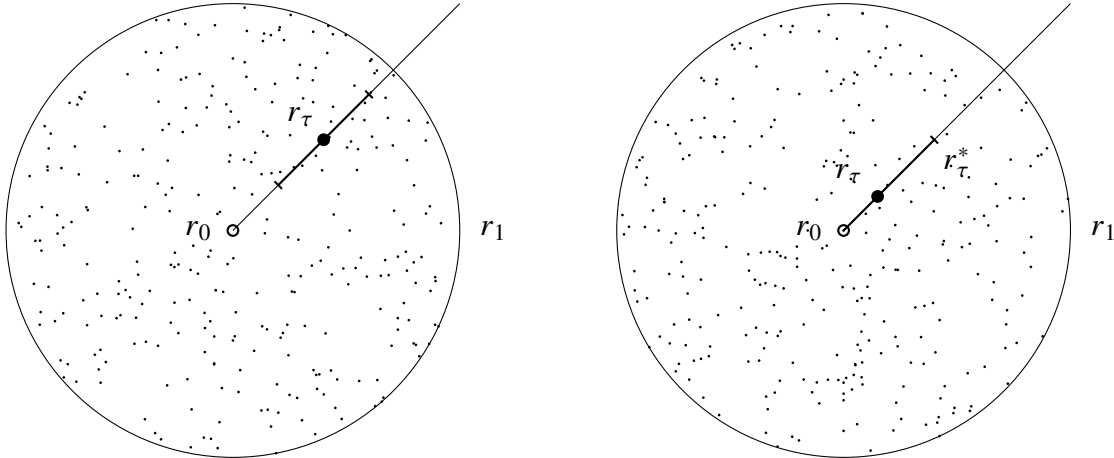


Figure 19 – The figure depicts an disk referring to the equilibrium measure when $r_0 = 0$. As the complex plane integral is reduced to a line integral, we draw the domain change and the point of interest r_τ where V has a critical point. Around this point we depict the region around the minima, which in the method that should account for the main order of the integral we solve for. The left image corresponds to the case when $r_{j/N} \gg N^{-\epsilon}$ and the right image corresponds to $r_{j/N} \ll N^{-\epsilon}$. In this case r_τ^* will define the contributing region to the integral.

Theorem 20. *For the annulus case, where $r_0 = 0$, we should have the following, asymptotically on N , for the free energy*

$$\ln \mathcal{Z}_N = -N^2 I_Q[\mu_Q] + \frac{1}{2} N \ln N + \frac{N}{2} (\ln(2\pi) - 2 - E_Q[\mu_Q]) - \frac{5}{12} \ln N \frac{\ln(2\pi)}{2} + \zeta'(-1) + F_V[D_{r_1}] + O(N^{-1/12} (\ln N)^3).$$

where we have set

$$I_Q[\mu_Q] := \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right],$$

$$E_Q[\mu_Q] := \frac{1}{2} \int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (s q'(s))' ds$$

and

$$F_V[D_{r_1}] := \frac{1}{6} \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{3} \ln(r_1) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)} \right) - \frac{1}{32} \frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12} \int_0^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr.$$

Again, we start by studying the asymptotic for h_j and then for the summation $\sum \ln(h_j)$. This theorem then should follow from the following lemmas concerning this two asymptotics. In the asymptotic expansions of h_j , one should distinguish the following two cases depending on a small constant $1/5 > \epsilon > 0$.

- Case 1: $r_{j/N} \gg N^{-\epsilon}$. For the annular droplet case where $r_0 > 0$, this case covers all $j = 0, 1, \dots, N-1$. On the other hand, for the disk droplet case where $r_0 = 0$, this case covers only $j = m_N, m_N + 1, \dots, N-1$ for $m_N = N^\epsilon$ with some $\epsilon > 0$;
- Case 2: $r_{j/N} \ll N^{-\epsilon}$. This covers the remaining disk droplet case with $j = 0, 1, \dots, m_N - 1$. Notably, the asymptotic expansion involves gamma functions in this case.

We separate this in two proofs, the first one will be regarding $j = m_N, m_N + 1, \dots, N-1$. One might assume, and would be mainly correct, that nothing is that different in the disk and annulus regimes if the points are distant from r_0 . The following results are direct adaptations of the previous arguments.

Theorem 21. *For $N-1 \geq j \geq m_N$, asymptotically on N ,*

$$h_j = \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left(1 + \frac{1}{N} \mathfrak{B}(r_{\tau(j)}) + \epsilon_{N,1} \right)$$

where

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3} \frac{5}{24},$$

and

$$\epsilon_{N,1} := O(j^{-\frac{3}{2}} (\ln N)^\alpha)$$

for some $\alpha > 0$.

To actually expand on the details, we need some useful facts. We will make a lot of use of the behavior for r_τ , expressed in Proposition 10.

Proposition 10. *Due to strict sub-harmonicity of Q in a neighborhood of the droplet, r_τ , defined as before, satisfies*

$$r_\tau = \left(\frac{2\tau}{q''(0)} \right)^{\frac{1}{2}} + O(\tau) = \left(\frac{4\tau}{V_0^{(2)}(0)} \right)^{\frac{1}{2}} + O(\tau)$$

or more simply

$$r_\tau = O\left(\tau^{\frac{1}{2}}\right)$$

as $\tau \rightarrow 0$.

Proof. Begin defining a multivariate function

$$F(\tau, r) = rq'(r) - 2\tau$$

such that

$$\begin{cases} F(0, 0) = rq'(r) - 2\tau = 0, \\ \partial_r F(\tau, r)|_{(0,0)} = q'(r) + q''(r) \cdot r|_{(0,0)} = 0 \end{cases}$$

since q has critical point at $r_\tau = 0$. This means that the assumptions in the Implicit Function Theorem 9 do not hold. We must use the *blow-up* technique. Consider a auxiliary variable $v := v(r)$ such that $v(r)\tau = r^2$ and look for solutions of the redefined

$$F(\sqrt{v(r)\tau}, \tau) := F'(v, \tau) = \frac{\sqrt{v\tau} \cdot q'(\sqrt{v\tau})}{2\tau} - 1$$

around $(v_0, 0)$, where v_0 is such that $F'(v_0, 0) = 0$. Note that, expanding $q'(r)$ in Taylor around $r = 0$ give us

$$q'(\sqrt{v\tau}) = q''(0) \cdot \sqrt{v\tau} + O(v\tau),$$

which, applied in the previous function,

$$F'(v, \tau) = \frac{q''(0)}{2} v - 1 + O(v^{3/2}\tau).$$

Now, we apply the Implicit Function Theorem on F' noting that the point v_0 must satisfy

$$v_0 = \frac{2}{q''(0)}$$

and that $q''(0) > 0$ by assumption of the global minimum. Then

$$\partial_v F'(v, \tau)|_{(v_0, 0)} = \frac{q''(0)}{2} > 0.$$

Then, the theorem states that there is a bijective function $v(r\tau) \in C^2$ defined in a neighborhood to $\tau = 0$, for which

$$F'(v, \tau) = 0 \quad \text{and} \quad v(0) = v_0.$$

However, $F'(v, \tau) = 0$ means precisely that

$$\sqrt{v\tau} \cdot q'(\sqrt{v\tau}) = 2\tau \quad \text{with} \quad r = r(\tau) = \sqrt{v(\tau)\tau}.$$

Leading us to believe that locally at v_0

$$r(\tau) = \left(\frac{2\tau}{q''(0)} \right)^{\frac{1}{2}} + O(\tau)$$

for some $\tau \rightarrow 0$. Next to zero, all higher powers in the expansion are just order τ . $\square$

To actually go on and prove the stated theorem we start as usual dividing the integral into two sections

$$\begin{aligned} h_j &= \int_0^\infty 2r^{2j+1} e^{-Nq(r)} dr = \int_0^\infty 2r e^{-NV_{\tau(j)}(r)} dr, \\ &= \int_{r_{\tau(j)-\delta_N}}^{r_{\tau(j)+\delta_N}} 2r e^{-NV_{\tau(j)}(r)} dr + \int_{|r-r_{\tau(j)}|>\delta_N} 2r e^{-NV_{\tau(j)}(r)} dr =: (1) + (2). \end{aligned} \quad (6.16)$$

Let's start with a result on the outer integral, which we should be able to control the growth of.

Lemma 8. *Consider the case $m_N \leq j \leq N-1$. If we choose M to be such that*

$$\int_{|r-r_j|>\delta_N} \left(e^{-q(r)} r^2 \right)^M r dr < \infty,$$

then, with (2) as in Equation 6.16, for the error-bound integral we have the asymptotic for N given by

$$(2) := \int_{|r-r_{\tau(j)}|>\delta_N} 2r e^{-NV_{\tau(j)}(r)} dr = e^{-NV_{\tau(j)}(r_{\tau(j)})} O\left(e^{-c_2(\ln(N))^2}\right).$$

for some positive constant c_2 .

Proof. Note that for $d \geq 1$, we can choose C_1 a constant uniformly for all τ and write

$$|V_\tau^{(d)}(r_\tau)| = |q^{(d)}(r_\tau) + (-1)^d 2\tau(d-1)! r_\tau^{-d}| \leq C_1(1 + \tau^{-\frac{d}{2}+1}).$$

Firstly, for $m_N \leq j < N$, we have

$$r_{\tau(j+1/2)} - r_{\tau(j)} = O((jN)^{-\frac{1}{2}}).$$

Since the potential is increasing in (r_τ, ∞) , the Taylor series expansion for V_τ is for all $r > r_{\tau(j)} + \delta_N$

$$\begin{aligned} V_{\tau(j+1/2)}(r) &\geq V_{\tau(j+1/2)}(r_{\tau(j)} + \delta_N), \\ &= V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) + \frac{1}{2} V_{\tau(j+1/2)}^{(2)}(r_{\tau(j+1/2)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1/2)})^2 + O(\tau(j)^{-1/2} \delta_N^3). \end{aligned} \quad (6.17)$$

where $O(\tau(j)^{-1/2} \delta_N^3) = O(N^{-1} j^{-1/2} (\ln(N))^3)$. Similarly, since V_τ is decreasing in $(0, r_\tau)$, we have for all $r < r_{\tau(j)} - \delta_N$

$$\begin{aligned} V_{\tau(j+1/2)}(r) &\geq V_{\tau(j+1/2)}(r_{\tau(j)} - \delta_N), \\ &= V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) + \frac{1}{2} V_{\tau(j+1/2)}^{(2)}(r_{\tau(j+1/2)})(r_{\tau(j)} - \delta_N - r_{\tau(j+1/2)})^2 + O(\tau(j)^{-1/2} \delta_N^3). \end{aligned} \quad (6.18)$$

Using the Taylor series for V_τ again, we have

$$\begin{aligned} V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) &= V_{\tau(j)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) - \frac{1}{N} \ln r_{\tau(j+1/2)}, \\ &= O((jN)^{-1}) - \frac{1}{N} \ln r_{\tau(j+1/2)}. \end{aligned}$$

Noting that

$$V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) = [V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) - V_{\tau(j+1/2)}(r)] + [V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)})]$$

one can rewrite using Equations 6.18 and 6.17,

$$V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)}) = -\frac{1}{N} \ln r_{\tau(j+1/2)} + O\left(\frac{\log(N)^2}{N}\right).$$

Thus, it follows from the last observations that

$$\begin{aligned} e^{NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_{\tau(j)}|>\delta_N} 2r^{2j+1} e^{-Nq(r)} dr &= \int_{|r-r_{\tau(j)}|>\delta_N} 2e^{-N(V_{\tau(j+1/2)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\ &= \int_{|r-r_{\tau(j)}|>\delta_N} 2e^{-(N-M)(V_{\tau(j+1/2)}(r)-V_{\tau(j)}(r_{\tau(j)}))} e^{-M(V_{\tau(j+1/2)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\ &\leq e^{-c_1(\ln(N))^2} e^{\frac{N-M}{N} \ln(r_{\tau(j+1/2)})} \int_{|r-r_{\tau(j)}|>\delta_N} 2e^{-M(V_{\tau(j+1/2)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\ &= O(e^{-c_2(\ln(N))^2}). \end{aligned}$$

for some constants $c_1, c_2 > 0$ and M such that the integral above converges. The details taken here are very similar to the ones in the annulus case. $\square$

Lemma 9. Consider the case $m_N \leq j \leq N-1$. For the value-contributing integral we have the asymptotic on N given by

$$(1) := \int_{r_{\tau(j)} - \delta_N}^{r_{\tau(j)} + \delta_N} 2r e^{-NV_{\tau(j)}(r)} dr = \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left(1 + \frac{1}{N} \mathfrak{B}(r_{\tau(j)}) + \epsilon_{N,1} \right)$$

where (1) is as in Equation 6.16, $\mathfrak{B}(r_{\tau(j)})$ is as in Theorem 20 and

$$\epsilon_{N,1} := O(j^{-\frac{3}{2}}(\ln N)^\alpha)$$

for some $\alpha > 0$.

Proof. We now need to examine the integral near the critical point. That is,

$$\begin{aligned} & \int_{r_{\tau(j)} - \delta_N}^{r_{\tau(j)} + \delta_N} 2r e^{-NV_{\tau(j)}(r)} dr \\ &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{-\delta_N}^{\delta_N} 2(r_{\tau(j)} + t) e^{-N \left(\frac{1}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})(t)^2 + \frac{1}{3!} V_{\tau(j)}^{(3)}(r_{\tau(j)})(t)^3 + \frac{1}{4!} V_{\tau(j)}^{(4)}(r_{\tau(j)})(t)^4 + O(\tau(j)^{-\frac{3}{2}} t^5) \right)} dt, \\ &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{-\sqrt{N}\delta_N}^{\sqrt{N}\delta_N} 2e^{-\frac{1}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})t^2} \left(r_{\tau(j)} + \frac{t}{\sqrt{N}} \right) \left(1 - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})t^3}{6\sqrt{N}} \right. \\ & \quad \left. - \frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})t^4}{24N} + \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2 t^6}{72N} + \epsilon_{N,1} \right) dt, \end{aligned}$$

where $\epsilon_{N,1} = O(j^{-\frac{3}{2}}(\ln N)^\alpha)$ for some $\alpha > 0$. From here, the expression can be directly identified with Equation 6.8 and the techniques for getting the result are one-to-one with the ones already done in that case. then, distributing the integral,

$$\begin{aligned} (1) = 2r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} & \left[\left(-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24N} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6r_{\tau(j)}N} \right) \int_{-\ln(N)}^{\ln(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} y^4 dy \right. \\ & \left. - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{72N} \int_{-\ln(N)}^{\ln(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} y^6 dy + \int_{-\ln(N)}^{\ln(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} dy + \epsilon_{N,1} \right]. \end{aligned}$$

At this point, the result can be directly evaluated using the Gaussian integrals in Proposition 9. So that we can finally substitute the values to get

$$\begin{aligned} (1) = \sqrt{2\pi} r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} & \left[\left(-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24N} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6r_{\tau(j)}N} \right) \frac{3}{16} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{5}{2}} \right. \\ & \left. - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{72N} \frac{15}{64} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{7}{2}} + \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} + \epsilon_{N,1} \right], \end{aligned}$$

and, rearranging the terms we should get our result. $\square$

Proof. (Theorem 21) This is now only a matter to notice that (2) is absorbed in to the error term of (1) and the result follows. $\square$

So far, nothing is considerably different from both cases as we are dealing with point distant from the origin. We must now deal with the more delicate case, for $j \leq m_N$. The behavior for h_j , in this regimen, is expressed by the following theorem.

Theorem 22. *For $0 \leq j < m_N$, we will have the following asymptotic for the norming constant*

$$h_j = e^{-Nq(0)} \left[N^{-(j+1)} \Gamma(j+1) \left(\frac{1}{2} q''(0) \right)^{-(j+1)} \left(1 + O(N^{-\frac{1}{2}} (j+1/2)^{\frac{3}{2}} (\ln N)^3) \right) + \epsilon_N(j) \right]$$

where Γ is the usual Gamma function and

$$\epsilon_N(j) := O \left(e^{-c_2(j+1)(\ln N)^2} \left(\frac{j+1}{N} (\ln(N))^2 \right)^{(j+1) \frac{N-M}{N}} \right)$$

with $c_2 > 0$ a constant.

Proof. For $j < m_N$, we need to define some $r_\tau^* := r_\tau \ln(N)$ from which we separate the integral in a important part close to the origin and another part which has a lesser order. Consider then the decomposition

$$h_j = \int_0^\infty 2r^{2j+1} e^{-Nq(r)} dr = \int_0^{r_{\tau(j+1)}^*} 2r^{2j+1} e^{-Nq(r)} dr + \int_{r_{\tau(j+1)}^*}^\infty 2r^{2j+1} e^{-Nq(r)} dr.$$

Due to strict sub-harmonicity of Q in a neighborhood of the droplet, r_τ satisfies $r_\tau = O(\tau^{\frac{1}{2}})$ as $\tau \rightarrow 0$, this is given by Proposition 10. Since V_τ has a global minimum at $r = r_\tau$ and increases for larger values, for all $r > r_{\tau(j+1)}^*$ we have

$$\begin{aligned} V_{\tau(j+1)/2}(r) &\geq V_{\tau(j+1)}(r)(r_{\tau(j+1)}^*) = q(r_{\tau(j+1)}^*) - \frac{2j+1}{N} \ln(r_{\tau(j+1)}^*), \\ &= q(0) + \frac{1}{2} q''(0) (r_{\tau(j+1)}^*)^2 - \frac{2j+1}{N} \ln(r_{\tau(j+1)}^*) + O(r_{\tau(j+1)}^*)^3. \end{aligned}$$

Note that $q'(0) = 0$ since $\Delta Q(z) \in (0, \infty)$ near the origin and $\Delta Q(z) \propto q'(0)/r + q''(0)$. Take

$$\begin{aligned} e^{-Nq(0)} \int_{r_{\tau(j+1)}^*}^\infty 2r^{2j+1} e^{-Nq(r)+Nq(0)} dr &= e^{-Nq(0)} \int_{r_{\tau(j+1)}^*}^\infty 2e^{(2j+1)\ln(r)-Nq(r)+Nq(0)} dr, \\ &= e^{-Nq(0)} \int_{r_{\tau(j+1)}^*}^\infty 2e^{-N(-\frac{2}{N}(j+\frac{1}{2})\ln(r)-q(r)+q(0))} dr, \\ &= \int_{r_{\tau(j+1)}^*}^\infty 2e^{-N(V_{\tau(j+1)/2}(r)-q(0))} dr \end{aligned}$$

and, using the inequality explicited above

$$\begin{aligned}
e^{-NV_{\tau(j+1)}(r)+Nq(0)} &= e^{-NV_{\tau(j+1)}(r)+Nq(0)+MV_{\tau(j+1)}(r)-MV_{\tau(j+1)}(r)}, \\
&= e^{-(N-M)V_{\tau(j+1)}(r)+Nq(0)-MV_{\tau(j+1)}(r)}, \\
&\leq e^{-(N-M)V_{\tau(j+1)}(r^*)+Nq(0)-MV_{\tau(j+1)}(r)}, \\
&= e^{-(N-M)(V_{\tau(j+1)}(r^*)-q(0))} e^{-M(V_{\tau(j+1)}(r)-q(0))}.
\end{aligned}$$

Then

$$\begin{aligned}
&\int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-N(V_{\tau(j+1)}(r)-q(0))} dr \\
&\leq e^{-(N-M)V_{\tau(j+1)}(r^*)} e^{(N-M)q(0)} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr, \\
&= e^{-(N-M)\left[q(r_{\tau(j+1)}^*)-\frac{2j+2}{N}\ln(r_{\tau(j+1)}^*)\right]} e^{(N-M)q(0)} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr, \\
&= e^{-(N-M)q(r_{\tau(j+1)}^*)} r_{\tau(j+1)}^{*(2j+2)\frac{N-M}{N}} e^{(N-M)q(0)} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr, \\
&= e^{-(N-M)\left[\frac{q''(0)}{2}(r_{\tau(j+1)}^*)^2+O(r_{\tau(j+1)}^*)^3\right]} r_{\tau(j+1)}^{*(2j+2)\frac{N-M}{N}} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr, \\
&\leq e^{-\frac{(N-M)}{2}q''(0)(r_{\tau(j+1)}^*)^2} r_{\tau(j+1)}^{*(2j+2)\frac{N-M}{N}} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr, \\
&= e^{-c_1\frac{(N-M)}{2}r_{\tau(j+1)}^2\ln(N)^2} r_{\tau(j+1)}^{*(2j+2)\frac{N-M}{N}} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr := \epsilon_N(j)
\end{aligned}$$

for some $c_1 > 0$ and M as given in the other proof such that the integral converges. Since $r_{\tau} = O(\tau^{\frac{1}{2}})$, there is a constant $c_2 > 0$ such that as N goes to infinity

$$\begin{aligned}
\epsilon_N(j) &= O\left(e^{-c_1\frac{(N-M)}{2}\tau(j+1)\ln(N)^2} r_{\tau(j+1)}^{*(2j+2)\frac{N-M}{N}}\right), \\
&= O\left(e^{-c_2(j+1)\ln(N)^2} r_{\tau(j+1)}^{*(2j+2)\frac{N-M}{N}}\right), \\
&= O\left(e^{-c_2(j+1)(\ln N)^2} \left(\frac{j+1}{N}(\ln N)^2\right)^{(j+1)\frac{N-M}{N}}\right).
\end{aligned}$$

Therefore we obtain

$$\begin{aligned}
h_j &= \int_0^{\infty} 2r^{2j+1} e^{-Nq(r)} dr = \int_0^{r_{\tau(j+1)}^*} 2r^{2j+1} e^{-Nq(r)} dr + e^{-Nq(0)} \epsilon_N(j), \\
&= \int_0^{r_{\tau(j+1)}^*} 2r^{2j+1} e^{-N\left(q(0)+\frac{1}{2}q''(0)r^2+O(N(r_{\tau(j+1)}^*)^3)\right)} dr + e^{-Nq(0)} \epsilon_N(j), \\
&= e^{-Nq(0)} \left[\int_0^{r_{\tau(j+1)}^*} 2r^{2j+1} e^{-\frac{1}{2}q''(0)r^2N} dr \left(1 + O(N(r_{\tau(j+1)}^*)^3)\right) + \epsilon_N(j) \right], \\
&= e^{-Nq(0)} \left[N^{-(j+1)} \Gamma(j+1) \left(\frac{1}{2}q''(0)\right)^{-(j+1)} \left(1 + O(N^{-\frac{1}{2}}(j+1/2)^{\frac{3}{2}}(\ln N)^3)\right) + \epsilon_N(j) \right]
\end{aligned}$$

where Γ is the usual gamma function with integral representation

$$\Gamma(j+1) = \int_0^\infty t^j e^{-t} dt$$

and we used $t = 1/2q''(0)r^2N$.

□

So now we need to return to Equation 6.4 and calculate each term for $\sum_{j=0}^{N-1} \ln h_j$ using the expansion in terms of Theorem 21 and 22 for the expression

$$\sum_{j=0}^N \ln(h_j) = \sum_{j=0}^{m_N-1} \ln(h_j) + \sum_{j=m_N}^{N-1} \ln(h_j).$$

Theorem 23. *We have the following asymptotics on N for the summation of $\ln(h_j)$ from 0 to $N-1$,*

$$\begin{aligned} \sum_{j=0}^{N-1} \ln(h_j) = & -N^2 I[\mu_Q] - \frac{1}{2} N \ln(N) - \frac{N}{2} (E_Q[\mu_Q] - \ln(2\pi)) - \frac{\ln(N)}{12} \\ & + \zeta'(-1) + F_V[D_{r_1}] + O(N^{-12}(\ln N)^3) \end{aligned}$$

where $I_Q[\mu_Q]$, $F_V[D_{r_1}]$ and $E_Q[\mu_Q]$ are as in Theorem 20.

To prove this theorem, we must give the asymptotics for both summations. This means we need both Lemmas 10, referring to the sum from powers 0 to m_N-1 , and 11, referring to the sum from m_N to $N-1$. We start by the lower power terms.

Lemma 10. *We have the following asymptotic for the summation of $\ln(h_j)$ from 0 to m_N-1 ,*

$$\begin{aligned} \sum_{j=0}^{m_N-1} \ln(h_j) = & -m_N N q(0) - \frac{m_N(m_N+1)}{2} \left(\ln(N) + \ln\left(\frac{1}{2}q''(0)\right) \right) + \frac{m_N^2 \ln(m_N)}{2} - \frac{3}{4} m_N^2 \\ & + \frac{\ln(2\pi)m_N}{2} - \frac{\ln(m_N)}{12} + \zeta'(-1) + O(m_N^{-2} + N^{-\frac{1}{2}(1-5\epsilon)}(\ln(N))^3). \end{aligned}$$

where ϵ is as in the definition for m_N .

Proof. The first term in the right is straightforwardly determined, we need only to write, using

Theorem 22,

$$\begin{aligned}
\sum_{j=0}^{m_N-1} \ln(h_j) &= -m_N N q(0) - \frac{m_N(m_N+1)}{2} \left(\ln(N) + \ln\left(\frac{1}{2} q^{(2)}(0)\right) \right) \\
&\quad + \ln(G(m_N+1)) + \mathcal{O}(N^{-\frac{1}{2}(1-5\epsilon)} (\ln(N))^3), \\
&= -m_N N q(0) - \frac{m_N(m_N+1)}{2} \left(\ln(N) + \ln\left(\frac{1}{2} q^{(2)}(0)\right) \right) + \frac{m_N^2 \ln(m_N)}{2} - \frac{3}{4} m_N^2 \\
&\quad + \frac{\ln(2\pi) m_N}{2} - \frac{\ln(m_N)}{12} + \zeta'(-1) + \mathcal{O}(m_N^{-2} + N^{-\frac{1}{2}(1-5\epsilon)} (\ln(N))^3)
\end{aligned}$$

where G is the Barnes G -function defined recursively as

$$G(z+1) = \Gamma(z) G(z),$$

and we have used the asymptotic for the G-Barnes function

$$\ln(G(z+1)) = \frac{z^2 \ln(z)}{2} - \frac{3}{4} z^2 + \frac{\ln(2\pi) z}{2} - \frac{\ln(z)}{12} + \zeta'(-1) + \mathcal{O}\left(\frac{1}{z^2}\right).$$

□

Lemma 11. *We have the following asymptotic for the summation of $\ln(h_j)$ from m_N to $N-1$,*

$$\begin{aligned}
\sum_{j=m_N}^{N-1} \ln(h_j) &= -N^2 I[\mu_Q] + \frac{(N-m_N)}{2} \ln \frac{2\pi}{N} + N m_N q(0) + \frac{3}{4} m_N^2 - \frac{N}{2} E_Q[\mu_Q] \\
&\quad - \frac{m_N}{2} (m_N+1) \ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{2} \left(m_N^2 - \frac{1}{6}\right) \ln\left(\frac{m_N}{N}\right) + F_V[D_{r_1}] + \mathcal{O}(N^{-12} (\ln N)^3),
\end{aligned}$$

where $I_Q[\mu_Q]$, $F_V[D_{r_1}]$ and $E_Q[\mu_Q]$ are as in Theorem 20.

The second summation is a bit more detailed, we use the previously defined asymptotics in Theorem 21,

$$\ln(h_j) = -N V_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \ln\left(\frac{8\pi r_{\tau(j)}^2}{N V_{\tau(j)}^{(2)}(r_{\tau(j)})}\right) + \frac{1}{N} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(j^{-\frac{3}{2}} (\ln(N))^\alpha)$$

and need to determine the asymptotic of three terms in order to get the full summation,

$$-N \sum_{j=m_N}^{N-1} V_{\tau(j)}(r_{\tau(j)}), \quad \sum_{j=m_N}^{N-1} \ln\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right) \quad \text{and} \quad \sum_{j=m_N}^{N-1} \ln(r_{\tau(j)}).$$

This is done in Proposition 11, 12 and 13.

Proposition 11. *The following asymptotic holds under the same context as Lemma 11.*

$$\begin{aligned} -N \sum_{j=m_N}^{N-1} V_{\tau(j)}(r_{\tau(j)}) &= -N^2 I_Q[\mu_Q] + N U_{\mu_Q}(0) + \frac{1}{6} \ln(r_1) + N m_N q(0) + \frac{3}{4} m_N^2 \\ &\quad - \frac{1}{2} \left(m_N^2 - m_N + \frac{1}{6} \right) \ln \left(\frac{4m_N}{N V_0^{(2)}(0)} \right) - \frac{m_N}{2} + \mathcal{O} \left(N^{-\frac{1}{2}(3-5\epsilon)} \right). \end{aligned}$$

Here, $I_Q[\mu_Q]$ is as in Theorem 20, ϵ is as in the definition for m_N , and

$$2U_{\mu_Q}(0) := q(r_1) - 2\ln(r_1) - q(0).$$

Proof. We proceed as before and use the Euler-Maclaurin formula to write

$$\begin{aligned} \sum_{j=m_N}^{N-1} V_{\tau(j)}(r_{\tau(j)}) &= \int_{m_N}^N V_{\tau(t)}(r_{\tau(t)}) dt - \frac{1}{2} (V_{\tau(N)}(r_{\tau(N)}) - V_{\tau(m_N)}(r_{\tau(m_N)})) + \\ &\quad \frac{1}{12} [\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=m_N}] + \mathcal{O}(N^{-3}). \end{aligned}$$

Then, we are able to rewrite, setting $s = r_{\tau(j)}$,

$$\begin{aligned} \int_{m_N}^N V_{\tau(t)}(r_{\tau(t)}) dt &= \int_{m_N}^N (q(r_{\tau(t)}) - 2\tau(t) \ln(r_{\tau(t)})) dt, \\ &= \frac{1}{2} \int_{r_{\tau(m_N)}}^{r_1} (q(s) - s q'(s) \ln(s)) s V^{(2)}(s) N ds, \\ &= N \left[\frac{1}{2} \int_{r_{\tau(m_N)}}^{r_1} q(s) s V^{(2)}(s) ds - \frac{1}{2} \int_{r_{\tau(m_N)}}^{r_1} s^2 q'(s) \ln(s) V^{(2)}(s) ds \right]. \end{aligned}$$

Using that $r_1 q'(r_1) = 2$, apply integration by parts on the second term and get

$$\begin{aligned}
& \frac{1}{2} \int_{r_{\tau(m_N)}}^{r_1} s^2 q'(s) \ln(s) V^{(2)}(s) ds \\
&= \frac{1}{2} \int_{r_{\tau(m_N)}}^{r_1} s^2 q'(s) \ln(s) \frac{1}{s} (s q'(s))' ds, \\
&= \frac{1}{4} \int_{r_{\tau(m_N)}}^{r_1} ((s q'(s))^2)' \ln(s) ds, \\
&= \frac{1}{4} (s q'(s))^2 \ln(s) \Big|_{r_{\tau(m_N)}}^{r_1} - \frac{1}{4} \int_{r_{\tau(m_N)}}^{r_1} (s q'(s))^2 \frac{1}{s} ds, \\
&= \ln(r_1) - \frac{1}{4} \int_{r_{\tau(m_N)}}^{r_1} s q'(s)^2 ds - \frac{m_N^2}{N^2} \ln(r_{\tau(m_N)}), \\
&= \ln(r_1) - \frac{1}{4} (s q'(s) q(s)) \Big|_{r_{\tau(m_N)}}^{r_1} + \frac{1}{4} \int_{r_{\tau(m_N)}}^{r_1} q(s) (s q'(s))' ds - \frac{m_N^2}{N^2} \ln(r_{\tau(m_N)}), \\
&= \ln(r_1) - \frac{1}{2} q(r_1) + \frac{1}{4} \int_{r_{\tau(m_N)}}^{r_1} q(s) (s q'(s))' ds - \frac{m_N^2}{N^2} \ln(r_{\tau(m_N)}) + \frac{m_N}{2N} q(r_{\tau(m_N)}).
\end{aligned}$$

Leaving us with

$$\begin{aligned}
& N \left[\frac{1}{4} \int_{r_{\tau(m_N)}}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) - \frac{m_N^2}{N^2} \ln(r_{\tau(m_N)}) + \frac{m_N}{2N} q(r_{\tau(m_N)}) \right] \\
&:= N I_Q[\mu_Q] - \frac{1}{4} \int_{r_0}^{r_{\tau(m_N)}} q(s) s V^{(2)}(s) ds - \frac{m_N^2}{N} \ln(r_{\tau(m_N)}) + \frac{m_N}{2} q(r_{\tau(m_N)}).
\end{aligned}$$

where we can expand some terms using the following results given by Proposition 10

$$\ln(r_{\tau(m_N)}) = \frac{1}{2} \ln \left(\frac{4\tau(m_N)}{V_0^{(2)}(0)} \right) + \mathcal{O}(\tau(m_N)^{\frac{1}{2}}) = \frac{1}{2} \ln \left(\frac{4\tau(m_N)}{V_0^{(2)}(0)} \right) + \mathcal{O}(N^{-\frac{1}{2}(1-\epsilon)}), \quad (6.19)$$

$$q(r_{\tau(m_N)}) = q(0) + \frac{1}{2} q''(0) (r_{\tau(m_N)})^2 + \mathcal{O}((\tau(m_N)^{\frac{3}{2}})) = q(0) + \tau(m_N) + \mathcal{O}(N^{-\frac{3}{2}(1-\epsilon)}), \quad (6.20)$$

and, using integration by parts we can solve the integral

$$\begin{aligned}
\frac{1}{4} \int_{r_0}^{r_{\tau(m_N)}} q(s) s V^{(2)}(s) ds &= \frac{1}{4} \int_{r_0}^{r_{\tau(m_N)}} q(s) (s q'(s))' ds, \\
&= \frac{1}{2} \tau(m_N) q(r_{\tau(m_N)}) - \frac{1}{4} \int_0^{r_{\tau(m_N)}} s (q'(s))^2 ds, \\
&= \frac{1}{2} \tau(m_N) q(r_{\tau(m_N)}) - \frac{1}{4} (q''(0))^2 \int_0^{r_{\tau(m_N)}} s^3 ds + \mathcal{O}(r_{\tau(m_N)}^5), \\
&= \frac{1}{2} \tau(m_N) q(r_{\tau(m_N)}) - \frac{1}{4} \tau(m_N)^2 + \mathcal{O}(N^{-5/2(1-\epsilon)}).
\end{aligned}$$

Therefore, for this first term, gathering the above and expanding using Proposition 10 and Equation 6.20,

$$\begin{aligned} \int_{m_N}^N V_{\tau(t)}(r_{\tau(t)}) dt &= NI_Q[\mu_Q] - \frac{1}{2} \tau(m_N) q(r_{\tau(m_N)}) - \frac{1}{4} \tau(m_N)^2 + O(N^{-5/2(1-\epsilon)}) \\ &\quad - \frac{m_N^2}{2N} \ln \left(\frac{4\tau(m_N)}{V_0^{(2)}(0)} \right) + O(N^{-\frac{3}{2}(1-\epsilon)}) + \frac{m_N}{2} (q(0) + \tau(m_N)) + O(N^{-\frac{1}{2}(1-\epsilon)}), \\ &= NI_Q[\mu_Q] - m_N q(0) - \frac{3}{4} \frac{m_N^2}{N} + \frac{m_N^2}{2N} \ln \left(\frac{4m_N}{NV_0^{(2)}(0)} \right) + O(N^{-\frac{1}{2}(3-5\epsilon)}). \end{aligned}$$

For the other terms in the Euler-Maclaurin formula we have

$$\begin{aligned} V_{\tau(N)}(r_{\tau(N)}) - V_{\tau(m_N)}(r_{\tau(m_N)}) &= q(r_1) - 2 \ln(r_1) - q(r_{\tau(m_N)}) + 2\tau(m_N) \ln(r_{\tau(m_N)}), \\ &= q(r_1) - 2 \ln(r_1) - q(0) - \frac{m_N}{N} + \frac{m_N}{N} \ln \left(\frac{4m_N}{NV_0^{(2)}(0)} \right) + O(N^{-3/2(1-\epsilon)}), \\ &:= 2U_{\mu_Q}(0) - \frac{m_N}{N} + \frac{m_N}{N} \ln \left(\frac{4m_N}{NV_0^{(2)}(0)} \right) + O(N^{-3/2(1-\epsilon)}), \end{aligned}$$

and,

$$\begin{aligned} \partial_t(V_{\tau(t)}(r_{\tau(t)}))|_{t=N} - \partial_t(V_{\tau(t)}(r_{\tau(t)}))|_{t=m_N} &= \frac{2}{N} (\ln(r_{\tau(m_N)}) - \ln(r_1)), \\ &= \frac{1}{N} \ln \left(\frac{4m_N}{NV_0^{(2)}(0)} \right) - \frac{2}{N} \ln(r_1) + O(N^{-1/2(3-\epsilon)}). \end{aligned}$$

Finally, gathering terms, we get the lemma. $\square$

Proposition 12. *We have following asymptotic*

$$\sum_{j=m_N}^{N-1} \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) = NE_Q(\mu_Q) - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - m_N \ln \left(\frac{V_0^{(2)}(0)}{4} \right) + O(N^{-1/2(1-\epsilon)}),$$

where $E_Q[\mu_Q]$ is as in Theorem 20 and ϵ is as in the definition for m_N .

Proof. Again, we start with the Euler-Maclaurin formula

$$\begin{aligned} \sum_{j=m_N}^{N-1} \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) &= \int_{m_N}^N \ln \left(\frac{V_{\tau(t)}^{(2)}(r_{\tau(t)})}{4} \right) dt - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})} \right) + \\ &\quad \frac{1}{12} \left[\partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=N} - \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=m_N} \right] + O(N^{-3}). \end{aligned}$$

We first tackle the integral on the right hand side. Changing variables $s = r_{\tau(t)}$ we have

$$\begin{aligned}
 \int_{m_N}^N \ln \left(\frac{V_{\tau(t)}^{(2)}(r_{\tau(t)})}{4} \right) dt &= \frac{N}{2} \int_{r_{\tau(m_N)}}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (sq'(s))' ds, \\
 &= \frac{N}{2} \int_{r_{\tau(m_N)}}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) s V^{(2)}(s) ds, \\
 &= \frac{N}{2} \left[\int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) s V^{(2)}(s) ds - \int_{r_0}^{r_{\tau(m_N)}} \ln \left(\frac{V^{(2)}(s)}{4} \right) s V^{(2)}(s) ds \right], \\
 &= \frac{N}{2} \left[2E_Q[\mu_Q] - \int_{r_0}^{r_{\tau(m_N)}} \ln \left(\frac{V^{(2)}(s)}{4} \right) s V^{(2)}(s) ds \right],
 \end{aligned}$$

where one can also compute

$$\begin{aligned}
 &\int_{r_0}^{r_{\tau(m_N)}} \ln \left(\frac{V^{(2)}(s)}{4} \right) s V^{(2)}(s) ds \\
 &= \int_{r_0}^{r_{\tau(m_N)}} \ln \left(\frac{V_0^{(2)}(0)}{4} + \frac{V_0^{(2)}(0)}{4} s + \mathcal{O}(s^2) \right) s \left(V_0^{(2)}(0) + V_0^{(2)}(0)s + \mathcal{O}(s^2) \right) ds, \\
 &= \int_{r_0}^{r_{\tau(m_N)}} \ln \left(\frac{V_0^{(2)}(0)}{4} \right) s V_0^{(2)}(0) ds + \mathcal{O}(r_{\tau(m_N)}^3), \\
 &= \ln \left(\frac{V_0^{(2)}(0)}{4} \right) r_{\tau(m_N)}^2 V_0^{(2)}(0) + \mathcal{O}(r_{\tau(m_N)}^3), \\
 &= \frac{2m_N}{N} \ln \left(\frac{V_0^{(2)}(0)}{4} \right) + \mathcal{O}(N^{-3(1-\epsilon)})
 \end{aligned}$$

noting the definition for V and Proposition 10. Furthermore, we know

$$\begin{aligned}
 \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)}) &= \partial_V \left(\ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right) \frac{1}{4} \partial_r \left(V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right) \partial_\tau(r_{\tau(t)}), \\
 &= \frac{2\partial_r V^{(2)}(r_{\tau(t)})}{Nr_{\tau(t)} \left(V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right)^2}, \\
 &= \mathcal{O}(N/m_N)^{-1/2}.
 \end{aligned}$$

Therefore, for the third term on the right hand side

$$\begin{aligned}
 \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=N} - \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=m_N} &= \frac{2\partial_r V^{(2)}(r_1)}{Nr_1 \left(V_1^{(2)}(r_1) \right)^2} - \mathcal{O}(N/m_N)^{-1/2}, \\
 &= \mathcal{O}(N/m_N)^{-1/2} = \mathcal{O}(N^{-1/2(1-\epsilon)}).
 \end{aligned}$$

Then, simply expanding the series on $V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})$, we can write,

$$\sum_{j=0}^{N-1} \ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) = NE_Q(\mu_Q) - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)} \right) - m_N \ln \left(\frac{V_0^{(2)}(0)}{4} \right) + \mathcal{O}(N^{-1/2(1-\epsilon)}).$$

□

Proposition 13. *The following asymptotic holds under the same context as Lemma 11.*

$$\sum_{j=m_N}^{N-1} \ln(r_{\tau(j)}) = N \left[-U_{\mu_Q}(0) + \frac{1}{2} \tau(m_N) \right] - \frac{1}{2} \ln(r_1) - \frac{2m_N - 1}{4} \ln \left(\frac{4\tau(m_N)}{V_0^{(2)}(0)} \right) + \mathcal{O}(N^{-\epsilon})$$

where $U_{\mu_Q}(0)$ is as in Proposition 11.

Proof. As usual, we start with the Euler-Maclaurin formula.

$$\sum_{j=m_N}^{N-1} \ln(r_{\tau(m_N)}) = \int_{m_N}^N \ln(r_{\tau(t)}) dt - \frac{1}{2} \ln \left(\frac{r_1}{r_{\tau(m_N)}} \right) + \frac{1}{12} [\partial_t \ln r_{\tau(t)}|_{t=N} - \partial_t \ln r_{\tau(t)}|_{t=m_N}]$$

where we can directly note that

$$\begin{aligned} \partial_t \ln r_{\tau(t)}|_{t=m_N} &= \partial_r \ln r_{\tau(t)} \partial_\tau r_{\tau(t)} \partial_t \tau(t)|_{t=m_N} = \frac{2}{r_{\tau(t)}^2 V_{\tau(t)}^{(2)}(r_{\tau(t)}) N} \Big|_{t=m_N} = \mathcal{O}(m_N^{-1}), \\ \partial_t \ln r_{\tau(t)}|_{t=N} &= \frac{2}{r_{\tau(t)}^2 V_{\tau(t)}^{(2)}(r_{\tau(t)}) N} \Big|_{t=N} = \mathcal{O}(N^{-1}). \end{aligned}$$

Now, for the other terms we start with the first term, the integral, to get using integration by parts

$$\begin{aligned} \int_{m_N}^N \ln(r_{\tau(t)}) dt &= \frac{N}{2} \int_{r_{\tau(m_N)}}^{r_1} \ln(s) (s' q(s))' ds, \\ &= \frac{N}{2} \left[\ln(s) s q'(s) \Big|_{r_{\tau(m_N)}}^{r_1} - \int_{r_{\tau(m_N)}}^{r_1} q'(s) ds \right], \\ &= \frac{N}{2} [\ln(r_1) r_1 q'(r_1) - \ln(r_{\tau(m_N)}) r_{\tau(m_N)} q'(r_{\tau(m_N)}) - q(r_1) + q(r_{\tau(m_N)})], \\ &= \frac{N}{2} [2 \ln(r_1) - 2 \tau(m_N) \ln(r_{\tau(m_N)}) - q(r_1) + q(r_{\tau(m_N)})]. \end{aligned}$$

Ultimately, using Equation 6.19 and Equation 6.20,

$$\begin{aligned} \int_{m_N}^N \ln(r_{\tau(t)}) dt - \frac{1}{2} \ln \left(\frac{r_1}{r_{\tau(m_N)}} \right) + \mathcal{O}(m_N^{-1}) &= N \left[\ln(r_1) - \frac{q(r_1) - q(0)}{2} + \frac{1}{2} \tau(m_N) \right] \\ &\quad - \frac{1}{2} \ln(r_1) - \frac{2m_N - 1}{4} \ln \left(\frac{4\tau(m_N)}{V_0^{(2)}(0)} \right) + \mathcal{O}(N^{-\epsilon}). \end{aligned}$$

□

Finally, we turn our attention to gathering these propositions and proving the original lemma.

Proof. (Lemma 11) We start by stating from Equation 6.15, with the new limits,

$$\begin{aligned} \frac{1}{N} \sum_{j=m_N}^{N-1} \mathfrak{B}(r_{\tau(j)}) = & -\frac{3}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_{\tau(m_N)} V_{\tau(m_N)}^{(3)}(r_{\tau(m_N)})}{V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})} \right) \\ & - \frac{1}{12} \int_{r_{\tau(m_N)}}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(N^{-2}). \end{aligned}$$

We should have by the asymptotic for h_j in Theorem 21, referring to $j > m_N$

$$\begin{aligned} \sum_{j=m_N}^{N-1} \ln(h_j) = & \sum_{j=m_N}^{N-1} \left(-NV_{\tau(j)}(r_{\tau(j)}) + \ln(r_{\tau(j)}) - \frac{1}{2} \ln \frac{1}{4} V_{\tau(j)}^{(2)}(r_{\tau(j)}) + \frac{1}{N} \mathfrak{B}(r_{\tau(j)}) \right) \\ & + \frac{(N-m_N)}{2} \ln \frac{2\pi}{N} + O(m_N^{-1/2} (\ln N)^\alpha) + O(N^{\frac{1}{2}(1-5\epsilon)}). \end{aligned}$$

By the lemmas just proved, we can write as $N \rightarrow \infty$,

$$\begin{aligned} \sum_{j=m_N}^{N-1} \ln(h_j) = & -N^2 I[\mu_Q] + \frac{(N-m_N)}{2} \ln \frac{2\pi}{N} + Nm_N q(0) + \frac{3}{4} m_N^2 - \frac{N}{2} E_Q[\mu_Q] - \frac{m_N}{2} (m_N + 1) \ln \left(\frac{4}{V_0^{(2)}(0)} \right) \\ & - \frac{1}{2} \left(m_N^2 - \frac{1}{6} \right) \ln \left(\frac{m_N}{N} \right) + \frac{1}{6} \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{3} \ln(r_1) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})} \right) \\ & - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_{\tau(m_N)} V_{\tau(m_N)}^{(3)}(r_{\tau(m_N)})}{V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})} \right) - \frac{1}{12} \int_{r_{\tau(m_N)}}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(N^{-12} (\ln N)^3). \end{aligned}$$

Using Proposition 10 we write that

$$r_{\tau(m_N)} = r_0 + O(\tau(m_N)^{\frac{1}{2}}) = O(N^{\frac{1}{2}(\epsilon-1)})$$

and

$$V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)}) = V_0^{(2)}(r_0) + O(N^{(\epsilon-1)}).$$

Then, we rewrite

$$\begin{aligned} \sum_{j=m_N}^{N-1} \ln(h_j) = & -N^2 I[\mu_Q] + \frac{(N-m_N)}{2} \ln \frac{2\pi}{N} + Nm_N q(0) + \frac{3}{4} m_N^2 - \frac{N}{2} E_Q[\mu_Q] \\ & - \frac{m_N}{2} (m_N + 1) \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{2} \left(m_N^2 - \frac{1}{6} \right) \ln \left(\frac{m_N}{N} \right) + F_V[D_{r_1}] + O(N^{-12} (\ln N)^3). \end{aligned}$$

where

$$F_V[D_{r_1}] := \frac{1}{6} \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{3} \ln(r_1) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)} \right) - \frac{1}{32} \frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12} \int_0^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr.$$

□

And here finally we return to the full summation.

Proof. (Theorem 23) Gathering Lemma 10 and Lemma 11, we get

$$\sum_{j=0}^{N-1} \ln(h_j) = -N^2 I[\mu_Q] - \frac{1}{2} N \ln(N) - \frac{N}{2} (E_Q[\mu_Q] - \ln(2\pi)) - \frac{\ln(N)}{12} + \zeta'(-1) + F_V[D_{r_1}] + O(N^{-12}(\ln N)^3).$$

Incredibly, all the terms depending on m_N cancels out. $\square$

We can finally prove Theorem 20.

Proof. (Theorem 20) We restate the previous result

$$\sum_{j=0}^{N-1} \ln(h_j) = -N^2 I_Q[\mu_Q] - \frac{1}{2} N \ln N - \frac{N}{2} (E_Q[\mu_Q] - \ln(2\pi)) - \frac{\ln N}{12} + \zeta'(-1) + F_V[D_{r_1}] + O(N^{-1/12}(\ln N)^3).$$

We can get the asymptotic expansion for the Free energy noting that all terms involving m_N vanish and using that

$$\ln(N!) = N \ln(N) - N + \frac{1}{2} \ln(N) + \frac{1}{2} \ln(2\pi) + O(N^{-1})$$

that is,

$$\begin{aligned} \ln \mathcal{Z}_N &= \ln N! + \sum_{j=0}^{N-1} \ln h_j \\ &= -N^2 I_Q[\mu_Q] + \frac{1}{2} N \ln N + N \left(\frac{\ln(2\pi)}{2} - 1 - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{5}{12} \ln N \frac{\ln(2\pi)}{2} \\ &\quad + \zeta'(-1) + F_V[D_{r_1}] + O(N^{-1/12}(\ln N)^3). \end{aligned}$$

$\square$

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